The Pennsylvania State University
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MONTE CARLO SIMULATION OF TERRESTRIAL GAMMA-RAY FLASHES PRODUCED BY STEPPING LIGHTNING LEADERS

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Abstract

Terrestrial gamma-ray flashes (TGFs), discovered in 1994 by the Burst and Transient Source Experiment (BATSE) detector aboard the Compton Gamma-Ray Observatory, are high-energy photon bursts originating from the Earth's atmosphere. Space-borne measurements indicate that TGFs typically consist of single or multiple pulses, last from a few tens of microseconds to a few milliseconds, and exhibit very hard energy spectra that extend up to a few tens of MeV. Recently, the observations by Astrorivelatore Gamma a Immagini Leggero (AGILE) have revealed photons in the high-energy part of TGF spectra with energies extending up to 100 MeV. Since their discovery, TGFs have been clearly identified to stem from bremsstrahlung emissions of energetic electrons in close association with lightning activity. Moreover, recent measurements of TGF-associated radio signals have correlated TGFs with the initial development stages of normal polarity intracloud lightning that transports negative charge upward (+IC). As one of the most powerful natural particle accelerators on the Earth, TGFs have attracted extensive observational and theoretical interests over the last two decades.

Although numerous TGFs have been observed and various models have been developed, the production mechanism of TGFs remains a subject of active research. Two main production mechanisms have been proposed and provided consistent results with satellite measurements. The first mechanism is relativistic runaway electron avalanches (RREA) driven by the large-scale thunderstorm electric fields. This mechanism is now challenged by the recent discovery of extremely energetic gamma-rays in the high-energy tail of TGF spectra. The other viable mechanism is based on production of thermal runaway electrons by streamer discharges during negative corona flash stages of stepping lightning leaders. Strong support of this mechanism comes from the good agreements obtained between theoretical modeling and recent AGILE measurements. The purpose of the present dissertation is to improve our understanding of TGFs and the related phenomena by study-

ing theoretically the underlying electron runaway mechanisms using Monte Carlo techniques.

The first scientific question addressed in the present dissertation is that of the TGF production altitude. To tackle this problem, Monte Carlo modeling is used to simulate typical energy spectra at satellite altitudes based on the mechanism of direct acceleration of thermal runaway electrons in the lightning leader field. By comparing modeling results with RHESSI measurements, a source altitude of 12.4 km is determined for TGFs. This altitude appears to be deeper in the Earth's atmosphere and reconciles modeling results with recent radio observations of TGFs. Moreover, the properties of the photon source produced in this mechanism, including the beaming geometry and fluence, have also been quantified and are in good agreement with observational data-based studies.

The recently discovered, but unexplained, high-energy tail (up to 100 MeV) of TGF spectra is the second issue addressed in the present dissertation. Modeling results suggest that this high-energy tail can be explained using the production mechanism of thermal runaway electrons accelerated by the highly inhomogeneous electric field produced during the stepping of lightning leaders if extreme electric fields and potential drops are present. We also investigate the fundamental differences in energy spectra between this nonequilibrium acceleration process and RREAs developing in large-scale thunderstorm electric fields.

Despite various satellite and radio signal measurements, little observational data are available characterizing the TGF source. Another scientific question studied is whether and which optical emissions are generated during the production of TGFs. Optical emissions resulting from the excitation of air molecules produced by the large population of electrons involved in TGF events have been theoretically quantified for both production mechanisms (RREAs and stepping lightning leaders). Detailed discussion of the role of excitation and ionization collisions on the formation of the electron energy distribution is also presented. Moreover, we show that electron energy distributions established from the two TGF production mechanisms are inherently different over the full energy range. The strong energy dependence of the capability of electrons to generate excited states responsible for optical emissions from neutral and ionized nitrogen molecules leads to intrinsic differences in optical emissions produced by different mechanisms of TGF production. It is found that TGFs are likely accompanied by detectable levels of optical emissions and that the distinct optical features are of significant interest for constraining and validating current TGF production models.

In addition to TGFs, bursts of neutrons have been discovered in close association with lightning discharges. Although the origin of these neutrons remains a mystery, it has been suggested that photonuclear interaction of the large fluxes of gamma-rays generated during the production of TGFs may be responsible for these

events. To examine this mechanism, Monte Carlo model is employed to simulate photonuclear interactions between nuclei of air molecules and energetic photons involved in the production of TGFs. Modeling results suggest that significant amount of neutrons can be produced via photonuclear interaction undergone by high-energy photons resulting from the acceleration of thermal runaway electrons in the electric field of TGF-producing lightning leaders. Moreover, the energy distribution of these neutrons is quantified.

Intense and brief bursts of X-ray emissions have been measured during both natural cloud-to-ground and rocket-triggered lightning flashes. The RREA mechanism has been clearly ruled out for these X-ray emissions. Based on the thermal electron runaway mechanism, we predict the photon energy spectra produced by 5 and 10 MV negative CG lightning leaders that would be measured from the ground using ideal detectors. The radial dependence of X-ray energy spectra received at ground level during the leader stepping process is also investigated. Furthermore, using full energy Monte Carlo simulations combined with an optical emission model, we quantify the optical emissions induced by the strong acceleration of thermal runaway electrons associated with these lightning leaders. Modeling results demonstrate that this acceleration process is accompanied with not only high-energy X-ray emissions but also detectable levels of low-energy optical emissions.

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Dedication

to my father and mother,

Youli Feng *and* Fengqiong Xu

and to my brother,

Xuhui Feng

Chapter 1

Introduction

1.1 Electron Runaway Phenomena

In 1924, the Nobel Prize Winner C.T.R. Wilson first predicted the electron runaway phenomena in strong electric fields such as those of thunderclouds: "The faster the β-particle moves the smaller is the rate at which it loses energy in collisions. A particle may thus acquire energy corresponding to the greater part of the whole potential difference between the poles of the thundercloud" [*Wilson*, 1924]. Because the average energy per unit length that electrons acquire from external electric fields is higher than the energy loss due to interactions with air molecules, they can be continuously accelerated to higher energies and "run away". *Wilson* [1925] also suggested the production of X-ray and gamma-ray radiation from the electron runaway process: "some X and γ radiation must be produced and the upper limit of frequency will depend on the maximum energy which is possible for a particle to acquire in the field".

Kunhardt et al. [1986] have defined runaway electrons [*Gurevich*, 1961] as: "An electron is runaway if it does not circulate through all energy states available to it at a given reduced electric field E/N, but on average moves toward high-energy states." The runaway phenomenon is caused by the decreasing probability of interactions between air molecules and electrons in the energy range between ~100 eV and ~1 MeV [*Gurevich*, 1961]. This phenomenon can be readily illustrated by a direct comparison between the electric force exerted to free electrons by external electric fields and the dynamic friction force that they experience when moving in

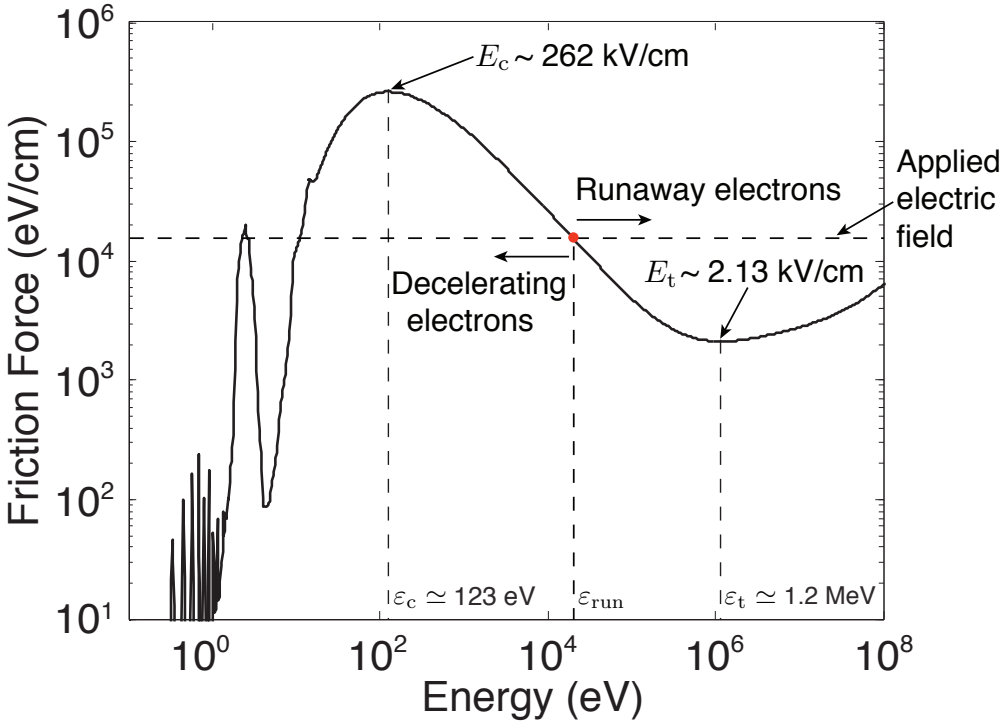

Figure 1.1. The dynamic friction force of electrons in air at ground level atmospheric density versus electron energy. The curve is calculated using the set of cross sections used in the Monte Carlo model of electrons for an air mixture of 80% N_2 and 20% O_2. E_c is the thermal runaway threshold electric field (∼262 kV/cm) and E_t is the relativistic runaway threshold electric field (∼2.13 kV/cm).

air (see Figure 1.1). The dynamic friction force is the mean energy loss of a free electron per unit length of travel due to collisions with air molecules and is highly energy-dependent:

$$\mathrm{F}_D(\varepsilon) = \sum_j N_j \sigma_j(\varepsilon) \delta\epsilon_j \tag{1.1}$$

where N_j is the partial density of target molecules, σ_j is the cross section of the j-th collision type, ε is the kinetic energy of incident electron, and $\delta\epsilon_j$ is the energy loss corresponding to the j-th type of collision. Dynamic friction force in air (80% N_2 and 20% O_2) at ground level atmospheric density calculated using the set of cross sections used in the present study is presented in Figure 1.1.

Gurevich [1961] first demonstrated that in the presence of extremely strong electric fields, with magnitudes greater than the thermal runaway threshold E_c

(∼262 kV/cm, see Figure 1.1), a large number of low energy electrons can be directly accelerated over the peak of dynamic friction force and even to relativistic energies, becoming thermal runaway electrons (see further discussion in Section 1.3.2). This mechanism is usually referred to as "thermal runaway" or "cold runaway". On the other hand, in 1992, *Gurevich et al.* [1992] discovered that when injected in an extended high-field region ($>E_t$, $\simeq$2.13 kV/cm), relativistic seed electrons (e.g., cosmic ray secondary electrons with energies above ε_t, see Figure 1.1) would undergo avalanche multiplications and result in a large population of relativistic runaway electrons and low-energy secondary electrons. This process was later named Relativistic Runaway Electron Avalanche (RREA) (see further discussion in Section 1.3.1).

The discovery of brief and intense gamma-ray bursts of atmospheric origin by [*Fishman et al.*, 1994a] in 1994 confirmed Wilson's predictions of electron runaway phenomena and energetic radiation from thunderstorms. These gamma-ray bursts originating from the Earth's atmosphere were later defined as Terrestrial Gamma-ray Flashes (TGFs). TGFs were first detected by the Burst and Transient Source Experiment (BATSE) aboard the Compton Gamma-ray Observatory (CGRO) [*Fishman et al.*, 1994a]. CGRO was launched into space in 1991 and BASTE was originally designed for the studies of cosmic gamma-ray bursts (GRBs) [*Mészáros*, 2006]. BATSE consisted of eight sets of scintillators, each with an effective area ∼2000 cm^2 facing outwards on the corners of the spacecraft. Each BATSE detector module comprised a Large Area Detector (LAD), a Spectroscopy Detector (SD), and the associated electronics, and was sensitive to X-rays and gamma-rays with energies ranging from ∼25 keV to ∼2 MeV [*Fishman et al.*, 1994b]. BATSE was capable of detecting hundreds of cosmic gamma-ray bursts per year and mapping their locations on the sky. TGFs were first identified by BATSE as bursts of gamma-rays with a duration much shorter than typical timescales of GRBs and much harder spectra.

1.2 Terrestrial Gamma-ray Flash Observations

After their discovery, TGFs have been further observed by the Reuven Ramaty High Energy Solar Spectroscopic Imager (RHESSI) [*Smith et al.*, 2005], the Fermi

Gamma-Ray Space Telescope [*Briggs et al.*, 2010], and the Astrorivelatore Gamma a Immagini Leggero (AGILE) satellite [*Marisaldi et al.*, 2010]. This section is devoted to a review of the literature related to satellite measurements of TGFs.

1.2.1 BATSE Observations

When the intensity of gamma-ray flux substantially exceeds the background noise, the detectors aboard CGRO would record their arrival time with an accuracy of ∼2 μs and the energy of incident photons in one of four energy channels (25–50 keV, 50–100 keV, 100–300 keV, and >300 keV) [*Fishman et al.*, 1994b]. During its 9-year lifetime in orbit (1991–2000), BATSE observed a total of 76 TGF events. BATSE measurements revealed that TGFs can consist of single or multiple pulses typically with a duration of ∼1 ms and a fluence ranging from 0.1 to 0.5 photons/cm^2, as shown in Figure 1.2. Figure 1.2 shows the characteristic time profiles of TGF events measured by BATSE. However, it has been later pointed out by *Grefenstette et al.* [2008] and *Gjesteland et al.* [2010] that BATSE measurements of TGFs suffer from significant dead time losses. Taking the dead time effect into account, *Gjesteland et al.* [2010] found that the true TGF source duration should be close to 250 μs.

BATSE observations of TGFs have provided crucial information to infer their relationship with lightning discharges. TGFs have been clearly identified to stem from the bremsstrahlung radiation of energetic electrons [e.g., *Fishman et al.*, 1994a; *Lehtinen et al.*, 1999; *Dwyer and Smith*, 2005]. Since the first measurement of TGFs, it has been noted by *Fishman et al.* [1994a] and later confirmed by *Inan et al.* [1996] that TGFs are closely correlated with lightning activity. This idea was further supported by *Cohen et al.* [2006] based on detailed analysis of the temporal and directional correlation between radio atmospherics and TGFs. As for the origin of this energetic phenomenon, although it was later shown by *Dwyer and Smith* [2005] to be unlikely, based on the analysis of BATSE measurements, *Inan et al.* [1996] suggested a possible TGF source of high-altitude runaway electron avalanches, possibly associated with a class of lightning-produced transient luminous events in the middle atmosphere, the so-called sprites [*Sentman et al.*, 1995].

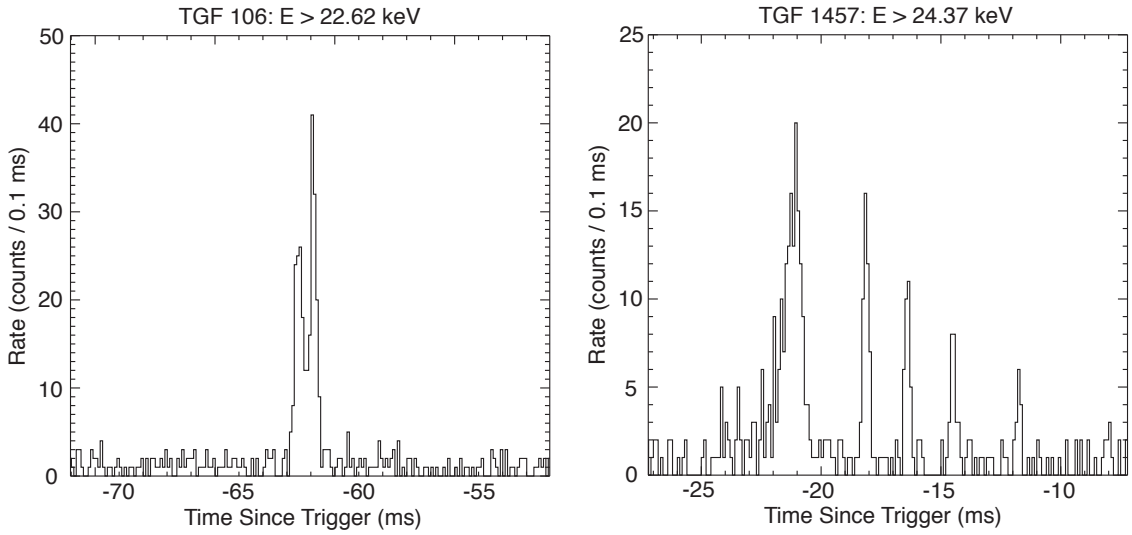

Figure 1.2. Time profiles of two TGF events observed by BATSE. The trigger numbers are TGF 106 and TGF 1457 in the BATSE catalog. Data are obtained from the Compton Observatory Science Support Center (COSSC) data archive at `http://gammaray.msfc.nasa.gov/batse/tgf`.

1.2.2 RHESSI Observations

The RHESSI satellite was a NASA Small Explorer spacecraft dedicated to studies of solar flares [*Smith et al.*, 2005]. RHESSI was launched on 5 February 2002, into an orbit with an inclination of 38° and an altitude of 600 km. An array of nine high-resolution germanium detectors was equipped in the tail of RHESSI in order to detect X-rays and gamma-rays from any direction in the sky [*Smith et al.*, 2005]. Since each arriving gamma-ray pulse was recorded individually and no onboard trigger time was necessary, RHESSI was more efficient in detecting TGFs than BATSE. As a result of this advantage, the frequency of TGF occurrence was found to be ∼50 per day globally [*Smith et al.*, 2005], which was more than an order of magnitude higher than that estimated from BATSE measurements. Moreover, even though the effective detecting area was smaller than BATSE, RHESSI was more sensitive to the energy of incident photons and was able to detect photons with energies in the range from 30 keV to 17 MeV. For this reason, TGFs were discovered, for the first time, to have gamma-rays with energies extending up to ∼20 MeV [*Smith et al.*, 2005].

Owing to the high efficiency of RHESSI in measuring TGFs, a cumulative TGF

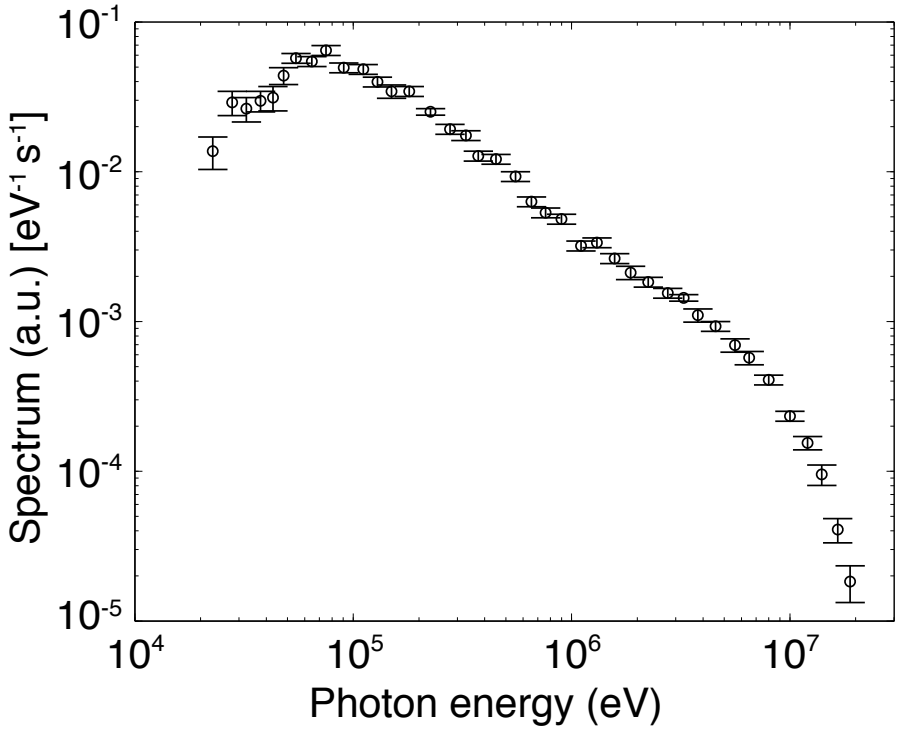

Figure 1.3. Cumulative TGF energy spectrum measured by RHESSI satellite. RHESSI data are obtained from [*Dwyer and Smith*, 2005].

energy spectrum was established [e.g., *Dwyer and Smith*, 2005], as shown in Figure 1.3. From this energy spectrum, one can trace back to the source altitude of this event considering that, for a given TGF source, the energy spectrum of gamma-rays at satellite altitude, especially the part below 10 MeV, is mainly defined by the propagation of gamma-rays in the Earth's atmosphere [e.g., *Dwyer and Smith*, 2005; *Xu et al.*, 2012]. Consequently, RHESSI measurements have been extensively used to constrain the source properties of TGFs. Based on the RREA mechanism, *Dwyer and Smith* [2005] have shown that the RHESSI spectrum is not consistent with a TGF source at altitudes above 24 km, but can be explained by a source in the range of 15–21 km. Modeling studies by *Carlson et al.* [2007] and *Hazelton et al.* [2009] support this finding by showing that the energy spectrum corresponding to high-altitude TGF source significantly deviates from RHESSI measurements. Furthermore, after correcting for dead time effects of BATSE measurements, *Østgaard et al.* [2008] and *Gjesteland et al.* [2010] have indicated that BATSE measurements

are also more consistent with low-altitude TGF source. More recently, *Xu et al.* [2012] determined a new source altitude of 12.4 km by comparing this combined RHESSI spectrum with modeling results based on the theory of direct acceleration of thermal runaway electrons produced by high-potential lightning leaders.

1.2.3 Fermi GBM Observations

The Fermi Gamma-ray Space Telescope is a space mission launched on June 11 2008 for the observation of GRBs [*Briggs et al.*, 2010]. It has an orbit at 565 km altitude with an inclination of 25.6°. The instruments on Fermi consist of two parts: the Large Area Telescope (LAT) and the Gamma-ray Burst Monitor (GBM). TGFs are mainly observed using GBM. GBM is composed of 12 uncollimated sodium iodide scintillators, which are capable of detecting photons with energies between ∼8 keV and ∼1 MeV, and two large bismuth germanate (BGO) scintillators to cover the photon energy range from ∼200 keV to ∼40 MeV [*Briggs et al.*, 2010]. Like BATSE, Fermi GBM also applies an onboard trigger algorithm with a 16 ms accumulation time and the temporal information of GBM measurements can achieve an accuracy of ∼20 μs. Moreover, a new analysis method, based on the downlinking of count data in GBM and offline searches of TGFs, has improved the detection frequency of TGFs by a factor of ten [*Briggs et al.*, 2013]. With this method, it is unveiled that Fermi can detect ∼850 TGFs per year and the global GBM-detectable TGF occurrence rate is approximately 400,000 TGF/year [*Briggs et al.*, 2013].

Measurements by Fermi GBM indicate that the representative brightness of TGFs is ∼0.7 photons/cm^2 and the most energetic gamma-rays for most TGFs are ∼30 MeV [*Briggs et al.*, 2010]. In addition, dedicated analysis on the time profiles of TGFs measured by Fermi GBM [*Fishman et al.*, 2011] have revealed that the duration of TGFs can be as brief as ∼50 μs (see the right panel of Figure 1.4) and is typically ∼100 μs. As shown in the left panel of Figure 1.4, Fermi GBM has also detected electron events originating from the Earth's atmosphere during TGFs [*Briggs et al.*, 2011], namely Terrestrial Electron Beams (TEB).

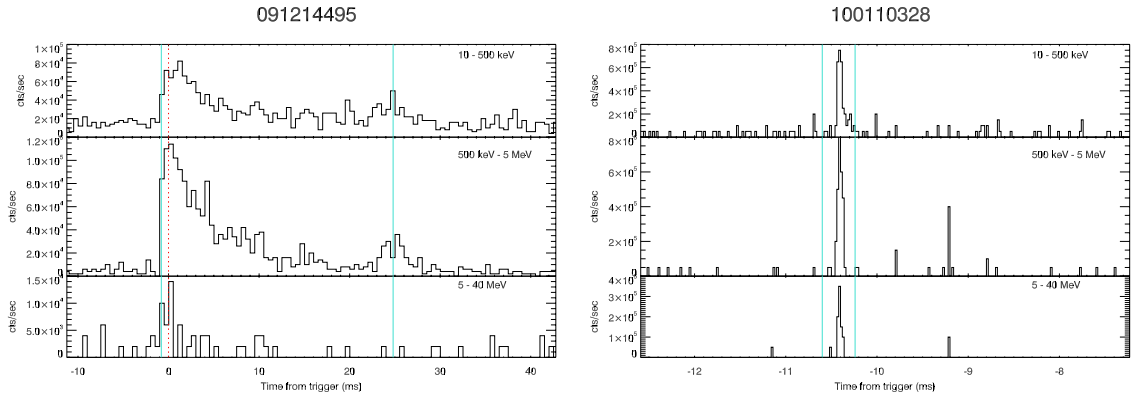

Figure 1.4. Light curves of two events measured by Fermi GBM. The left panel (TGF091214495) shows a possible electron TGF event reported in [*Briggs et al.*, 2011]. The right panel (TGF100110328) shows a brief TGF event with a duration of ∼50 μs [*Fishman et al.*, 2011]. Data are obtained from the Fermi Science Support Center data archive at http://gammaray.nsstc.nasa.gov/gbm/science/tgf.

1.2.4 AGILE Observations

AGILE is a space mission of the Italian Space Agency (ASI) that was launched on 23 April 2009 into a low-Earth orbit at 550 km altitude with 2.5° inclination [*Marisaldi et al.*, 2010]. AGILE consists of four instruments: a tungsten-silicon tracker (ST), a silicon based X-ray detector, a CsI(TI) mini calorimeter (MCAL), and an anticoincidence (AC) system. MCAL is the main instrument that detects TGFs. MCAL contains 30 scintillator bars that enable detecting extremely energetic photons in the range from 300 keV to 100 MeV [*Marisaldi et al.*, 2010]. The onboard time-tagging accuracy for photons is ∼1 μs and the data acquisition is based on a sub-millisecond trigger logic.

AGILE measurements have significantly improved our understanding regarding the energetics of gamma-rays in TGFs. The most energetic gamma-rays in TGFs as measured by AGILE were first found to be approximately 40 MeV [*Marisaldi et al.*, 2010]. Additionally, *Tavani et al.* [2011] discovered photons in the high-energy tail (>30 MeV) of TGFs with energies up to 100 MeV that can be fit by a power-law function. These measurements are shown in Figure 1.5. Although the RREA theory usually provides a very good agreement with satellite measurements at lower energies [*Dwyer and Smith*, 2005], it is now challenged by AGILE measurements (see Figure 1.5). However, *Celestin et al.* [2012] have shown that

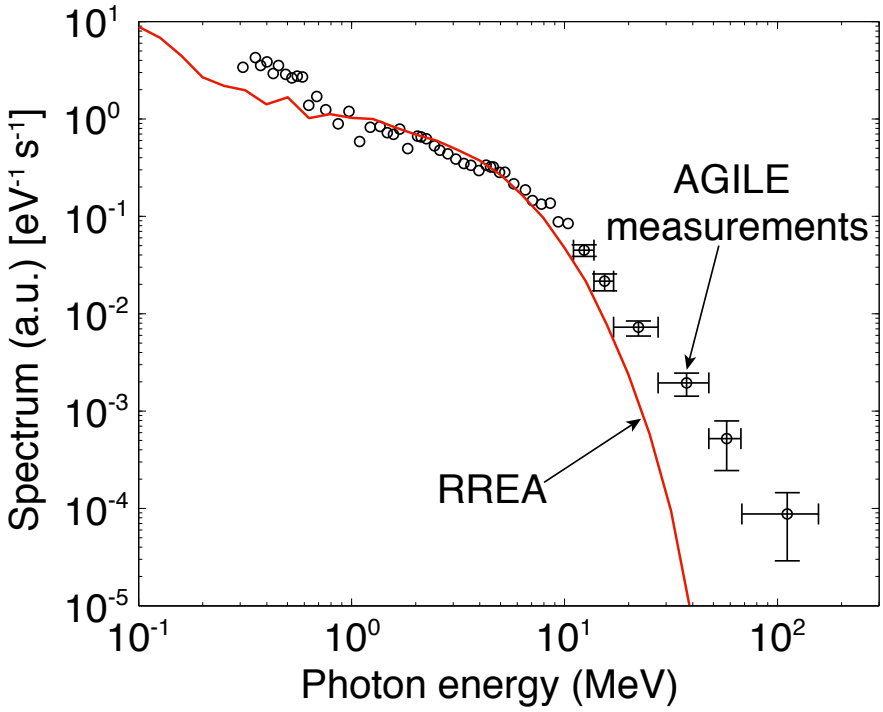

Figure 1.5. The cumulative energy spectrum of 130 TGFs detected by AGILE (Adapted from [*Tavani et al.*, 2011]). Also shown is the typical energy spectrum of TGFs derived from RREA theory.

this newly discovered high-energy tail can be explained by the nonequilibrium acceleration of thermal runaway electrons in the highly inhomogeneous electric field produced near the tip region by stepping lightning leaders.

1.3 TGF production mechanisms

Two main mechanisms have been proposed and developed for interpreting TGF observations. The first mechanism is that of RREA, which involves acceleration and multiplication of relativistic seed electrons under the application of large spatial scale electric fields within thunderstorms [*Gurevich et al.*, 1992; *Dwyer and Smith*, 2005]. The other proposed mechanism is based on production of thermal runaway electrons [*Gurevich*, 1961] during the negative corona flash stages of stepping intracloud lightning leaders [*Moss et al.*, 2006; *Celestin and Pasko*, 2011].

1.3.1 Relativistic Runaway Electron Avalanches

The initiation of RREA processes requires both pre-existing high-energy seed electrons (∼1.17 MeV, see Figure 1.1) and a large-scale (hundreds of meters [e.g., *Dwyer et al.*, 2012]) electric field that is strong enough to overcome the dynamic friction force of seed electrons. Natural background radiation or extensive cosmic-ray air showers could provide relativistic seed electrons to initiate RREAs. The electric field that is essential for triggering RREAs is called relativistic runaway threshold electric field E_t and its value in air at ground level, calculated using the set of cross sections adopted in present study, is approximately 2.13 kV/cm. We note that it is consistent with that reported in [e.g., *Dwyer et al.*, 2012]. However, this magnitude is the threshold field for avalanches of those relativistic seed electrons that travel along the electric field lines. Elastic scattering with air molecules causes significant deflection in the momentum of electrons and broadens the electron beam. Thus, the electric field that is required for the assembly of relativistic seed electrons to run away is approximately 30% larger than E_t, and has a magnitude of ∼2.8 kV/cm [e.g., *Symbalisty et al.*, 1998; *Dwyer*, 2003; *Babich et al.*, 2004].

Owing to the production of secondary electrons, the electron energy distribution function associated with the RREA process has a characteristic high-energy (>7 MeV) cutoff that produces a clear signature on the associated photon energy spectrum (see Figure 1.5). This energy cutoff is weakly dependent on the magnitude of driving electric field and ambient gas density [e.g. *Dwyer and Smith*, 2005]. This energy spectrum has so far successfully explained TGF observations in the energy region below 20 MeV as measured by RHESSI and BATSE [*Dwyer and Smith*, 2005; *Carlson et al.*, 2007]. However, since natural background radiation or extensive air showers have been shown to provide insufficient number of seed electrons for the RREA theory to explain the TGF fluences observed from satellites, secondary processes involving relativistic feedback mechanisms of positrons and X-rays have been invoked [e.g., *Dwyer*, 2008, 2012].

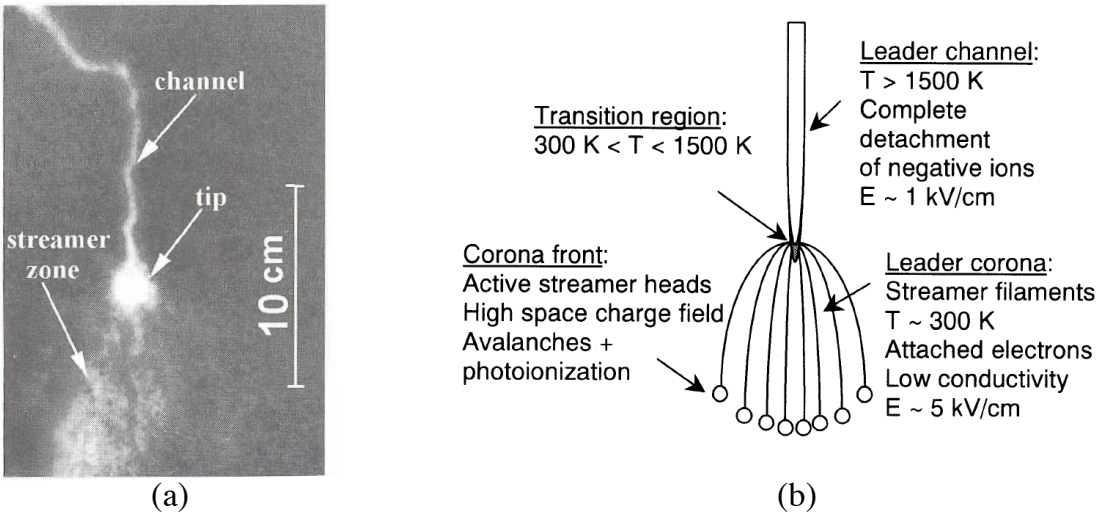

Figure 1.6. (a) An instantaneous photograph (0.1 μs exposure) of a leader channel and its tip [*Bazelyan and Raizer*, 2000, p. 66]. (b) A schematic of streamer-to-leader transition [*Comtois et al.*, 2003].

1.3.2 Thermal Runaway Processes

Thermal runaway processes [*Gurevich*, 1961] can be distinguished from RREAs by the necessity of an extremely intense electric field, with a magnitude exceeding the critical electric field E_c, corresponding to the peak value of dynamic friction force due to collisions with air molecules at ~123 eV (~262 kV/cm, see Figure 1.1). We note that this value is consistent with that documented in [*Moss et al.*, 2006]. An electric field with a magnitude greater than E_c is able to surmount the peak dynamic friction force and energize free low-energy electrons (<100 eV) to regimes where they can be continuously accelerated as long as the electric field is present. The high electric field around tips of propagating streamers during a negative corona flash stage of the development of negative stepped leader is one of the unique naturally occurring circumstances when an electric field higher than the critical electric field can be dynamically produced and sustained for relatively extended periods of time in air [e.g., *Moss et al.*, 2006].

1.3.3 Atmospheric Discharges

Streamers are narrow filamentary discharges that are omnipresent in atmospheric discharges. They are driven by highly nonlinear space charge waves [e.g., *Raizer*, 1991, p. 327]. At ground level, the streamer has a radius of 10^{-1}–10^{-2} cm and propagates with a velocity of 10^5–10^7 m/s. The dynamics of a streamer is mostly controlled by a highly enhanced field region, known as the streamer head, and different minimum electric fields are required for the propagation of positive and negative streamers [e.g., *Pasko*, 2008, and references therein]. Note that the polarity of a streamer discharge is determined by the polarity of the charge in the streamer head.

The breakdown in air gaps of many meters and in lightning discharges occurs via the growth of a *leader* (see Figure 1.6(a)): a thin channel that is conducting, with conductivity orders of magnitude higher than the streamer channel [*Raizer*, 1991, p. 327]. The key processes contributing to sustaining the conducting leader channel are accelerated detachment reactions and associative ionization by the elevated gas temperature in the channel [*Bazelyan and Raizer*, 2000, p. 59]. The head of the highly ionized and conducting leader channel is normally preceded by a streamer zone looking as a diverging column of diffuse glow, which is filled with highly branched streamers [e.g., *Bazelyan and Raizer*, 1998, p. 203, 253].

Many streamers connect to the tip of the leader and draw a large total current from it, and this large current can heat the gas in the leader channel to the required temperature for streamer-to-leader transition (see discussion in [*Pasko*, 2008]). As a self-consistent system, due to its high conductivity, the leader channel can be considered as equipotential and therefore plays the primary role in the focusing of the electric field in the streamer zone where relatively weakly conducting streamer coronas propagate [e.g., *Raizer*, 1991, p. 364]. Leaders of positive polarity attract electron avalanches, while in those of negative polarity the avalanching electrons move in the same direction as the leader head. In large experimental gaps (>100 m) and in thunderclouds, the electric fields required for propagation of positive and negative polarity leaders are known to be nearly identical (∼1 kV/cm).

The internal structure of streamer zones of positive and negative leaders, which is closely associated with the direction of electron avalanches, is very different (see [e.g., *Raizer*, 1991, p. 375; *Bazelyan and Raizer*, 1998, p. 253]). As opposed to

positive leaders, the propagation of negative leaders is dynamically characterized by a stepwise elongation observed in experiments and in natural lightning [*Bazelyan and Raizer*, 2000, p. 84]. One of the key components of the stepping process is the formation of a 'space leader', which originates near the external boundary of the negative streamer zone. The space leader propagates as a bi-directional discharge, whose positive end propagates toward the negative leader head [*Bazelyan and Raizer*, 2000, p. 85]. The junction of the space leader with the negative leader head closely resembles a miniature return stroke accompanied by a strong illumination of part of the entire leader channel [*Bazelyan and Raizer*, 2000, p. 87]. In fact, the tip of the main leader 'jumps' over to a new space, which was previously occupied by the space leader, and delivers to it the high potential of the previous leader head. The sudden rise of the space leader potential causes a high electric field leading to the inception of many streamers produced from the leader tip that constitute the negative corona flash.

Celestin and Pasko [2011] have shown that most of the thermal runaway electrons emitted from negative streamers during negative corona flashes are able to obtain an energy close to ∼65 keV. Moreover, the authors have suggested that the further acceleration of these runaway electrons in the electric field produced near the tip region of lightning leaders could generate sufficient number of photons to explain TGFs during intracloud lightning flashes (ICs), and X-ray bursts during negative cloud-to-ground lightning flashes (CGs). This TGF production mechanism is schematically illustrated in Figure 1.7. Modeling studies by *Celestin et al.* [2012] and *Xu et al.* [2012] support this mechanism by showing that the energy spectra of bremsstrahlung photons resulting from the accelerating process of thermal runaway electrons in lightning leader field are consistent with satellite measurements.

1.4 Problem Formulation

In this section, we list the research questions addressed in this dissertation. These research questions include the source altitude of TGFs, the recently discovered high-energy part of TGFs by AGILE, optical emissions associated with TGFs, neutron production by high-potential stepping lightning leaders, X-ray emissions

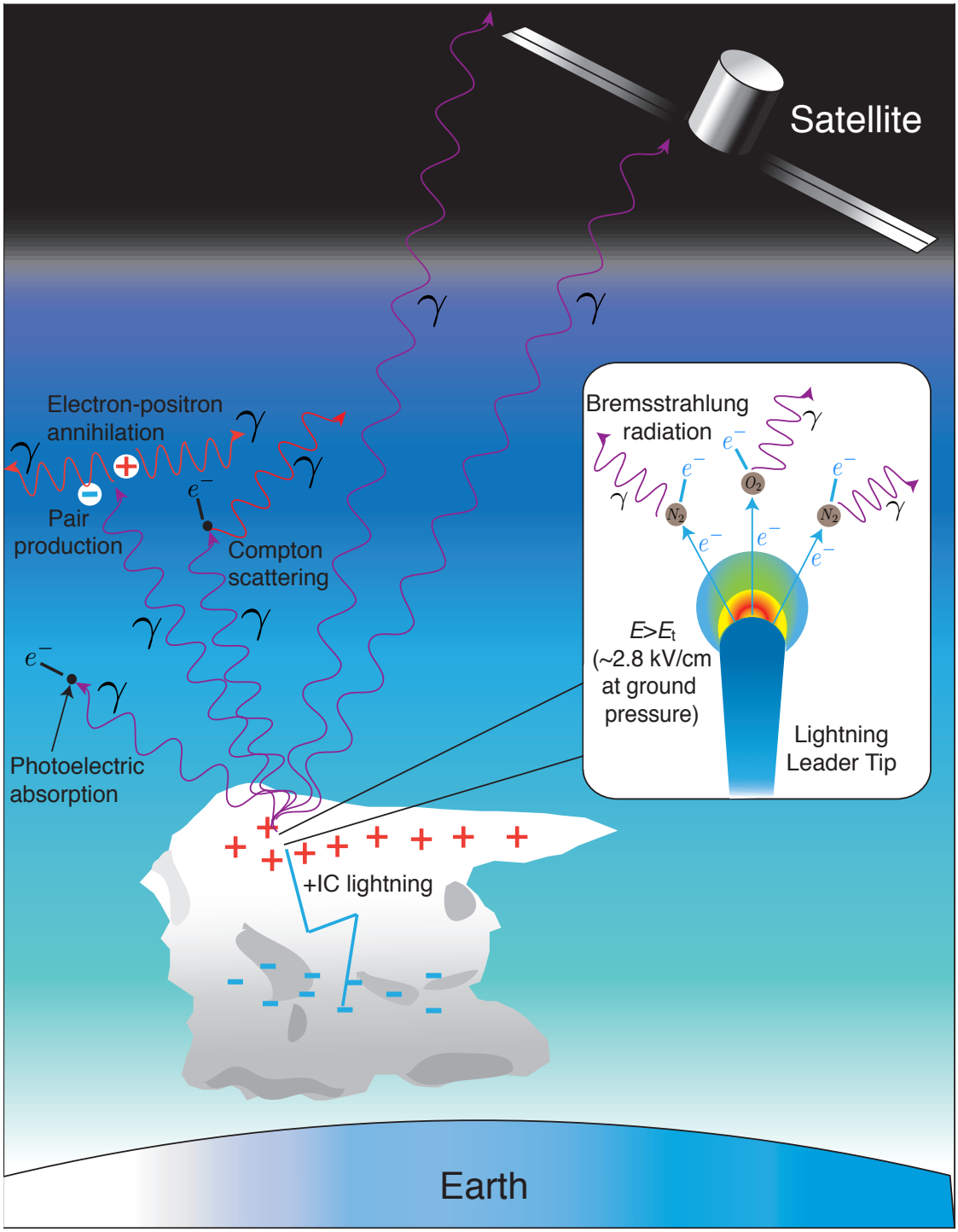

Figure 1.7. Schematic of a TGF production, including the generation mechanism, the photon transport in the Earth's atmosphere from the top of a thundercloud to satellite altitude, and detection of TGFs by satellite.

produced by stepping lightning leaders during CGs, and optical emissions associated with energetic electrons produced by stepping leaders in CGs.

1.4.1 Source Altitude of TGFs

Although numerous TGFs have been observed [e.g., *Fishman et al.*, 1994a; *Smith et al.*, 2005; *Grefenstette et al.*, 2009; *Briggs et al.*, 2010; *Marisaldi et al.*, 2010] and various models have been developed [e.g., *Dwyer and Smith*, 2005; *Carlson et al.*, 2007; *Østgaard et al.*, 2008; *Dwyer*, 2012], the production mechanism of TGFs is still uncertain. TGFs are associated with thunderstorm activities and originate from bremsstrahlung emission by energetic electrons [*Fishman et al.*, 1994a]. Although RREA theory [*Gurevich et al.*, 1992] has provided a very good agreement with satellite observations [*Dwyer and Smith*, 2005], it is now challenged by the recent measurements of the high energy part (>30 MeV) of the TGF spectrum measured by AGILE [*Tavani et al.*, 2011]. Moreover, observations have shown that TGFs are closely related to initial development stages of IC flashes [e.g., *Stanley et al.*, 2006; *Williams et al.*, 2006; *Shao et al.*, 2010; *Lu et al.*, 2010, 2011; *Cummer et al.*, 2011], suggesting a production altitude lower than originally thought. Specifically, *Stanley et al.* [2006] found two TGF-related intracloud lightning discharges at 13.6 km and 11.5 km. Analysis of TGF-related sferics by *Shao et al.* [2010] suggested a TGF source in the altitude range 10.5–14.1 km. *Lu et al.* [2010] also found that TGFs are produced during the initial development of intracloud lightning between a negative charge region centered at about 8.5 km and a positive region at 13 km altitude. These measurements motivate the present study to determine the TGF source altitude using the combined RHESSI TGF spectrum by simulating the transport of photons produced by energetic electrons generated during the stepping of a high-potential lightning leader.

1.4.2 Unexplained AGILE Measurements

Recently, *Tavani et al.* [2011] have reported that the high-energy part (>30 MeV) of the TGF spectrum, measured by the AGILE mission for the first time, significantly deviated from spectra corresponding to well-established model of RREA, which so far provided a very good agreement with observations at lower energies

[*Dwyer and Smith*, 2005]. Additionally, *Tavani et al.* [2011] discovered photons in the high-energy tail of terrestrial gamma-ray flashes with energies up to 100 MeV (see Figure 1.5). Theoretical understanding of the discrepancy between RREA processes and the nonequilibrium acceleration of energetic electrons produced during the stepping phase of lightning leaders and model-based interpretation of the recently discovered high-energy tail of TGF spectrum by AGILE are two important goals in this dissertation work. This dissertation work makes an original contribution by demonstrating that strong acceleration of a large number of thermal runaway electrons [*Celestin and Pasko*, 2011] in very strong and highly inhomogeneous electric fields of lightning discharge can reproduce recent AGILE measurements [*Tavani et al.*, 2011]. These results directly support the mechanism of TGF production by stepping lightning leaders.

1.4.3 Optical Emissions Associated with TGFs

Space-borne measurements indicate that TGFs typically consist of single or multiple pulses, last from a few tens of microseconds to a few milliseconds [e.g., *Fishman et al.*, 2011], and exhibit energy spectra that may extend up to 100 MeV [e.g., *Tavani et al.*, 2011]. In addition to satellite observations, radio signals radiated by TGF-associated lightning flashes have been used to infer the conditions that are required for the production of this high-energy phenomenon. Observational evidence shows that TGFs are closely associated with the initial development stages of normal polarity intra-cloud lightning that transports negative charge upward (+IC) [e.g., *Stanley et al.*, 2006; *Shao et al.*, 2010; *Lu et al.*, 2010]. Moreover, *Marshall et al.* [2013] have performed dedicated analyses on the initial breakdown (IB) stage of 10 IC flashes that emitted radio signals similar to TGF-producing ones and speculated that avalanches of relativistic runaway electrons caused IB pulses.

Although photon energy spectra established from the two production mechanisms described in Section 1.3 show good agreement with satellite measurements [e.g., *Dwyer and Smith*, 2005; *Xu et al.*, 2012; *Celestin et al.*, 2012], there are notable differences between these two mechanisms. Lightning flashes do not play a critical role in the RREA theory and the high-energy parts of the electron distribu-

tion and corresponding photon spectrum are weakly dependent on the magnitude of the homogeneous electric field driving the RREAs [i.e., *Babich et al.*, 2004; *Dwyer and Smith*, 2005], whereas the energy spectrum of gamma-rays emitted during the acceleration of thermal runaway electrons in the highly inhomogeneous electric fields around lightning leader tips is influenced by the physical properties and spatial electric field configurations associated with lightning stepped leaders [*Xu et al.*, 2012; *Celestin et al.*, 2012].

In spite of numerous experimental and theoretical efforts, little observational evidence is available about the TGF source. One of the most important unknowns is whether and which measurable optical emissions are generated during the production of TGFs. Optical emissions usually provide insightful knowledge about the energetics of electrons and the driving electric field in atmospheric discharges [e.g., *Celestin and Pasko*, 2010a] and are, therefore, extensively used in the studies of thunderstorm related Transient Luminous Events (TLEs) at high altitudes in the Earth's atmosphere [e.g., *Kuo et al.*, 2005, 2009; *Liu et al.*, 2006; *Adachi et al.*, 2006]. Recently, *Østgaard et al.* [2013] have reported optical emissions, detected for the first time, by the Lightning Imaging Sensor (LIS) on the Tropical Rainfall Measuring Mission (TRMM), from TGF-associated IC flashes. It is also worth mentioning that, based on high-speed camera observations, it has been suggested in [*Stolzenburg et al.*, 2013] that the impulsive breakdown associated with initial leaders during IB stages of cloud-to-ground (CG) and IC flashes can generate considerable amount of visible light. Furthermore, *Dwyer et al.* [2013] quantified optical emissions produced by relativistic feedback discharges based on the RREA mechanism. The present study aims at the development of an experimentally testable framework that would allow to distinguish between different production mechanisms of TGF events by quantifying the optical emissions generated by the large population of high- and low-energy electrons.

1.4.4 Neutron Production by High-potential Lightning Leaders

In 1985, *Shah et al.* [1985] first reported measurements of neutron generation during lightning discharges. Since their discovery, neutron flux increases have also been

observed in association with lightning activities either from the ground [*Shyam and Kaushik*, 1999; *Chilingarian et al.*, 2010; *Gurevich et al.*, 2012] or by satellite [*Bratolyubova-Tsulukidze et al.*, 2004]. Using solid state nuclear track recorder, *Shyam and Kaushik* [1999] estimated that neutrons with a number up to 10^9 can be produced during an atmospheric lightning discharge. Moreover, fluxes of neutrons, electrons, and gamma-rays, with a duration of several minutes, have been measured from the ground in correlation with thunderstorm activities [*Chilingarian et al.*, 2010]. More recently, *Gurevich et al.* [2012] have also reported the registration of high fluxes of low-energy neutrons, with a magnitude on the order of $\sim$0.03 neutrons/cm^2/s, during thunderstorms at the Tien-Shan Mountain Cosmic Ray Station. In addition to ground-based observations, discrete bursts of neutron originating from the Earth's atmosphere have also been detected by satellites [*Bratolyubova-Tsulukidze et al.*, 2004].

Despite of the above-mentioned measurements [*Shah et al.*, 1985; *Shyam and Kaushik*, 1999; *Bratolyubova-Tsulukidze et al.*, 2004; *Chilingarian et al.*, 2010; *Gurevich et al.*, 2012], the origin of these neutrons is still under debate. Even before their discovery, *Libby and Lukens* [1973] had suggested the possibility of neutron generation by nuclear reactions, primarily deuterium/deuterium fusion, during lightning discharge. However, *Babich* [2006] has shown that the efficiency of nuclear fusion reaction is too low to explain typical neutron observations and instead proposed that photonuclear interaction of energetic photons could be responsible for these measurements [*Babich and Roussel-Dupré*, 2007]. On the basis of this mechanism, *Babich et al.* [2010] have further shown that neutrons originating from photonuclear interaction of energetic photons in RREAs produced at altitudes between 8 km and 12 km could provide consistent results with ground-based measurements documented in [*Shah et al.*, 1985]. Furthermore, using GEANT4 simulations of photonuclear interactions and source properties of gamma-rays in TGFs, *Carlson et al.* [2010] have suggested that significant amount of neutrons is generated during the production of TGFs. One of the purposes of this dissertation work is to examine the production of neutron bursts through photonuclear interactions of energetic photons by high-potential stepping lightning leaders. Particularly, we investigate the characteristics of the resultant neutron bursts.

1.4.5 X-ray Emissions Produced by Stepping Lightning Leaders

Intense and very brief bursts of X-ray emissions have been detected from the ground during natural cloud-to-ground (CG) lightning flashes [*Moore et al.*, 2001] and rocket-triggered lightning flashes [*Dwyer et al.*, 2003]. These bursts are mainly composed of X-rays with energies extending up to a few hundreds of keV [e.g., *Dwyer et al.*, 2004a]. Measurements at the International Center for Lightning Research and Testing (ICLRT) have revealed that discrete and intense bursts of X-rays were time correlated with the formation of leader steps during CGs [*Dwyer et al.*, 2005]. This finding was extended by *Howard et al.* [2008], who have spatially and temporally co-located X-ray emissions with leader step electric field changes for both downward negative natural and rocket-triggered lightning flashes.

More recently, ground-based observations have provided valuable insights to the properties of these X-rays. By imaging X-ray emissions from triggered lightning dart leaders using a pinhole-type high-speed X-ray camera (XCAM), *Dwyer et al.* [2011] have demonstrated that the leader front is the source of the X-ray emissions. Moreover, from the analysis of X-ray measurements by the Thunderstorm Energetic Radiation Array (TERA) located at the ICLRT, *Schaal et al.* [2012] have indicated that energetic electrons, responsible for the production of these X-rays, exhibit a characteristic energy less than 3 MeV and the maximum production rate of electrons is on the order of 10^{17} electrons/s. Additionally, on the basis of XCAM measurements of the structure of X-ray emissions from triggered lightning leaders, *Schaal et al.* [2014] have found that X-rays can originate from either diffuse or compact source regions and the electrical charge contained within the compact source region, whose typical radius is between 2 and 3 m, is on the order of 10^{-4} C.

Despite the above-mentioned observational efforts, the exact production mechanism of X-rays during the stepping phase of lightning leaders is still not well understood. The mechanism of relativistic runaway electron avalanches (RREAs) in large-scale thunderstorm electric fields has been clearly ruled out as it cannot take place in the relatively small region in the vicinity of the leader tip and it is not capable of explaining the observed energy spectra [*Dwyer*, 2004]. On the

other hand, *Celestin and Pasko* [2011] have shown theoretically how the large flux of thermal runaway electrons generated by streamers during the negative corona flash stage of stepping lightning leaders could instead be responsible for these X-ray bursts during negative CGs, and for terrestrial gamma-ray flashes (TGFs) [*Fishman et al.*, 1994a] during intra-cloud lightning flashes (ICs). Our goal is to investigate the energy spectra of X-rays stemming from the acceleration of thermal runaway electrons in the electric field produced around lightning leader tips during CGs.

1.4.6 Optical Emissions Associated with Energetic Electrons Produced by Stepping Leaders in CGs

Recent high-speed video observations of *Stolzenburg et al.* [2013] have revealed bursts of light during the initial breakdown (IB) stage of both CGs and ICs and the authors suggested that the impulsive breakdown associated with initial leaders causes these visible light emissions. Spectroscopic analysis of optical emissions is usually used in the studies of Transient Luminous Events (TLEs) to infer the underlying electric field associated with these events [e.g., *Kuo et al.*, 2005; *Liu et al.*, 2006; *Celestin and Pasko*, 2010a]. One of the purposes of this dissertation work is to quantify theoretically the optical emissions induced by the acceleration process of thermal runaway electrons in the electric field present near the tip region of stepping CG lightning leaders with particular emphasis on the morphological and spectroscopic features.

1.5 Organization of the Dissertation

Chapter 1 is a review of the literature related to electron runaway phenomena, satellite observations of TGFs, and the two viable production mechanisms proposed for explaining TGFs. We also formulate the scientific questions addressed in this dissertation and briefly summarize the main scientific contributions of the work presented in this dissertation.

Chapter 2 is devoted to an introduction of the Method of Moment, the Monte Carlo model of electrons, the Monte Carlo model of photons, and the optical

emission model that are developed in the present dissertation to study TGFs.

In Chapter 3, we model the acceleration process of thermal runaway electrons under lightning leader electric field and investigate the influence of source altitude on the measured TGF energy spectrum in order to answer the scientific question described in Section 1.4.1.

In Chapter 4, in order to answer the scientific question described in Section 1.4.2, we model energetic electrons accelerating in inhomogeneous electric field produced during stepping of lightning leaders and the transport of bremsstrahlung photons in the Earth's atmosphere. We show that the energy spectrum obtained from this acceleration mechanism can accurately reproduce the recently discovered high-energy tail of TGF if extreme values of electric field and lightning leader potential are assumed.

The goal of Chapter 5 is to answer the scientific question described in Section 1.4.3. In particular, we quantify theoretically optical emissions resulting from the excitation of air molecules produced by the large population of electrons involved in TGF events based on two production mechanisms: RREAs and acceleration of thermal runaway electrons produced by high-potential IC lightning leaders. We also discuss the intrinsic differences in optical emissions generated by different mechanisms of TGF production.

In Chapter 6, we present modeling results of neutron production by energetic bremsstrahlung photons generated by high-potential stepping lightning leaders. The purpose of the work presented in this chapter is to answer the scientific question described in Section 1.4.4.

Chapter 7 is related to the scientific question described in Section 1.4.5. We predict theoretically the energy spectra of X-rays produced during the stepping process of lightning leaders in CG events. We also report simulation results of the photon energy spectra produced by 5 and 10 MV negative CG lightning discharges that would be measured from the ground using ideal detectors and estimated the associated radiation exposure to humans.

In Chapter 8, we answer the research question described in Section 1.4.6 by quantifying, from first principles, optical emissions induced by the strong acceleration of thermal runaway electrons in the highly inhomogeneous electric field produced by stepping lightning leaders. We mainly focus on the morphological

and spectroscopic features.

In Chapter 9, we provide suggestions of future research directions for the continuation of the work presented in this dissertation.

1.6 Scientific Contributions

Major advances related to the above research questions have been recently achieved and documented in [*Xu et al.*, 2012, 2014, 2015a, b; *Celestin et al.*, 2012], which can be summarized as follows:

- The source altitude of TGFs is determined based on the mechanism of direct acceleration of electrons in the lightning leader field. Modeling results suggest a TGF source altitude of 12.4 km for lightning leaders producing an electrical potential drop of 100 MV in the tip region. This altitude lies within the typical range of observations of TGF-producing +IC lightning and reconciles recent observations and measurements [*Xu et al.*, 2012]. The beaming geometry of source photons consistently calculated from this mechanism is in excellent agreement with previously published results.

- The recently observed high-energy part (>30 MeV) of TGF spectrum can be explained by thermal runaway electrons strongly accelerated in highly inhomogeneous fields of +IC lightning leader. This result provides the first direct evidence that TGFs are produced by lightning and not over large distances in weak thunderstorm electric fields [*Celestin et al.*, 2012].

- The characteristic electron energy distributions over the full energy range have been quantified for two TGF production mechanisms: RREAs and production of thermal runaway electrons during the negative corona flash stages of lightning leaders. The inherent differences in electron energy distribution between the two production mechanisms have been pointed out. It is also demonstrated that the strong energy dependence of the capability of electrons to generate excited states responsible for optical emissions from neutral and ionized nitrogen molecules leads to intrinsic differences in optical emissions produced by different mechanisms of TGF production. Moreover, TGFs are likely accompanied by detectable levels of optical emissions [*Xu et al.*, 2015b].

- Photon energy spectra produced by 5 and 10 MV negative CG lightning discharges that would be measured from the ground using ideal detectors have been predicted. The radial dependence of X-ray energy spectra received at ground level during the leader stepping process has also been pointed out. Furthermore, the ground radiation generated during this leader stepping process is evaluated to be harmless to humans [*Xu et al.*, 2014].

- Optical emissions induced by the strong acceleration of thermal runaway electrons in the highly inhomogeneous electric field produced by stepping lightning leaders, including the morphological and spectroscopic features, have also been quantified. Modeling results demonstrate that this acceleration process is accompanied with not only X-ray emissions but also detectable levels of low-energy optical emissions [*Xu et al.*, 2015a].

Chapter 2

Model Formulation

This chapter contains a description of the numerical models used in the present study, including the method of moments that is used to calculate the highly inhomogeneous electric field produced in the vicinity of lightning leader tips during the stepping process of CGs and ICs, a Monte Carlo model that is developed to simulate the propagation, dynamics, and collisions of electrons in air over the full energy range under various configurations of electric field, a Monte Carlo model that is developed to simulate the production of bremsstrahlung photons and their subsequent transport in the Earth's atmosphere, and an optical emission model that is developed to quantify optical emissions resulting from radiative relaxation of excited species of neutral and ionized molecules produced by energetic electrons.

2.1 Method of Moments

We use the method of moments [*Balanis*, 1989, p. 670] to calculate the electric field in the vicinity of negative lightning leader tip during the stage of negative corona flashes (see [*Celestin and Pasko*, 2011] and references therein for discussion of the related phenomenology). The negative corona flash corresponds to the production of a powerful streamer corona right after the completion of a new leader step [*Bazelyan and Raizer*, 2000, p. 199] and the extreme electric fields produced in the streamer heads during the corona flash are responsible for the production of thermal runaway electrons [*Moss et al.*, 2006; *Celestin and Pasko*, 2011].

Specifically, the method of moments is used to invert the integral equation of

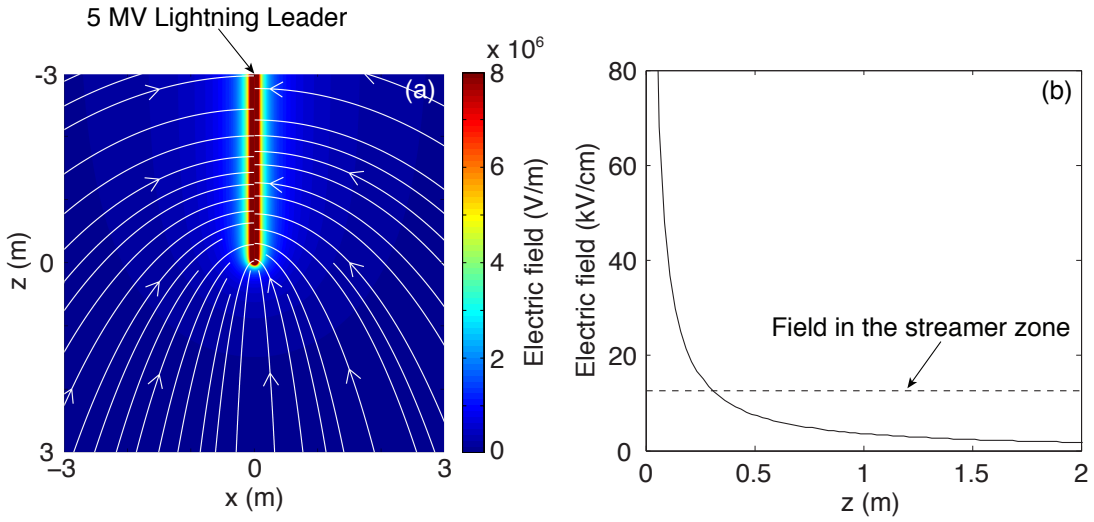

Figure 2.1. (a) Simulation results of the inhomogeneous electric field near the tip region of a lightning leader producing an electrical potential drop of 5 MV calculated using the method of moments. The leader channel has a length of 1 km with a radius of 1 cm and the ambient thundercloud electric field is 0.1 kV/cm. The leader tip is located at z=0. (b) Magnitude of the lightning leader electric field along the z axis. Also shown is the electric field in this region when the streamer zone is formed (∼12.5 kV/cm).

the electric potential in order to solve for the charge distribution along the leader channel. The charge is distributed so that the electric potential in the leader compensates the potential variation due to the ambient electric field, preserving equipotentiality of the leader channel. Once the charge density is obtained, the electric field produced by the leader channel can be further calculated. The electric potential of the lightning leader tip with respect to the ambient potential can be approximated by V_0=$E_0 l/2$ [*Bazelyan and Raizer*, 2000, p. 54] at the leader tip, where l is the length of the unbranched leader channel and E_0 is the ambient electric field.

Figure 2.1(a) shows simulation results of the inhomogeneous electric field near the tip region produced by a 5 MV lightning leader. In particular, we assume an ambient large-scale thunderstorm electric field of E_0=0.1 kV/cm [e.g., *Marshall et al.*, 2001]. The length of the lightning leader is considered to be l=1 km and the radius of the leader channel is 1 cm [*Rakov and Uman*, 2003, Section 4.4.6, p. 134]. The electric field is highly inhomogeneous because the electric field is intense close to the leader tip, but weak far away from it. Figure 2.1(b) shows the

characteristic magnitudes of the electric field along the z axis. We also show in this figure the electric field in the negative streamer zone after the streamer zone has been formed. The magnitude is roughly equal to the stability field of negative streamers propagation ($\sim$12.5 kV/cm) [*Bazelyan and Raizer*, 2000, p. 69].

The method of moments is a quasi-static approximation, which is valid only if the electric potential in the new leader step rises fast enough so that the electric field is not shielded. The timescale associated with the establishment of the new equipotential section of the leader limits the magnitude of the extreme electric field produced very close to the leader tip. We account for a timescale of 30 ns of the development of the new equipotential section of the leader [*Celestin and Pasko*, 2011]. The extreme electric field very close to the leader tip obtained from the method of moments is greater than the maximum sustainable local electric field during this timescale [*Celestin and Pasko*, 2011, Figure 7] and it is therefore is considered as unphysical. When inputing modeling results of electric field produced by lightning leaders in the Monte Carlo code, thermal runaway electrons are initially placed at proper distances from the leader tip in order to avoid accelerating them in the unphysical electric field region.

The electrical potential drop that stepping lightning leaders create in the tip region is a critical parameter in the present study as it directly controls the maximum energy that runaway electrons can gain from the leader field. Considering that bremsstrahlung photons are defined by the energetics of runaway electrons, the electrical potential difference also determines the energy spectrum of photons, especially the high-energy part. Indeed, we observe in the simulation that stepping lightning leaders in ICs with an electrical potential of 100 MV are capable of producing intense gamma-rays with energies up to a few tens of MeV, resembling typical satellite measurements of TGFs, whereas a 10 MV CG lightning leader leads to X-rays with energies less than 2 MeV, similar to ground-based measurements of X-ray bursts produced by lightning discharges. In addition, given that photonuclear interaction of energetic photons is strongly energy-dependent, the electrical potential difference is also critical in the studies of neutron production. For example, the threshold energy for photonuclear interaction with nitrogen nuclei is approximately 10.5 MeV [e.g., *Berman and Fultz*, 1975] and a lightning leader with an electrical potential drop of 10 MV would therefore be unable to generate

neutrons through this mechanism.

2.2 Monte Carlo Model of Electrons

In this section, we introduce a Monte Carlo model that is developed to simulate the propagation and collisions of electrons in air (80% N_2 and 20% O_2). In particular, we first introduce the set of cross sections adopted in the Monte Carlo model for describing collisions between electrons and air molecules in Section 2.2.1. We describe the relativistic binary-encounter-Bethe model developed to calculate ionization cross sections in Section 2.2.2 and the particle remapping technique employed to simulate the electron runaway processes with increasing number of particles in Section 2.2.3. Section 2.2.4 contains a description about the null collision technique and the execution of the Monte Carlo model. Validation of the Monte Carlo model is presented in Section 2.2.5.

2.2.1 Collisions with N_2 and O_2 Molecules

Monte Carlo simulation is distinguished from other numerical methods by the usage of random sampling to reconstruct physical or mathematical systems based on the knowledge of the stochastic processes undertaken in the system. Cross sections, describing the probabilities of collision occurrences, that are usually obtained either from theoretical calculations or experimental measurements are essential to Monte Carlo modeling of particles. The Monte Carlo model we have developed is three-dimensional (3-D) in the velocity space, 3-D in the configuration space, relativistic, and simulates the propagation of electrons from sub-eV to GeV and their collisions with air molecules. A total of 42 collisional processes is taken into account corresponding to a gas mixture of 80% N_2 and 20% O_2, as a representation of the composition of the Earth's atmosphere.

The interactions between electrons and N_2 molecules that are considered in the present Monte Carlo model, including elastic collision, rotational, vibrational, and electronic excitation collisions, and ionization collisions, are summarized in Table 2.1. Table 2.2 shows the 17 collision processes between electrons and O_2 molecules. For all collisions except ionization, the cross sections are taken from

Table 2.1. Collision Processes between Electrons and N_2 Molecules

Collision Type	Reaction	Threshold Energy (eV)
Elastic	$e + N_2 \rightarrow e + N_2$	-
Rotational	$e + N_2 \rightarrow e + N_2(rot)$	0.02
Vibrational	$e + N_2 \rightarrow e + N_2(v = 1)$	0.29
	$e + N_2 \rightarrow e + N_2(v = 1)$	0.291
	$e + N_2 \rightarrow e + N_2(v = 2)$	0.59
	$e + N_2 \rightarrow e + N_2(v = 3)$	0.88
	$e + N_2 \rightarrow e + N_2(v = 4)$	1.17
	$e + N_2 \rightarrow e + N_2(v = 5)$	1.47
	$e + N_2 \rightarrow e + N_2(v = 6)$	1.76
	$e + N_2 \rightarrow e + N_2(v = 7)$	2.06
	$e + N_2 \rightarrow e + N_2(v = 8)$	2.35
Electronic	$e + N_2 \rightarrow e + N_2(A^3\Sigma_u^+, v = 1–4)$	6.17
	$e + N_2 \rightarrow e + N_2(A^3\Sigma_u^+, v = 5–9)$	7.00
	$e + N_2 \rightarrow e + N_2(B^3\Pi_g)$	7.35
	$e + N_2 \rightarrow e + N_2(W^3\Delta_u)$	7.36
	$e + N_2 \rightarrow e + N_2(A^3\Sigma_u^+, v = 10+)$	7.80
	$e + N_2 \rightarrow e + N_2(B'^3\Sigma_u^-)$	8.16
	$e + N_2 \rightarrow e + N_2(a'^1\Sigma_u^-)$	8.40
	$e + N_2 \rightarrow e + N_2(a^1\Pi_g)$	8.55
	$e + N_2 \rightarrow e + N_2(w^1\Delta_u)$	8.89
	$e + N_2 \rightarrow e + N_2(C^3\Pi_u)$	11.03
	$e + N_2 \rightarrow e + N_2(E^3\Sigma_g^+)$	11.88
	$e + N_2 \rightarrow e + N_2(a''^1\Sigma_g^+)$	12.25
	$e + N_2 \rightarrow e + N_2^*$ (sum of singlet states)	13.00
Ionization	$e + N_2 \rightarrow e + e + N_2^+(X^2\Sigma_g^+ + A^2\Pi_u)$	15.58

the BOLSIG+ database [*Hagelaar and Pitchford*, 2005]. The cross sections of ionization with N_2 and O_2 molecules over the full energy range are calculated using the relativistic binary-encounter-Bethe (RBEB) model based on the description in [*Kim et al.*, 2000; *Kim and Desclaux*, 2002; *Santos et al.*, 2003; *Celestin and Pasko*, 2010b]. However, cross sections in the BOLSIG+ database are provided between the threshold energy and 1 keV. In order to perform full energy simulations, excitation cross sections are logarithmically extrapolated to high energies. We note that this approximation is valid because values of cross sections for energies >1 keV are small and their effects are negligible. The elastic scattering of high-

Table 2.2. Collision Processes between Electrons and O_2 Molecules

Collision Type	Reaction	Threshold Energy (eV)
Elastic	$e + O_2 \rightarrow e + O_2$	-
Rotational	$e + O_2 \rightarrow e + O_2(rot)$	0.02
Vibrational	$e + O_2 \rightarrow e + O_2(v = 1, \varepsilon < 4\ eV)$	0.19
	$e + O_2 \rightarrow e + O_2(v = 2, \varepsilon < 4\ eV)$	0.38
	$e + O_2 \rightarrow e + O_2(v = 3)$	0.57
	$e + O_2 \rightarrow e + O_2(v = 4)$	0.75
	$e + O_2 \rightarrow e + O_2(v = 1, \varepsilon > 4\ eV)$	0.19
	$e + O_2 \rightarrow e + O_2(v = 2, \varepsilon > 4\ eV)$	0.38
Electronic	$e + O_2 \rightarrow e + O_2(a'^1\Delta_g)$	0.977
	$e + O_2 \rightarrow e + O_2(b^1\Sigma_g^+)$	1.627
	$e + O_2 \rightarrow e + O_2(c^1\Sigma_u^-)$	4.50
	$e + O_2 \rightarrow e + O(^3P) + O(^3P)$	6.00
	$e + O_2 \rightarrow e + O(^3P) + O(^1D)$	8.40
	$e + O_2 \rightarrow e + O(^1D) + O(^1D)$	9.97
Ionization	$e + O_2 \rightarrow e + e + O_2^+(X^2\Pi_g)$	12.07
Three-body Attachment	$e + O_2 + A \rightarrow O_2^- + A$	-
Two-body Attachment	$e + O_2 \rightarrow O^- + O$	-

energy (>500 eV) electrons by nitrogen and oxygen molecules is modeled using a shielded-Coulomb cross section, similar to that described in [*Dwyer*, 2007].

Differential cross sections are usually utilized to find the energy loss and scattering angle of particles after collisions. Specifically, the scattering angles of primary and secondary electrons after ionization collision are obtained from the relativistic equations of conservation of momentum and energy after finding the energy of the secondary electron using the RBEB differential cross section (see Section 2.2.2). As for any other collisional processes, the elastic angular differential cross section documented by *Shyn et al.* [1972] and *Kambara and Kuchitsu* [1972] is used to find out the scattering angles of primary electrons with energies lower than 500 eV. Concerning higher energies, we use the shielded-Coulomb differential cross section to calculate the scattering angle of primary electrons.

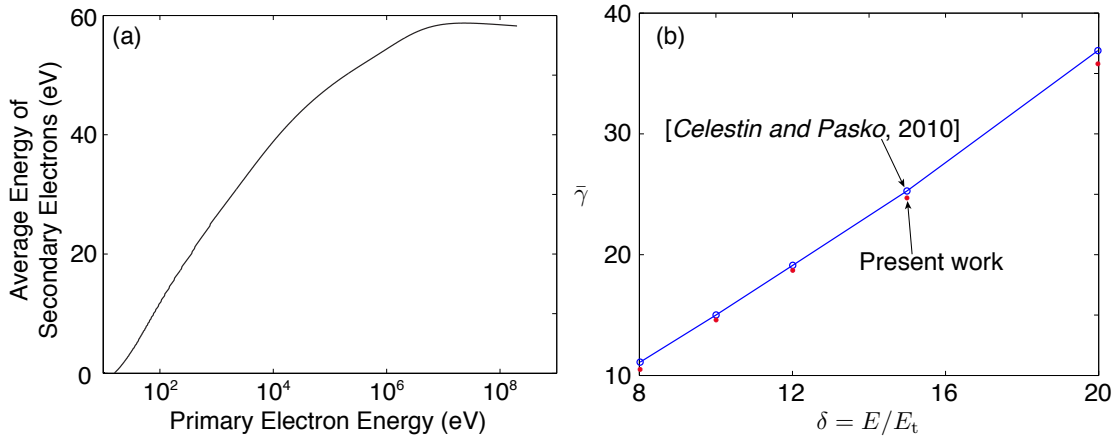

Figure 2.2. (a) Average energy of secondary electrons generated in ionization collisions as a function of the primary electron energy. (b) Comparison of RREA rates between present work and modeling results documented in [*Celestin and Pasko*, 2010b]. Results in the present work are calculated using the ionization cross sections derived from RBEB model.

2.2.2 Relativistic Binary Encounter-Bethe Model

Critical to the study of electron runaway phenomenon is an accurate and continuous description of ionization collisions over many decades of electron energy, i.e., from sub-electronvolts to several megaelectronvolts. The RBEB model [*Kim et al.*, 2000], owing to its analytical calculation of ionization cross section using the free-free Møller cross section and the leading dipole part of the relativistic Bethe cross section, is chosen here. This model is the relativistic extension of the binary-encounter-Bethe (BEB) model developed by *Kim and Rudd* [1994], which provides very good results in the low energy region when applied to neutral molecules [e.g., *Hwang et al.*, 1996]. This model only requires the binding energy of the target electron and the average orbital kinetic energy to provide an accurate description of the ionization collision.

The average energy of a secondary electron emerging from an ionization collision can be calculated using:

$$\langle \varepsilon_s (\varepsilon_p) \rangle = \frac{1}{\sigma_i (\varepsilon_p)} \int_0^{\frac{\varepsilon_p - \varepsilon_{iz}}{2}} \varepsilon_s \sigma (\varepsilon_p, \varepsilon_s) \, d\varepsilon_s \tag{2.1}$$

where ε_p is the energy of the primary electron, ε_s is the energy of the secondary

electron, ε_{iz} is the ionization energy, σ_i is the total ionization cross section, and $\sigma(\varepsilon_p, \varepsilon_s)$ is the singly differential cross section of ionization collision derived from the RBEB model. The ion is assumed to be at rest after the ionization collision. Therefore, the kinetic energy transferred to the ion is neglected and the energies of incident and secondary electrons must sum to $\varepsilon_p - \varepsilon_{iz}$ and by symmetrical about $\frac{\varepsilon_p - \varepsilon_{iz}}{2}$ [e.g., *Opal et al.*, 1971]. The quantity $\langle \varepsilon_s(\varepsilon_p) \rangle$ calculated using equation (2.1) is presented in Figure 2.2(a).

RREA rate is a parameter used for describing the avalanche and multiplication processes in RREAs. In a homogeneous electric field, the number of runaway electrons N_{run} increases exponentially in time and can be approximated by the relation $N_{\text{run}}{=}N_0 \exp(\gamma t)$, where N_0 is the initial number of relativistic runaway electrons used in the simulation, and γ is the RREA growth rate. Figure 2.2(b) shows the RREA rate presented in terms of $\bar{\gamma}{=}\gamma(ns^{-1}){\times}171.53$ ns with respect to δ, where $\delta{=}E/E_{\text{t}}$, E is the applied electric field, and E_{t} is the RREA threshold electric field taken as 2.13 kV/cm [*Celestin and Pasko*, 2010b]. The RREA rates computed using the current RBEB model are in good agreement with those presented by *Celestin and Pasko* [2010b] that used a purely high-energy model with collisional friction (stopping power) based on the Bethe's formula.

2.2.3 Particle Remapping Technique

TGFs are believed to be a result of electron runaway processes that are characterized by abundant generation of large quantities of secondary electrons. Ionization of air molecules by runaway electrons continuously increases the total number of particles in Monte Carlo simulations. To model this rapidly growing system, we employ a two-bin remapping technique [*Moss et al.*, 2006], similar to the statistical weighting described in [*Kunhardt and Tzeng*, 1986].

The particle remapping technique is applied once the total number of particles in the system exceeds a predetermined value. Specifically, this technique is implemented as follows: the assembly of electrons is first sorted in order of increasing energy and then partitioned into two groups, a high-energy group and a low-energy group. The boundary between these two groups depends on the accuracy that is required for high-energy electrons. The high-energy group is preserved in order

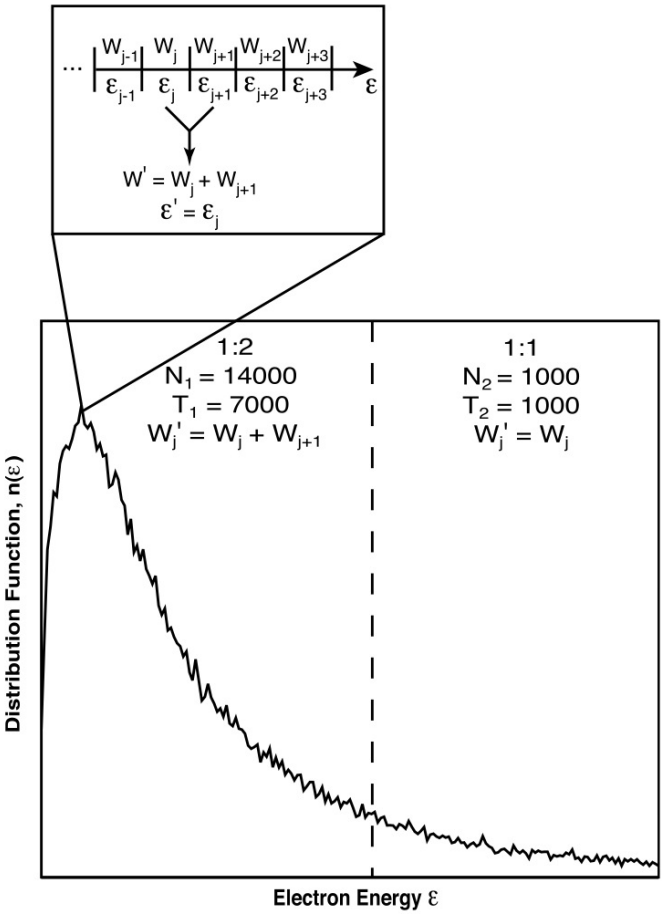

Figure 2.3. Illustration of the particle remapping technique. Adapted from [*Moss et al.*, 2006, Figure 10].

to maintain the statistics of high-energy electrons while the low-energy particles are remapped in order to limit the number of particles to that need to be tracked. Two adjacent particles in the low-energy group are merged into a new particle with a statistical weight assigned to represent the real number of electrons contained (super particle). This technique ensures precise description of runaway electrons, which is critical in the studies of TGF events.

Figure 2.3 is adapted from [*Moss et al.*, 2006] and illustrates this remapping technique. As shown in this figure, the ensemble of electrons is first sorted in energy and then separated into two groups. Only the 14000 electrons in the low-energy group are remapped. The statistics to the right of the dashed line is maintained. We note that this technique has been validated by calculating the RREA rates

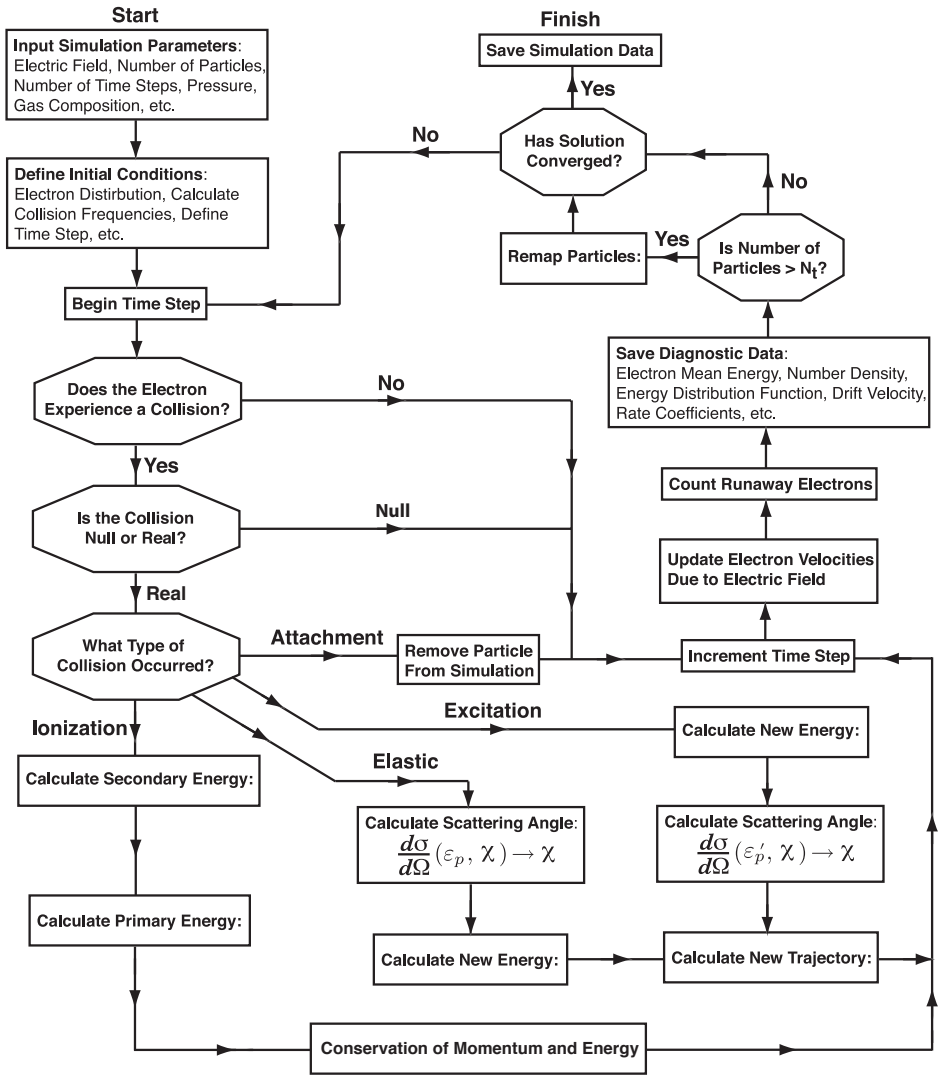

Figure 2.4. Flow chart depicting the execution of the Monte Carlo model of electrons. Adapted from [*Moss et al.*, 2006, Figure 12].

and typical RREA spectra and comparing with previously published results (see Figure 2.2(b) and Figure 2.5(c)).

2.2.4 Null Collision Technique

The null collision technique [e.g. *Moss et al.*, 2006] allows us to keep track of an ensemble of particles evolving with a unique time parameter by predefining a constant time step in order to cover the maximum collisional frequency of all

electrons modeled.

The constant time step is determined using the total collision frequency corresponding to all collision processes of interest. The total collision frequency is defined as the sum of all elastic and inelastic (see Tables 2.1 and 2.2) collisional frequencies and can be calculated using:

$$\nu_t(\varepsilon) = \sum_{j=1}^{n} \nu_j(\varepsilon) = \sum_{j=1}^{n} N_j \sigma_j(\varepsilon) v(\varepsilon) \tag{2.2}$$

where ε is the energy of incident electron, j represents a specific type of collision, ν_j is the frequency of the j-th type of collision, N_j is the partial density of target molecules, σ_j is the cross section of the j-th collision type, and v is the velocity of the electron. The constant time step $\triangle t$ is further calculated using:

$$\triangle t = \frac{\delta}{\max\left[\nu_t(\varepsilon)\right]} \tag{2.3}$$

where δ is an arbitrary number lower than 1. In this study, it is assumed to be equal to 0.1.

However, during each time interval $\triangle t$, an electron may or may not experience a collision with air molecules and the type of collision that it experiences largely depends on the energetics of this electron. For each electron, three random numbers are drawn and compared with the probabilities of collisions. The first random number is compared with the probability of the occurrence of collision over $\triangle t$, P_{coll}, to determine if a collision occurs. The second random number is compared with the probability of a real collision, P_{real}, to determine if the collision is real or null. The third random number is compared with the probabilities of all collisional processes, P_j, to determine the type of collision that occurs. P_{coll}, P_{real}, and P_j are calculated using the following formulas:

$$\begin{aligned} P_{\text{coll}} &= 1 - e^{-\delta} \\ P_{\text{real}} &= \frac{\nu_t(\varepsilon)}{\max[\nu_t(\varepsilon)]} \\ P_j &= \frac{\nu_j(\varepsilon)}{\nu_t(\varepsilon)} \end{aligned} \tag{2.4}$$

Figure 2.4 is a flow chart adapted from *Moss et al.* [2006] that shows the null collision method and the execution of the current Monte Carlo model. After the type of collision is determined, the energy loss and deflection angle of the electron are further calculated.

2.2.5 Model Validation

In order to perform a validation of this Monte Carlo model, we have calculated the mean energies of electrons driven by different magnitudes of uniform electric field and compared modeling results with those calculated using Boltzmann equation solvers ELENDIF [*Morgan and Penetrante*, 1990] and BOLSIG+ [*Hagelaar and Pitchford*, 2005], and Monte Carlo models presented by *Moss et al.* [2006]. The comparison is shown in Figure 2.5(a). One sees in this figure that, for a wide range of electric field values, the present results of electron mean energies are in excellent agreement with ELENDIF and BOLSIG+ calculation, and results reported by *Moss et al.* [2006].

In addition, we have calculated the energy distribution of relativistic runaway electrons in the full energy range driven by an electric field of 70 Td ($\sim$18.8 kV/cm at $N\simeq 2.688\times10^{25}$ m^{-3}, 1 Td=10^{-21} Vm^2) in RREAs and compared modeling results with BOLSIG+ calculation and results documented in [*Colman et al.*, 2010], as shown in Figure 2.5(b). The inset is a zoom-in view of the energy range below 20 eV. In spite of the noise in the energy distribution, it is clearly shown that over six orders of magnitude, the low-energy part (<100 eV) is consistent with recently published full energy range model results [*Colman et al.*, 2010, Figure 11] and BOLSIG+ calculation [*Hagelaar and Pitchford*, 2005]. Moreover, the high-energy part is also consistent with typical energy cutoff in RREAs. We note that the ratio between the number of electrons with energy below and above 1 MeV is found to be 9365, in agreement with the analytical calculation documented in [*Dwyer and Babich*, 2011]. Based on these comparisons, we conclude that our results on the electron energy distribution are verified over more than 15 orders of magnitude of the dynamic range.

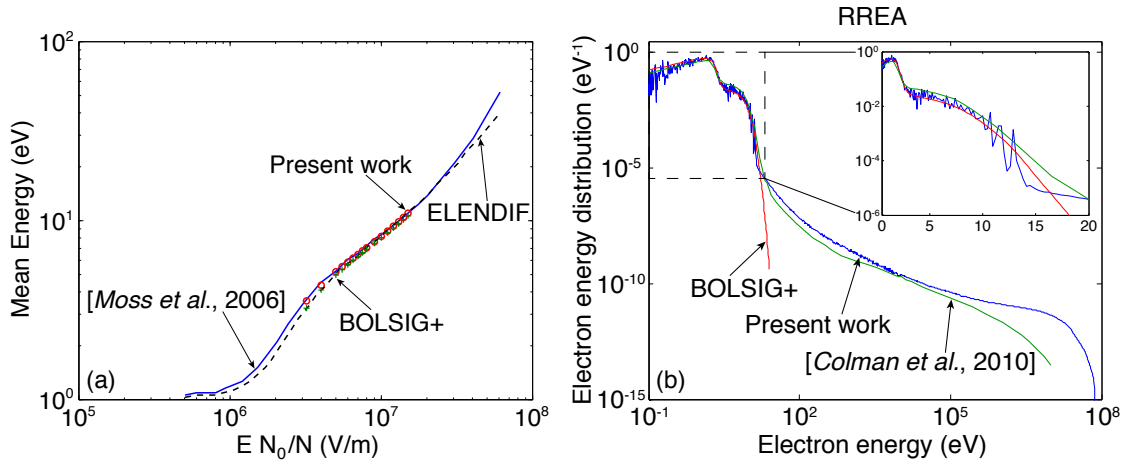

Figure 2.5. (a) Comparison of electron mean energies for different applied electric fields when steady-state is reached, between present calculation, ELENDIF [*Morgan and Penetrante*, 1990] calculation, BOLSIG+ [*Hagelaar and Pitchford*, 2005] calculation, and results reported in [*Moss et al.*, 2006]. (b) Distribution of electrons in the full energy range produced by the acceleration and multiplication of relativistic runaway electrons in a large-scale homogeneous electric field of 70 Td (18.8 kV/cm). The results are obtained in air at ground level ($N_0 \simeq 2.688 \times 10^{25}$ m^{-3}). Also presented are BOLSIG+ calculations [*Hagelaar and Pitchford*, 2005] and modeling results presented in [*Colman et al.*, 2010, Figure 11]. The inset is a zoom-in view of the energy range below 20 eV.

2.3 Monte Carlo Model of Photons

2.3.1 Bremsstrahlung Radiation

When an electron passes through the field of a nucleus (or atom), it is briefly accelerated and can emit radiation. This process is called bremsstrahlung radiation (or braking radiation). In the present study, we model this process using a factorization of the angular and energy parts of the differential cross section [*Lehtinen*, 2000, p. 45–49]. The doubly differential cross section of bremsstrahlung radiation is:

$$\frac{\partial^2 \sigma}{\partial \varepsilon_{ph} \partial \Omega} = \Phi(\theta) \frac{d\sigma}{d\varepsilon_{ph}} \tag{2.5}$$

where θ is the angle between the momenta of the incident electron and the emitted photon, Ω is the solid angle in the photon momentum space, $\frac{d\sigma}{d\varepsilon_{ph}}$ is the singly differential cross section for producing a photon with an energy of ε_{ph} in a unit interval of ε_{ph} that is integrated over all directions of the produced photon momen-

tum, and $\Phi(\theta)$ characterizes the angular distribution of bremsstrahlung radiation and is normalized so that $\int_{4\pi}\Phi(\theta)d\Omega$=1. The function $\Phi(\theta)$ is given by [*Jackson*, 1975, p. 705]:

$$\Phi(\theta)=\frac{3}{16\pi}\frac{1}{\gamma^2(1-\beta\cos\theta)^2}\left[1+\frac{(\cos\theta-\beta)^2}{(1-\beta\cos\theta)^2}\right] \tag{2.6}$$

where γ=1+$\varepsilon/(mc^2)$ is the electron relativistic factor, ε is the electron energy, m is the electron rest mass, c is the speed of light, and β=v/c is the dimensionless velocity.

The singly differential cross section used for finding the energy of emitted photon is calculated based on Born approximation [*Heitler*, 1954, p. 245] without considering the screening effect [*Lehtinen*, 2000, p. 48]:

$$\frac{d\sigma}{d\varepsilon_{ph}}=\chi_0\frac{4}{k}\left\{\left[1+\left(\frac{\gamma'}{\gamma}\right)^2-\frac{2}{3}\frac{\gamma'}{\gamma}\right]\log(183Z_a^{-1/3})+\frac{1}{9}\frac{\gamma'}{\gamma}\right\} \tag{2.7}$$

where χ_0=$\frac{Z_a(Z_a+1){r_0}^2}{137}$, Z_a is the atomic number, r_0 is the classical electron radius, k=$\varepsilon_{ph}/(mc^2)$ is the dimensionless photon energy, γ and γ' are, respectively, the relativistic factors of electron before and after collisions. It is important to note that the bremsstrahlung cross section has been validated by the good agreement obtained with previous calculations of TGF energy spectra in RREAs (see Figure 3.2(a)).

2.3.2 Photon Transport in the Earth's Atmosphere

The Monte Carlo model developed to simulate photon transport through the Earth's atmosphere is similar to that described in [*Østgaard et al.*, 2008]. Photons with energies between 10 keV and 100 MeV can be simulated. Four main types of collisions with air molecules are considered: photoelectric absorption, Compton scattering, electron-positron pair production, and photonuclear interaction, as shown in Figure 2.6.

Photoelectric absorption, in which a photon is absorbed by electrons bound in the molecules and an electron is kicked out, is the most probable collision process for photons with energies below 30 keV. The analytical formula documented in

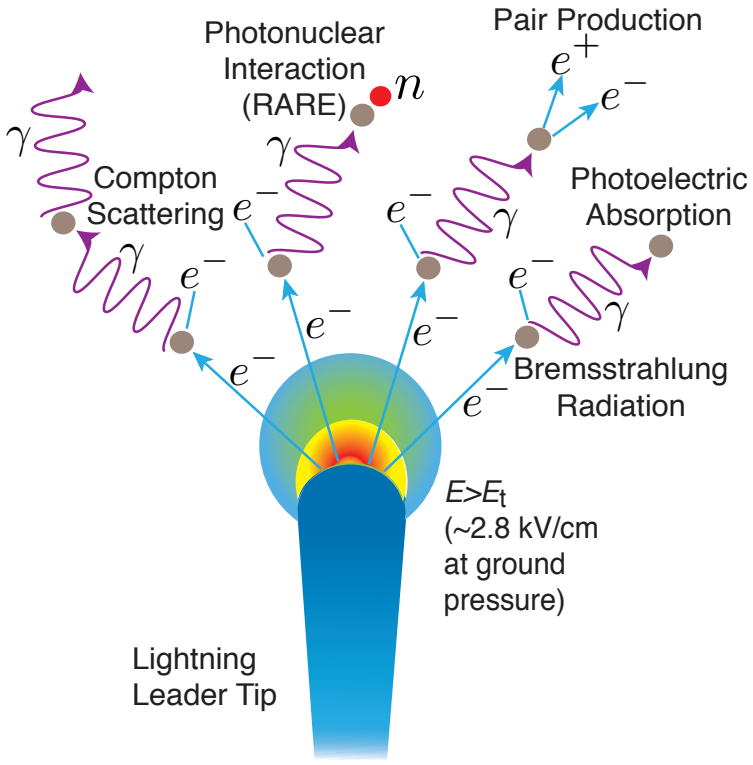

Figure 2.6. Schematic of the four types of collisions between photons with energies from 10 keV to 100 MeV and air molecules: photoelectric absorption, Compton scattering, electron-positron pair production, and photonuclear interaction.

[e.g., *Lehtinen*, 2000, Section 3.2.2; *Heitler*, 1954, p. 208] is a good approximation of the atomic photoelectric cross section:

$$\begin{aligned}\sigma_p =& \sigma_0 F \frac{3}{2}\frac{Z_a^5}{137^4}\left(\frac{mc^2}{\varepsilon_{ph}}\right)^5 \left(\gamma^2-1\right)^{3/2} \\ & \left[\frac{4}{3}+\frac{\gamma\left(\gamma-2\right)}{\gamma+1}\left(1-\frac{1}{2\gamma}\sqrt{\gamma^2-1}\right)\log\frac{\gamma+\sqrt{\gamma^2-1}}{\gamma-\sqrt{\gamma^2-1}}\right]\end{aligned} \tag{2.8}$$

where $\sigma_0{=}(8/3)\pi {r_0}^2$, r_0 is the classical electron radius, m is the electron rest mass, c is the speed of light, ε_{ph} is the energy of the incident photon, and $\gamma{=}1{+}\varepsilon_{ph}/(mc^2)$ is the photoelectron relativistic factor. The absorption edge factor is

$$F = 2\pi\sqrt{\frac{\varepsilon_{ion}}{\varepsilon_{ph}}}\frac{e^{-4\xi cot^{-1}\xi}}{1-e^{-2\pi\xi}}; \qquad \xi = \sqrt{\frac{\varepsilon_{ion}}{\varepsilon_{ph}-\varepsilon_{ion}}},$$

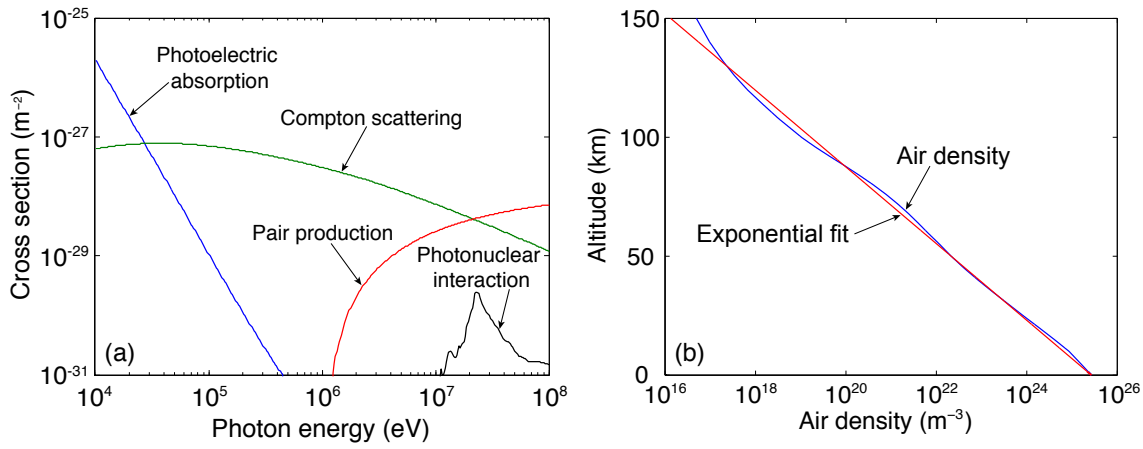

Figure 2.7. (a) Cross sections used in the Monte Carlo model for photons. (b) Air density with respect to the altitude.

where $\varepsilon_{ion}=\frac{Z_a^2 mc^2}{2\times 137^2}$ is the K-electron ionization energy. The cross section obtained is in very good agreement with that provided in the NIST/XCOM library (available at http://www.nist.gov/pml/data/xcom/).

Compton scattering is the dominant collision process for photons with energies between ∼30 keV and ∼30 MeV. Compton scattering, as the inelastic scattering of a photon by quasi-free electrons, is characterized by an energy loss, a deflection in the momentum, and a possible ejection of an electron from the target molecule. The cross section of Compton scattering used in the present study is taken from the NIST/XCOM library (available online at http://www.nist.gov/pml/data/xcom). The resultant energy of the scattered photon is calculated using [*Lehtinen*, 2000, Appendix A.6.2]:

$$\gamma' = \frac{\gamma}{1 + sX + (2\gamma - s)\, X^3}; \qquad s = \frac{\gamma}{1 + 0.5625\gamma} \tag{2.9}$$

where $\gamma=\varepsilon_{ph}/mc^2$ is the energy of the incident photon normalized to the electron rest energy, m is the electron rest mass, c is the speed of light, and X is a random number uniformly distributed between 0 and 1. The deflection angle between the incident photon and the scattered photon is obtained using the conservation of momentum and energy:

$$\theta = \arccos\left(1 + \frac{1}{\gamma} - \frac{1}{\gamma'}\right) \tag{2.10}$$

where γ' is the energy of the scattered photon normalized to the electron rest energy. The deflection in the azimuthal angle is assumed to be uniformly distributed between 0 and 2π.

Electron-positron pair production is the main collision process for photons with energies above ~30 MeV. In this process, the incident photon interacts with a nucleus to create an electron-positron pair in the Coulomb field of the nucleus. The cross section of this process is taken from the NIST/XCOM library (available at http://www.nist.gov/pml/data/xcom).

At energies above ~10 MeV, another photon-nucleus process can occur in which the incident photon excites the nucleus and a neutron emerges. Photonuclear interaction is a rare event compared to the other three processes presented above, as evident in Figure 2.7(a). The singly differential cross section of photonuclear interaction is obtained from the Evaluated Nuclear Data File (ENDF-7) library [*Chadwick et al.*, 2006] at the National Nuclear Data Center (NNDC) (available at http://www.nndc.bnl.gov/exfor/endf00.jsp). The set of differential cross sections is carefully interpolated over photon energies in order to be incorporated in the Monte Carlo model. The threshold energy of photonuclear interaction with nitrogen molecules is approximately ~10.5 MeV, as shown in Figure 2.7(a), and the cross section peaks around ~23 MeV, namely the giant dipole resonance. Both features are in fairly good agreements with the description in [*Berman and Fultz*, 1975; *Carlson et al.*, 2010].

Different from the null collision method used in the Monte Carlo simulation of electrons, the cumulative probability method [*Fano et al.*, 1959] is employed in the Monte Carlo simulation of photons. This method tracks the propagation of each particle individually until the end of its lifetime (i.e., being absorbed or arriving at detectors) by finding the free path length between collisions. The free path length can be calculated using the formula presented in [*Østgaard et al.*, 2008]:

$$s\pm = \frac{-1}{a\cos\alpha}\ln\left[1\pm\frac{\ln(R)a\cos\alpha}{\sigma_T\rho_0}\right] \tag{2.11}$$

where a is the coefficient used in the exponential fit of air density and is taken as $1/7$ km^{-1} (see Figure 2.7(b)), α is the angle between the photon momentum and vertical direction, σ_T is the sum of cross sections for all collisional processes, ρ_0 is

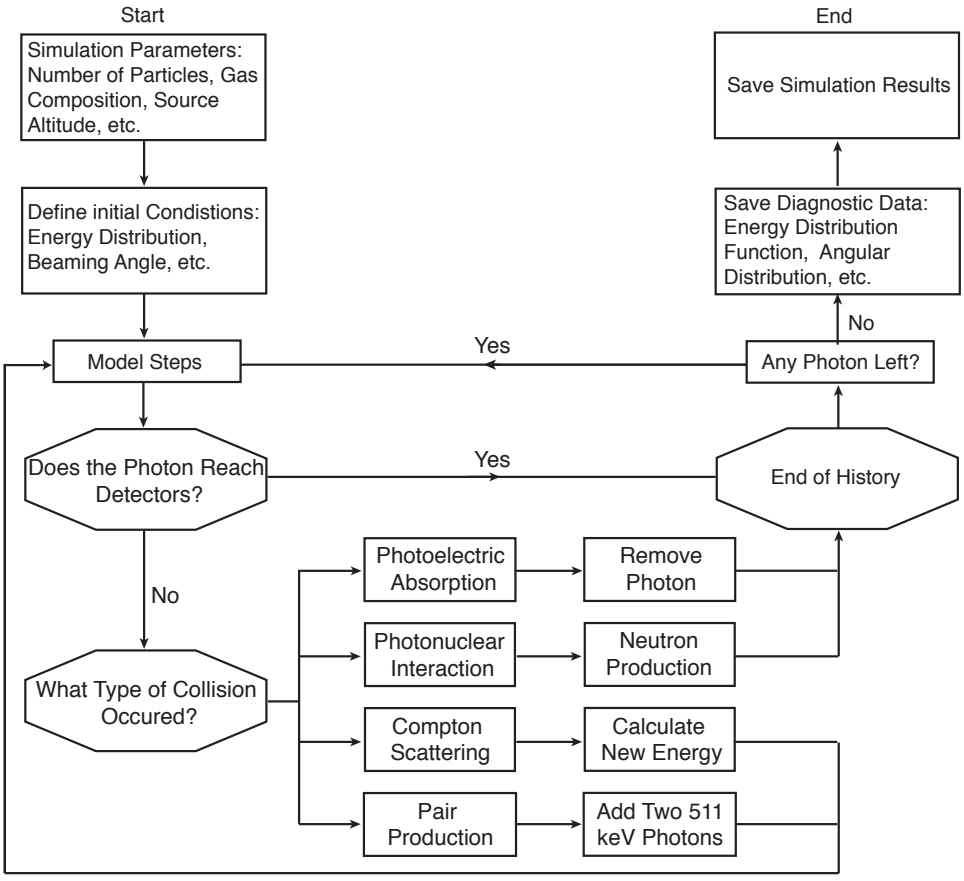

Figure 2.8. Flow chart depicting the execution of the Monte Carlo model of photons.

the air density at the initial position, and R is a random number that is uniformly distributed between 0 and 1.

After obtaining the free path length, a random number is drawn and compared with the probability of all collisional processes in order to determine the type of collision that occurred. If the type of collision is photoelectric absorption, the photon is removed from the ensemble of photons. If the type of collision is Compton scattering, the photon momentum is changed according to equations (2.9) and (2.10) [e.g., *Lehtinen*, 2000, p. 113]. If the type of collision is electron-positron pair production, the positron is assumed to annihilate locally and two photons with an energy of 511 keV in opposite directions are added to the pool of photons. The secondary Compton-produced electron bremsstrahlung is not taken into account. If the type of collision is photonuclear interaction, a neutron is generated according to

the differential cross section and the incident photon is removed from the ensemble of photons. These simplifications are justified by the good agreement obtained with previously published results [e.g., *Dwyer and Smith*, 2005].

Figure 2.8 is a flow chart representing the execution of the Monte Carlo simulation for photons. First, the user inputs basic simulation parameters such as air density, source altitude, and the number of particles. Then, this model needs information about the energy distribution and the beaming geometry of the photon source to initiate simulations. Each photon is propagated through the Earth's atmosphere until the end of its history, i.e., reaching the ground or escaping the atmosphere. The simulation is terminated when the particle number in the system is zero. In the last step, properties of those photons that arrive at detectors, including energy distribution and radial distribution, are saved for further processing and analysis.

We note that, in the studies of photon energy spectra (Chapters 3, 4, and 7), we consider that photonuclear interaction has a negligible impact on the ensemble of photons and is therefore not included in the Monte Carlo simulations. This simplification is valid considering that the cross section of this process is much smaller than the other three processes (see Figure 2.7(a)). Photonuclear interaction is taken into account in the Monte Carlo simulation results presented in Chapter 6 when studying neutron production by high-potential stepping lightning leaders.

2.4 Optical Emission Model

2.4.1 Full Energy Monte Carlo Simulation of Electrons

The main energy loss for runaway electrons occurs through ionization collisions with air molecules, resulting in low-energy secondary electrons. These primary and secondary electrons are able to generate excited species via impact excitation, and optical emissions can be generated by the radiative relaxation of these excited species. Optical emissions are quantified with the knowledge about the energy and number density distributions of the large population of electrons involved. Due to the fact that secondary low-energy electrons are efficient in exciting nitrogen molecules, in the studies of optical emissions (Chapter 5 and Chapter 8), we focus

on providing a rigorous description of the electron energy distribution in the full energy range by keeping complete track of both runaway electrons and the their low-energy secondary electrons using the Monte Carlo model described in Section 2.2.

Monte Carlo simulation encounters difficulty when modeled electrons cover many decades in the energy space, for instance, from sub-eV to GeV, for mainly two reasons. First, the copious amount of low-energy secondary electrons generated via ionization collisions severely increases the complexity of the simulation. Second, the simulation time that is required for the ensemble of electrons to achieve a statistical steady state is dramatically prolonged because of the high collision rate of low-energy electrons. In order to overcome these difficulties, the calculation of electron energy distribution is divided into two parts: a low-energy part (from sub-eV to 10 keV) and a high-energy part (from 10 keV to GeV). After steady state is reached in both parts, distribution function of electrons over the full range of energy is then obtained by normalizing the high-energy part calculation in order to connect to the low-energy part at 10 keV. This technique relies on adopting two characteristic time steps to maintain the collisional frequencies of high-energy and low-energy electrons separately. It is supported by the similarity of the low-energy parts of energy distributions for electron swarms initiated by either low-energy electrons or high-energy runaway electrons [*Colman et al.*, 2010]. This technique not only allows precise descriptions for both high- and low-energy electrons, but also greatly shortens the computation time. We have carefully validated this technique by comparing the results with full energy range simulations for representative test cases.

2.4.2 Optical Emissions Model

The model employed for evaluating optical emissions is similar to that documented by *Liu and Pasko* [2004]. Modeling optical emissions from the first positive band system of N_2 ($B^3\Pi_g \rightarrow A^3\Sigma_u^+$) ($1PN_2$) is complex, particularly for cases at high air pressures with continuous excitation by electrons. Populating of $N_2(B^3\Pi_g)$ involves processes of direct impact excitation by electrons, cascading from higher electronic states of N_2, intersystem collisional transfer with adjacent states [e.g., *Morrill and*

Benesch, 1996], and energy pooling reactions [e.g., *Piper*, 1988]. Besides, the de-excitation of $N_2(B^3\Pi_g)$ by N_2 molecules is strongly vibrational-level-dependent [e.g., *Piper*, 1988; *Morrill and Benesch*, 1996]. Given that the focus of the present study is the fluorescence light produced by TGF sources at low altitudes where $N_2(B^3\Pi_g)$ is heavily quenched and $1PN_2$ emissions are weak, in the present work, we mainly focus on optical emissions originating from the second positive band system of N_2 ($C^3\Pi_u \rightarrow B^3\Pi_g$) ($2PN_2$) and the first negative band system of N_2^+ ($B^2\Sigma_u^+ \rightarrow X^2\Sigma_g^+$) ($1NN_2^+$).

The intensity of optical emissions in Rayleighs is given by the expression [*Chamberlain*, 1978, p. 213]:

$$I_k = 10^{-6} \int_{L'} A_k n_k dl \tag{2.12}$$

where n_k [$1/cm^3$] is the number density of excited species in state k, A_k is the radiation transition rate and the typical values for optical emissions from $2PN_2$ and $1NN_2^+$ are, respectively, 2×10^7 s^{-1} and 1.4×10^7 s^{-1} [*Vallance Jones*, 1974, p. 119]. The integral is taken along L' [cm], representing the horizontal line of sight.

For $2PN_2$ and $1NN_2^+$, the quantity n_k is governed by processes including direct excitation by electrons and collisional quenching by air molecules. The number density n_k can be calculated using the relation [*Sipler and Biondi*, 1972]:

$$\frac{\partial n_k}{\partial t} = -\frac{n_k}{\tau_k} + \nu_k n_e \tag{2.13}$$

where τ_k=$[A_k + \alpha_1 N_{N_2} + \alpha_2 N_{O_2}]^{-1}$ is the total lifetime of state k, α_1 and α_2 are the quenching rate coefficients due to collisions with N_2 and O_2 molecules, respectively, N_{N_2} and N_{O_2} are, respectively, the number densities of N_2 and O_2 molecules, ν_k is the excitation frequency for generating the upper excited states responsible for emissions from $2PN_2$ and $1NN_2^+$, and n_e is the number density of electrons. For the sake of simplicity, steady state assumption is applied in order to find out n_k. This is valid given that the lifetimes of the upper excited states responsible for $2PN_2$ and $1NN_2^+$ emissions are very short (less than several μs [e.g., *Vallance Jones*, 1974, p. 119]) when compared to the characteristic time scale of the TGF sources considered in this study.

The excitation frequency, describing contributions to n_k from direct electronic

excitation by electrons, is consistently calculated using Monte Carlo simulation results. The quantity ν_k, defined as the number of molecular species j in the excited state k produced per unit time by one electron, is given by:

$$\nu_k = N_j \int \sigma_k(\varepsilon) v(\varepsilon) f(\varepsilon) d\varepsilon \tag{2.14}$$

where N_j is the number density of unexcited species j, $\sigma_k(\varepsilon)$ is the cross section for the excitation of state k, which depends on the energy of the incident electron ε, $v(\varepsilon)$ is the velocity of the colliding electron with an energy ε defined as $v(\varepsilon)=c\sqrt{1-[1+\varepsilon/(mc^2)]^{-2}}$, where c is the speed of light and m is the rest mass of electron, and $f(\varepsilon)$ is the electron energy distribution function, which is normalized so that $\int_0^{+\infty} f(\varepsilon)d\varepsilon=1$. Electron-impact excitation cross sections used for determining the excitation frequency of the upper excited states of $2PN_2$ are taken from BOLSIG+ database [*Hagelaar and Pitchford*, 2005]. These cross sections are provided between the threshold energy and 1 keV. However, for the energy distributions considered here, their values for energies >100 eV are small and their effects are negligible. Concerning $1NN_2^+$, it is considered that an $N_2^+(B^2\Sigma_u^+)$ ion is produced if one of the two $(2\sigma_u)$ electrons from the nitrogen molecule is removed [e.g., *Van Zyl and Pendleton*, 1995]. Thus, an accurate representation for the ionization cross section of $(2\sigma_u)$ electrons from the threshold energy to hundreds of MeVs is needed. The RBEB model (see Section 2.2.2), due to its orbital-based description for the differential ionization cross section, is therefore well suited for the calculation of the excitation frequency of $N_2^+(B^2\Sigma_u^+)$ leading to $1NN_2^+$ emissions. Indeed, the branching-ratio fraction for producing $N_2^+(B^2\Sigma_u^+)$ during ionization reactions of N_2 molecules by 100 eV electrons calculated using the RBEB model yields 0.163 and agrees well with the value 0.145 ± 0.019 suggested by *Van Zyl and Pendleton* [1995].

Collisional quenching processes are critical at the source altitudes of TGFs ($\leq$15 km) [e.g., *Dwyer and Smith*, 2005; *Xu et al.*, 2012], which are far below the quenching altitudes of both excited states responsible for $2PN_2$ and $1NN_2^+$ that are, respectively, $\sim$30 km and $\sim$50 km [e.g., *Liu et al.*, 2006]. In the present study, we consider that $N_2(C^3\Pi_u)$ can be quenched by collisions with N_2 and O_2 molecules with rate coefficients of 10^{-11} cm^3/s [*Kossyi et al.*, 1992] and 3×10^{-10}

cm^3/s [*Vallance Jones*, 1974, p. 119], respectively. The primary quenchers of $N_2^+(B^2\Sigma_u^+)$ are N_2, with a rate coefficient of 4.53×10^{-10} cm^3/s [e.g., *Mitchell*, 1970; *Kuo et al.*, 2005; *Pancheshnyi et al.*, 1998], and O_2, with a rate coefficient of 7.36×10^{-10} cm^3/s [e.g., *Mitchell*, 1970; *Kuo et al.*, 2005; *Pancheshnyi et al.*, 1998]. It is important to note that optical emissions evaluated using the modeling approach adopted in this study significantly rely on the choice of quenching scheme and corresponding rate coefficients.

As in [*Dwyer et al.*, 2013], aurora spectra of $2PN_2$ and $1NN_2^+$ presented in [*Vallance Jones*, 1974] are used as an approximate spectral distribution for fluorescence photons to investigate the wavelength dependence of optical features. We note that our present calculations do not take into account the effects of radiative transfer between the source of emission and the observer.

2.4.3 Model Validation

In order to perform a validation of the optical emission model, fluorescence emissions produced by energetic electrons are first calculated in the framework of Monte Carlo simulation and compared to laboratory observations of *Davidson and O'Neil* [1964] and *Nagano et al.* [2004]. One of the advantages that is afforded by the current Monte Carlo model is that it is capable of accurately describing and recording the spatial and temporal information for all the collisions taking place in the system. Owing to this advantage, an analog of photograph of fluorescence from air generated by a continuous beam of 50 keV electrons at ground pressure can be derived from first principles and is presented in Figure 2.9. As clearly shown in this figure, these fluorescence emissions exhibit a conical shape, following the spatial distribution of electrons, and the diameter of this beam, best represented by the strongest emissions from $2PN_2$, is approximately 4 cm. The morphological features of these optical emissions, including the conical shape and the size of the illuminated region, are in excellent agreement with experimental observations [*Davidson and O'Neil*, 1964].

In the absence of external electric fields, the energy distribution, especially in the low energy range (up to a few tens of eV), is characterized by the dynamic friction force of electrons. This point can be readily illustrated by a direct com-

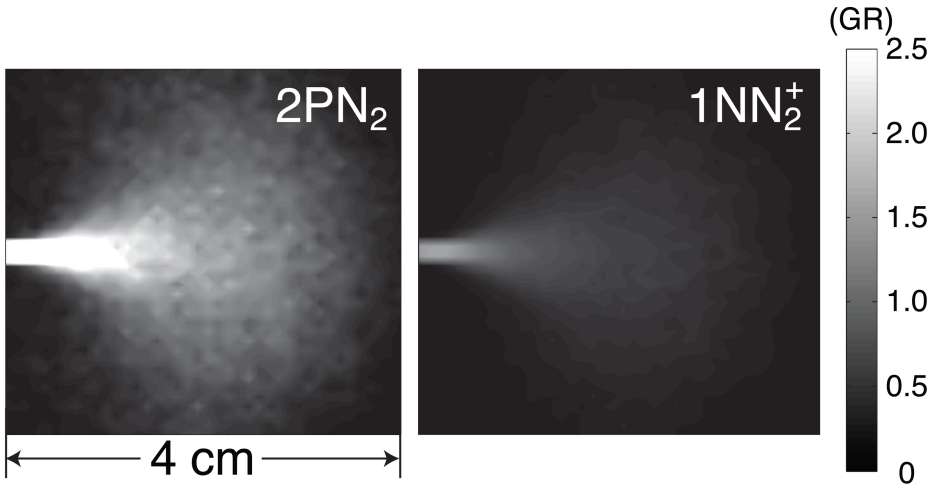

Figure 2.9. Simulation results of fluorescence radiation from air at ground pressure ($N_0 \simeq 2.688 \times 10^{25}$ m^{-3}) excited by a continuous beam of 50 keV electrons. Fluorescence emissions exhibit a conical shape. The diameter of this beam, best represented by the strongest emissions from $2PN_2$, is approximately 4 cm.

parison of the dynamic friction force of electrons resulting from collisions with air molecules at ground pressure and the steady-state electron energy distribution produced by the continuous beam of 50 keV electrons, as shown in Figure 2.10. This energy distribution is calculated by adopting a convolution technique. In particular, we first model the temporal evolution of the deceleration process undergone by an ensemble of 50 keV electrons due to collisions with air molecules in the case that they are injected instantaneously into air at 600 Torr. Namely, we obtain the impulse response of the system. The steady state energy distribution of the continuous 50 keV electron beam is then calculated by convolving over the temporal evolution of this electron pulse. We have verified that directly modeling a continuous beam of 50 keV electrons in the simulation leads to identical results, while being a lot more time-consuming. From Figure 2.10, we see that the depletion in the electron energy distribution around 2.5 eV precisely pinpoints the first peak of the dynamic friction force, defined by strong vibrational excitations of nitrogen molecules, while the peak of electron energy distribution around 4 eV coincides with a trough in the dynamic friction force. Furthermore, enhancement around 14.1 eV in the dynamic friction force (marked by a circle in Figure 2.10), that mainly comes from energy losses to the electronic excitation of $N_2(C^3\Pi_u)$, is

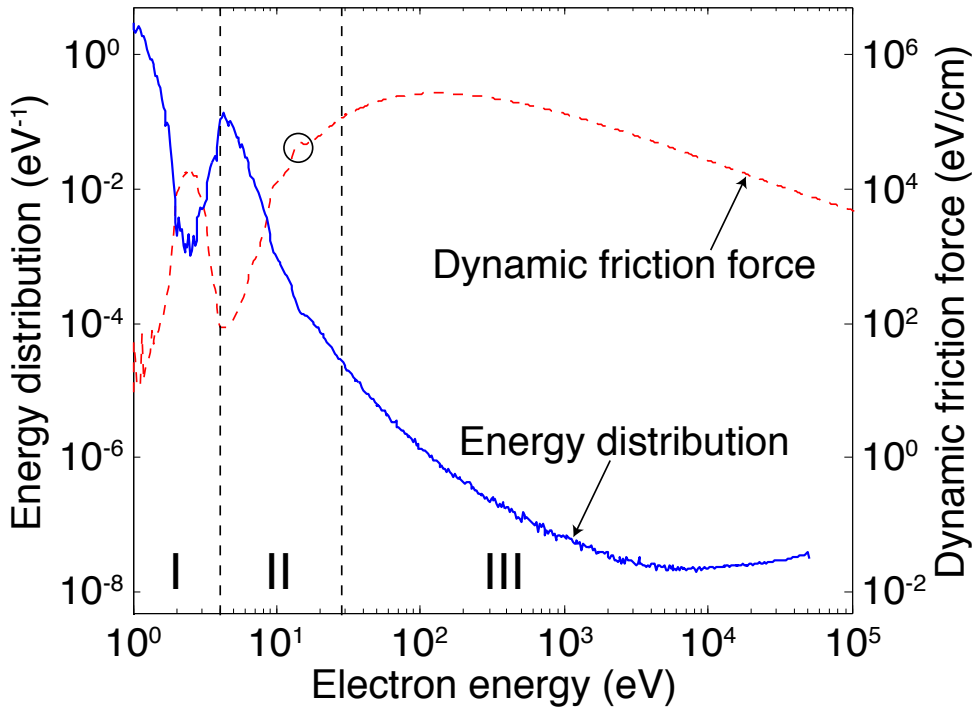

Figure 2.10. Steady state energy distribution of a continuous beam of 50 keV electrons in air at 600 Torr. Also shown is the dynamic friction force of electrons at ground pressure due to collisions with air molecules computed using the set of cross sections adopted in Monte Carlo simulation (red dashed line). The dominant energy loss processes in regions I, II, and III are vibrational and rotational excitations, electronic excitation, and ionization collisions, respectively. The circle around 14.1 eV represents the enhancement in the dynamic friction force, mainly arising from energy losses to the electronic excitation of $N_2(C^3\Pi_u)$, that is also reflected in the electron energy distribution.

reflected in the energy distribution.

In the region I marked in Figure 2.10, where the electron energy is below ∼4 eV, vibrational and rotational excitations are the dominant energy loss processes. In particular, vibrational excitations facilitate fast removal of electrons from this energy region and accordingly cause the precipitous decrease in the energy distribution from 1 eV to ∼2.5 eV. Region II is delineated by electrons with energy between ∼4 eV and ∼28 eV. Electronic excitation is the principal channel of energy loss in this region and the cross sections of electronic excitation collisions for generating the upper excited states of $2PN_2$ peak in this region. In region III, electrons possess energy in excess of the binding energy of electrons in N_2 and O_2

molecules and mostly convert their kinetic energy into the generation of secondary electrons, namely ionization collisions.

Fluorescence efficiency is defined in the studies of extensive air showers [*Bunner*, 1967] as the fraction of the total energy deposited during the passage of electrons in a gas that is eventually transferred into fluorescence photons. For the comparison with [*Nagano et al.*, 2004], fluorescence efficiency is calculated by dynamically recording the number of fluorescence photons generated and the amount of energy deposited by 850 keV electrons in Monte Carlo simulations. However, in contrast with the thin target technique used in [*Nagano et al.*, 2004], *Davidson and O'Neil* [1964] employed a thick target technique to measure the fluorescence efficiency of 50 keV electrons by injecting a continuous electron beam into the gas chamber. In order to compare with the experimental measurements in [*Davidson and O'Neil*, 1964], we first calculate the production of fluorescence photons in the wavelength range between 300 and 430 nm and the energy deposition into air molecules by an ensemble of 50 keV electrons in the case that they are injected instantaneously into air at 600 Torr. Fluorescence emissions corresponding to the steady state of continuous injection are then obtained by convolving the fluorescence pulse produced by the instantaneous injection of 50 keV electrons and assuming the same injection current as that in [*Davidson and O'Neil*, 1964]. Fluorescence efficiency is further determined as the ratio of the energy released in fluorescence photons to the energy deposited by the incident electron beam.

The comparison with wavelength-resolved measurements [*Davidson and O'Neil*, 1964; *Nagano et al.*, 2004] is shown in Figure 2.11 and documented in Table 2.3. Our modeling results are generally in very good agreement with laboratory measurements of *Nagano et al.* [2004] for both $2PN_2$ and $1NN_2^+$ band systems. This agreement also validates our use of aurora spectra in order to quantify the wavelength dependence of optical emissions produced by energetic electrons. In addition, we note that the mean energy loss of 850 keV electrons derived from Monte Carlo simulation is approximately 1.74 keV/cm, close to the theoretical value defined by the dynamic friction force at this pressure. The slight difference observed for the (0-0) transition of $1NN_2^+$ between calculated and experimental results, at 391 nm, is likely due to three-body deactivation processes as pointed out in [*Mitchell*, 1970; *Nagano et al.*, 2004], which are able to weaken the efficiency

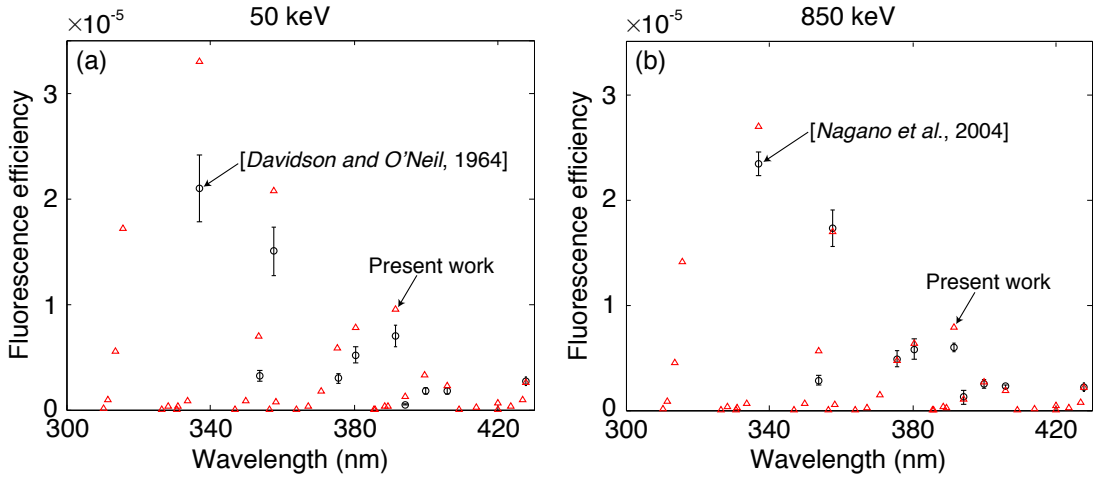

Figure 2.11. Comparison of fluorescence efficiency from air at 600 Torr for major emission bands of $2PN_2$ and $1NN_2^+$, in the wavelength range between 300 nm and 430 nm, excited by (a) 50 keV electrons between experimental measurements of *Davidson and O'Neil* [1964] and present modeling results; (b) 850 keV electrons between experimental measurements of *Nagano et al.* [2004] and present modeling results.

Table 2.3. Comparison of fluorescence efficiency ($\times 10^{-5}$) from air at 600 Torr for major emission bands of $2PN_2$ and $1NN_2^+$, in the wavelength range between 300 nm and 430 nm, excited by energetic electrons between experimental measurements of *Davidson and O'Neil* [1964] (50 keV electrons), *Nagano et al.* [2004] (850 keV electrons), and present modeling results. The systematic error for the measurements of *Davidson and O'Neil* [1964] is ±15%.

Wavelength (Å)	*Davidson and O'Neil* [1964] 50 keV	*Nagano et al.* [2004] 850 keV	50 keV	850 keV
3371, $2PN_2$(0-0)	2.10	2.23–2.45	3.306	2.703
3537, $2PN_2$(1-2)	0.32	0.237–0.331	0.695	0.568
3577, $2PN_2$(0-1)	1.50	1.56–1.90	2.076	1.697
3756, $2PN_2$(1-3)	0.30	0.415–0.563	0.582	0.476
3805, $2PN_2$(0-2)	0.52	0.480–0.680	0.783	0.640
3914, $1NN_2^+$(0-0)	0.70	0.558–0.638	0.951	0.791
3943, $2PN_2$(2-5)	0.05	0.058–0.188	0.131	0.107
3998, $2PN_2$(1-4)	0.18	0.212–0.286	0.333	0.272
4059, $2PN_2$(0-3)	0.18	0.209–0.243	0.230	0.188
4278, $1NN_2^+$(0-1)	0.27	0.181–0.259	0.265	0.220

by approximately 20%. As for the fluorescence efficiency produced by continuous injection of 50 keV electrons, our modeling results also agree with experimental measurements of *Davidson and O'Neil* [1964] for $1NN_2^+$, while the agreement is not as good for $2PN_2$ and we note that, for some bands of $2PN_2$ (i.e., $2PN_2$(1-2)), as large as a factor of two differences are observed.

Chapter 3

Source Altitude of Terrestrial Gamma-ray Flashes Produced by Lightning Leaders

In this chapter, we apply the method of moments introduced in Section 2.1, the Monte Carlo model of electrons introduced in Section 2.2, and the Monte Carlo model of photons introduced in Section 2.3 to simulate characteristic energy spectra of TGFs at satellite altitudes based on the production mechanism of thermal runaway electrons by high-potential lightning leaders. We calculate consistently the beaming geometry of source photons produced in this mechanism and we investigate the dependence of TGF energy spectra measured by satellites on the production altitude. Using the cumulative energy spectra of TGFs measured by RHESSI [*Dwyer and Smith*, 2005], we estimate a characteristic production altitude of TGFs. The results presented in this chapter have been published in [*Xu et al.*, 2012].

3.1 Modeling Results

We use the Method of Moments [*Balanis*, 1989, p. 670] described in Section 2.1 in order to calculate the electric field in the vicinity of a +IC lightning negative leader tip during the corona flash associated with the stepping of the leader. The ambient

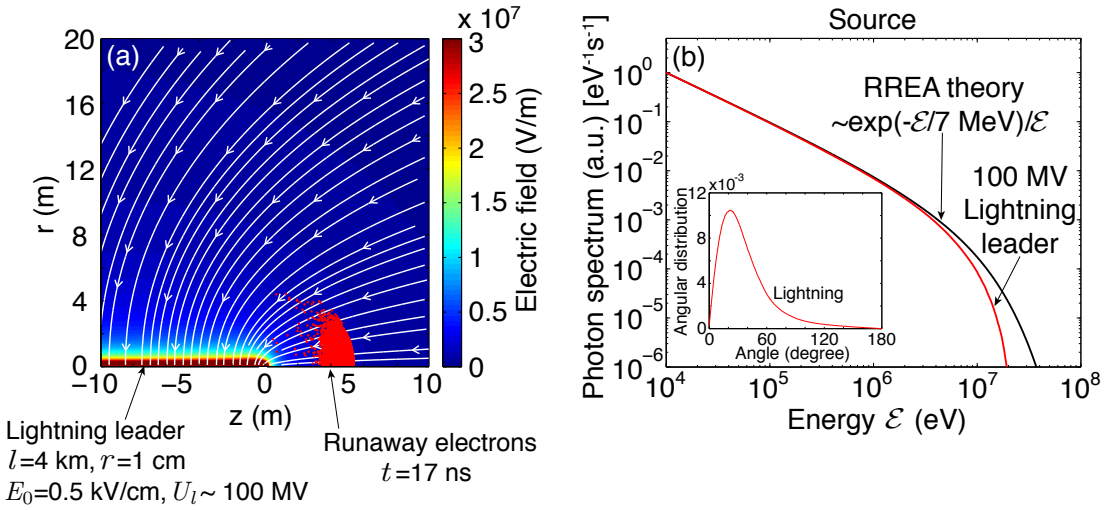

Figure 3.1. (a) Illustration of the electric field in the vicinity of a 100 MV lightning leader. The panel also shows the positions of runaway electrons at t=17 ns obtained from our Monte Carlo model. (b) RREA and lightning bremsstrahlung photon spectra. The inset in panel (b) shows the angular distribution of source photons produced by bremsstrahlung emission of energetic electrons accelerated in the lightning leader field.

large-scale thunderstorm electric field is taken as E_0=5×10^4 V/m [e.g., *Marshall et al.*, 2001]. The radius of the leader channel is chosen as 1 cm [*Rakov and Uman*, 2003, Section 4.4.6, p. 134] and the IC lightning length is l=4 km. This leads to a difference of 100 MV of leader tip potential with respect to ambient potential, which is 10 times of the electric potential used in [*Celestin and Pasko*, 2011]. Indeed, in our simulations, we have observed that a 100 MV lightning leader produces electrons with a maximum energy of a few tens of MeVs, as typically observed in TGFs.

TGF spectra depend on the electron energy distribution at the source, the electron beam geometry and the source altitude [*Dwyer and Smith*, 2005]. In this study, electron energy distribution and the electron beam geometry are consistently obtained from the acceleration of electrons in the electric field of a stepping lightning leader.

Figure 3.1(a) shows the geometry of the electric field lines calculated by the method of moments and the acceleration of thermal runaway electrons [e.g., *Celestin and Pasko*, 2011] in the vicinity of the lightning leader tip. The time integrated (over 108 ns) photon spectra generated in the high-potential difference

produced by the lightning leader and in the case of RREA theory (large-scale, homogeneous, and weak ambient field) are shown in Figure 3.1(b). One sees that the photon spectrum produced by the thermal runaway electrons in the lightning leader region is softer than that in the case of the RREA theory, which demonstrates the typical 7 MeV cutoff signature [e.g., *Dwyer*, 2008]. The inset in Figure 3.1(b) shows the angular distribution of the lightning-produced source photons with energies between 10 keV and 100 MeV.

Figure 3.2(a) shows the simulated spectra obtained at an altitude of 500 km (i.e., low Earth orbit) corresponding to the two mechanisms with different source altitudes that give the best match to the combined RHESSI spectrum. Due to the limited spatial extent of the detector, the measured spectrum is different from the real photon spectrum. Indeed, some photons will partially release their energy in the detector, for example by Compton scattering before exiting it. In order to directly compare our simulation results with RHESSI measurement, the detector response is taken into account by applying the RHESSI detector response matrix [*Dwyer and Smith*, 2005; *Grefenstette et al.*, 2009] (available at `http://scipp.ucsc.edu/~dsmith/tgflib_public/data/`).

In this study, the simulation is conducted to model a TGF produced by one specific lightning discharge while RHESSI TGF spectrum is a spectrum accumulated over many TGF events. It is possible that the TGF spectrum generated by lightning discharge does not exactly match the averaged RHESSI data in the high energy part, which is established by the specific potential of the lightning event. It is important to note that for a higher potential lightning discharge, the related TGF spectrum is able to reproduce entirely the averaged RHESSI spectrum. However, for the purpose of determining source altitudes of TGFs, it is enough to observe the effects of atmospheric attenuation on the low energy part of the spectra. Indeed, high-energy photons have a reduced probability of collision, and the corresponding part of the photon spectrum directly depends on the available potential difference due to lightning leader. We have verified that different electric potentials of lightning discharges produce different high energy parts of the spectrum, while the low energy part is mostly defined by the source altitude (see inset in Fig. 3.2(a)). In this context, we consider that 100 MV lightning leaders are among the least energetic discharges that could produce TGFs as observed by

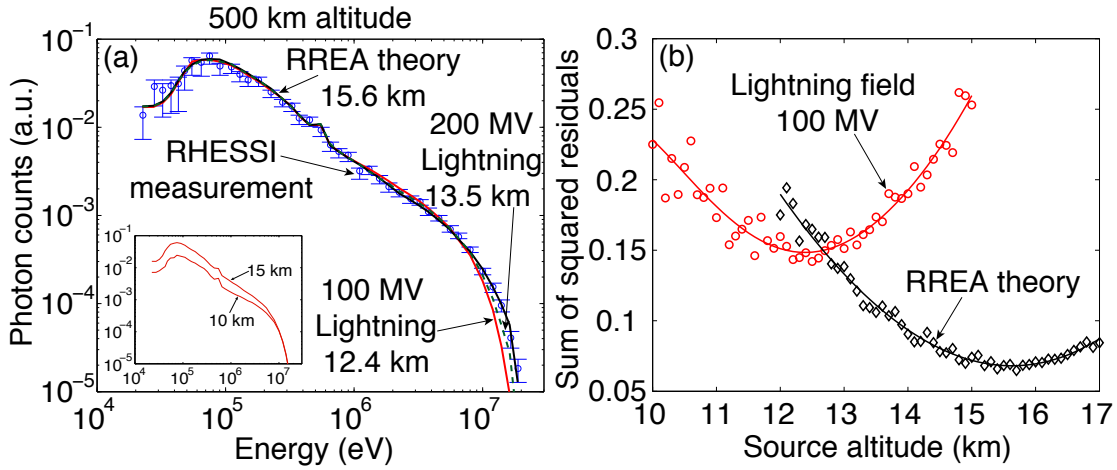

Figure 3.2. (a) Comparison between RHESSI measurements and simulated spectra. The source altitude of lightning-produced TGF is estimated from the low energy part of the spectra. An RREA source at 15.6 km generates a spectrum consistent with RHESSI data. The spectrum produced by a lightning leader also matches RHESSI data but for a source located at 12.4 km. The combined RHESSI data are obtained from [*Dwyer and Smith*, 2005]. The inset in panel (a) illustrates variations in lightning-produced TGF spectrum for source altitudes between 10 and 15 km. (b) Sum of squared residuals calculated between RHESSI data and simulated spectra of both mechanisms for an altitude range from 10 km to 17 km. Solid lines show third-order polynomial fits to the data.

RHESSI. The transition point between low and high energy parts is approximately 10 MeV. The source altitude is estimated by matching the low energy part with the averaged RHESSI data.

An RREA-produced TGF at 15.6 km altitude provides a spectrum consistent with RHESSI spectrum. This altitude is consistent with that found in [e.g., *Dwyer and Smith*, 2005; *Carlson et al.*, 2007]. Figure 3.2(a) also shows that a 100 MV lightning leader at 12.4 km altitude is able to explain the RHESSI TGF spectrum as well.

The least square method is used to quantify the agreement between the simulated TGF spectra corresponding to different source altitudes and RHESSI data. Since the normalization of the spectrum is a free parameter, each point of Figure 3.2(b) corresponds to the normalization factor that provides the minimum of the sum of squared residuals for a given altitude. Using the photons produced by the electrons accelerated in the field produced by the stepping of the 100 MV +IC lightning leader, we find that a TGF source located at 12.4 km would give the best

match to the RHESSI data. For a 200 MV lightning leader, we find that RHESSI measurements can be explained by a source at 13.5 km altitude.

It is important to note that a function of altitude $N(z)=N_0\exp(z/7\text{ km})$, with $N_0=2.688\times10^{25}$ m^{-3}, which fits the U.S. standard atmosphere [*Committee on Extension to the Standard Atmosphere*, 1976], is used in the present work as the neutral air density profile for simulating photon transport in the Earth's atmosphere. Collisions between photons and air molecules mainly occur below ∼100 km altitude and we have verified that the latitudinal, longitudinal, and annual variations of neutral air density are not significant in this altitude range.

3.2 Discussion

The source photon beam geometry is determined from the energy and beaming geometry of the source electrons. In the lightning leader case, these quantities directly follow from the layout of the electric field calculated using the method of moments. The momentum distribution of photons after their generation through bremsstrahlung collisions is calculated from the momentum distribution of electrons in our model. This consistent simulation of momentum distribution for both electrons and photons provides us with a direct quantitative description of the beaming geometry of the photons source without requirement of any additional assumptions. The momentum distribution of source photons is found to be a broad beam with an average angle of 37.7° for photons (see inset in Figure 3.1(b)) with energies between 10 keV and 100 MeV, in agreement with previously published results on the beaming of TGF sources [e.g., *Carlson et al.*, 2007; *Gjesteland et al.*, 2011].

From the theory developed in [*Celestin and Pasko*, 2011], we can estimate that the 100 MV lightning leader used in this study produces ∼10^{18} energetic electrons as an upper limit. Using the simulation results obtained in the present study, we estimate that these electrons and the secondary energetic electrons would generate a maximum of ∼10^{19} energetic photons through bremsstrahlung emissions. At satellite altitude, this corresponds to a maximum fluence of 3 photons/cm^2 at 200 km distance from the subsatellite point. This is consistent with measurements of TGF fluences at low Earth orbit [*Dwyer and Smith*, 2005; *Østgaard et al.*, 2008;

Carlson et al., 2009].

As shown in Figure 3.1(b), the source photon spectrum for the lightning leader case is softer than that produced by RREA. This difference is due to the fact that the the electric field driving runaway electrons has a limited spatial extent in the case of a stepping leader. Indeed, electrons can only gain a part of the potential energy available and by the time they have stopped in the regions where electric field is too low to sustain their acceleration, the time averaged electron energy distribution appears to be not as energetic as in the RREA case (Figure 3.1(b)). We observe that 100 MV lightning leaders should be among the least energetic discharges that could produce TGFs observed by RHESSI. In fact, RREA-producing fields are extended over long enough distances to enable the distribution of electrons to reach the RREA steady state. In the same context, while lower potential lightning leaders would lead to too few energetic photons to be observed by a satellite, they would produce an even softer electron distribution than that obtained in the present study. The related bremsstrahlung radiation manifests itself in X-ray bursts observed at close range from negative cloud-to-ground lightning discharges [e.g., *Moore et al.*, 2001; *Dwyer et al.*, 2005] (see Chapter 7).

At the source, the leader-produced photon spectrum has a lower high energy cutoff than that of the RREA spectrum (Figure 3.1(b)). The source of photons needs to be placed at lower altitudes for the simulated spectrum to match RHESSI measurements. Indeed, the deeper in atmosphere a given source spectrum is placed, the harder is the observed spectrum at satellite altitude. The inset in Figure 3.2(a) illustrates this point by showing variation of lightning-produced TGF spectrum observed at satellite altitudes for source altitudes between 10 and 15 km. It is worth mentioning that a lightning source at 8 km is ruled out since the related spectrum is outside the error bars of the RHESSI measurements.

We note that an altitude of 12.4 km of the TGF source found in this study lies within the typical range of observations of TGF-producing +IC lightning. Indeed, recent studies of TGF-related lightning processes suggest that TGF sources are located at altitudes lower than originally thought. In fact, *Stanley et al.* [2006] found two specific TGF-related intracloud lightning discharges at 13.6 km and 11.5 km. Further study of TGF-related sferics by *Shao et al.* [2010] suggests a TGF source in the altitude range 10.5–14.1 km. *Lu et al.* [2010] also found that

TGFs are produced during the initial development of intracloud lightning between a negative charge region centered at about 8.5 km and a positive region at 13 km altitude.

It is also interesting to note that a 12.4 km TGF source altitude happens to be closer to the altitudes of commercial flights and a large amount of TGF electrons and photons in the source region may potentially lead to significant radiation doses received by aircraft passengers [*Dwyer et al.*, 2010]. However, we emphasize that in the proposed leader based model the electron beam does not extend over a large distance (few tens of meter), and further studies on the probability for an aircraft to be exposed to radiation from TGF source need to be carried out.

3.3 Conclusions

The main contributions of this chapter can be summarized as follows:

1. We show that the large fluxes of thermal runaway electrons produced by streamer discharges during the negative corona flash stage of high-potential stepping lightning leaders [*Celestin and Pasko*, 2011] are capable of explaining the cumulative energy spectra of TGFs measured by RHESSI [e.g., *Dwyer and Smith*, 2005]. More precisely, we find that potential drops of 100 MV formed in the vicinity of lightning leader tips are among the weakest that could produce TGFs.

2. Based on the modeling results presented in this chapter, we suggest a TGF source altitude of 12.4 km for lightning leaders with an electrical potential drop of 100 MV, and 13.5 km for lightning leaders with an electrical potential drop of 200 MV. These source altitudes lie within the range of observations of TGF-producing +IC lightning [e.g., *Stanley et al.*, 2006; *Shao et al.*, 2010; *Lu et al.*, 2010]. In the case of TGF production by RREA processes driven by the large-scale thunderstorm electric field, the source altitude is found to be 15.6 km, in good agreements with that suggested by *Dwyer and Smith* [2005].

3. The source photons are found to have a broad beam geometry with an average angle of 37.7°, in good agreements with previous analysis [e.g., *Carlson et al.*, 2007; *Gjesteland et al.*, 2011]. The 100 MV lightning leader is capable of generating a maximum of approximately $\sim 10^{19}$ energetic photons with energies greater than 10 keV. The corresponding fluence of gamma-rays is found to be lower

than 3 photons/cm^2 at 200 km distance from the subsatellite point at satellite altitudes, also in good agreements with satellite measurements [e.g., *Dwyer and Smith*, 2005; *Østgaard et al.*, 2008; *Carlson et al.*, 2009].

Chapter 4

Terrestrial Gamma-ray Flashes with Energies up to 100 MeV Produced by Non Equilibrium Acceleration of Electrons in Lightning

In this chapter, the Monte Carlo models of electrons and photons described in Chapter 2 are used in order to investigate the transient nonequilibrium acceleration of thermal runaway electrons in the highly inhomogeneous electric field produced near the tip region of high-potential lightning leaders. The simulation results give insight into the fundamental differences between the nonequilibrium acceleration of thermal runaway electrons produced during stepping of lightning leaders and RREA processes developing in weak homogeneous electric fields. We also provide a modeling-based explanation to the high-energy tail of TGF spectra recently discovered by *Tavani et al.* [2011]. The results presented in this chapter have been published in [*Celestin et al.*, 2012].

4.1 Motivation

Measurements have correlated TGFs with initial development stages of normal polarity intracloud lightning that transports negative charges upward (+IC) [*Stanley*

et al., 2006; *Shao et al.*, 2010; *Lu et al.*, 2010, 2011]. *Connaughton et al.* [2010] and *Cummer et al.* [2011] have shown a close association between TGFs detected by the Fermi-GBM and fast lightning processes within several tens of microseconds. Moreover, *Tavani et al.* [2011] have recently reported that the high-energy part (>30 MeV) of the TGF spectrum, measured by the AGILE mission for the first time, significantly deviated from spectra corresponding to well-established model of relativistic runaway electron avalanches (RREAs), which so far provided a very good agreement with observations at lower energies [*Dwyer and Smith*, 2005]. Additionally, *Tavani et al.* [2011] discovered photons in the high-energy tail of terrestrial gamma-ray flashes with energies up to 100 MeV (see Figure 4.1(b)).

X-ray bursts from lightning discharges have been clearly identified to stem from the production of thermal runaway electrons [*Dwyer et al.*, 2004b] and TGFs have been suggested to be related as well to thermal runaway processes in lightning leader tips [*Moss et al.*, 2006]. However, until recently, TGF spectra were very well reproduced by RREA models involving acceleration of runaway electrons in large-scale (hundreds of meters) weak electric fields in thunderstorms [*Dwyer et al.*, 2005]. The RREA processes have then been understood to provide an amplification factor of $\sim 10^5$ in the number of energetic electrons to reach a total number of $\sim 10^{17}$ energetic electrons (calculated at ~15 km altitude) believed to produce TGFs through the bremsstrahlung process [*Dwyer and Smith*, 2005; *Carlson et al.*, 2009]. The great majority of secondary electrons produced by electron-impact ionization are low energy electrons, therefore, such an amplification in the total number of energetic electrons ($\sim 10^5$) is necessarily accompanied by a marginalization of the most energetic electrons, which drive the relativistic avalanche. This produces a characteristic high-energy (~10 MeV) cutoff on the electron energy distribution function that creates a clear signature on the photon spectrum associated with the RREA process, which is weakly influenced by ambient conditions such as electric field or gas density [e.g., *Babich et al.*, 2004; *Dwyer and Smith*, 2005]. The lower energy part of the photon spectrum (<10 MeV) mostly depends on the air density (altitude) at the electron source location [*Dwyer and Smith*, 2005].

In the present study, we demonstrate that strong acceleration of a large number of thermal runaway electrons [*Celestin and Pasko*, 2011] in very strong and highly inhomogeneous electric fields of lightning discharge could naturally explain the

significant deviation from the RREA equilibrium at energies >30 MeV recently reported by *Tavani et al.* [2011]. We show that the spectrum of bremsstrahlung photons propagated in the atmosphere up to satellite altitude can reproduce the recent observations in the high-energy range. This analysis provides the first direct evidence that TGFs are produced by lightning and not over large distances in weak thunderstorm electric fields.

4.2 Modeling Results

The Method of Moments described in Section 2.1 is used to calculate the electric field in the vicinity of a +IC lightning negative leader tip during the corona flash associated with the stepping of the leader. Specifically, the ambient large-scale thunderstorm electric field is taken as E_0=2×10^5 V/m [e.g., *Marshall et al.*, 1995]. The radius of the leader channel is chosen as 10 cm [e.g., *Rakov and Uman*, 2003, Section 4.4.6, p. 134], and the IC lightning length is l=3.5 km. Beyond the initial location of electrons in our code ($r \geq 0.15$ m, where r is the distance from the leader tip) the electric field can be approximated by a A/r function. For the set of parameters chosen in this study, $A \simeq 27.5$ MV represents correctly the electric field within 20% and is especially accurate in the highly inhomogeneous region close to the lightning leader tip. More generally, we find that $A(U_l)=0.04E_0l=0.04 \times 2U_l$ offers a good representation (within 20%) of the electric field in the lightning leader tip region for a wide range of voltages (or correspondingly product of ambient electric field and lightning length) from $U_l \simeq 50$ MV to ∼500 MV, with E_0 varying from 0.5 kV/cm to 3 kV/cm, with a better agreement as l increases from 2 km to 3 km. It is important to note that cloud-to-ground lightning discharges usually exhibit extensive branching that leads to a significant reduction of the lightning electric potential [*Bazelyan and Raizer*, 2000, p. 166], preventing these discharges from producing the extremely energetic radiation studied in the present article [*Celestin and Pasko*, 2011].

The unexpected strong deviation from the RREA spectral signature in the high-energy part of the spectrum (>30 MeV) observed by AGILE satellite [*Tavani et al.*, 2011] as illustrated in Figure 4.1(b) is challenging RREA-based theories of TGF production, since the TGF spectrum was precisely the main evidence of the RREA

mechanism. In the following, we show that the dynamics of electrons accelerated in the vicinity of lightning leader tips reproduces the high-energy spectrum recently uncovered [*Tavani et al.*, 2011], while acceleration of electrons in homogeneous large-scale weak electric fields (RREA process) is only able to fit the spectrum for photons with energies lower than ∼30 MeV [e.g., *Dwyer and Smith*, 2005].

Figure 4.1(a) illustrates the fundamental reason behind this effect. The acceleration of electrons up to energies on the order of hundreds of MeV in large-scale weak electric fields present in thunderstorm is theoretically possible, however, as energetic electrons accelerate, they produce a large number of lower energy secondary electrons by impact ionization. In Figure 4.1(a), this mechanism is represented with a characteristic length λ_i. Thus, a relationship exists between the total distance of acceleration undergone by the electrons and the quantity of the secondary electrons produced. In a weak electric field, the population of energetic electrons is promptly overcome by lower energy electrons that create a fast drop, or energy cutoff, in the electron energy distribution (see Figure 4.2(a)). Therefore, in the case of large-scale weak electric fields present in thunderstorms, the RREA mechanism cannot deviate from this particular electron distribution [*Babich et al.*, 2004; *Dwyer and Smith*, 2005; *Dwyer and Babich*, 2011]. Contrary to the large-scale weak electric field scenario, the inhomogeneous high electric fields naturally present in compact regions around negative lightning leader tips during stepping processes can accelerate thermal runaway electrons to high energy over a short distance, corresponding to a much lower number of impact ionization lengths (see Figure 4.1(a)) and much lower corresponding number of low-energy secondary electrons. Due to the rapid change of the electric field seen by the electrons as they are progressing in the inhomogeneous field, the equilibrium is not reached and the ensemble of electrons is intrinsically in non-equilibrium. The characteristics of electron energy distribution during transient acceleration stage near leader tips should therefore be significantly different from the RREA distribution and these characteristics are examined in the analysis that follows.

Figure 4.2(a) presents the electron energy distribution of an ensemble of energetic electrons that propagate under a homogeneous electric field of magnitude 12.5 kV/cm in air at ground pressure calculated by a Monte Carlo model at different instants of time. In order to emphasize the time evolution of the energy

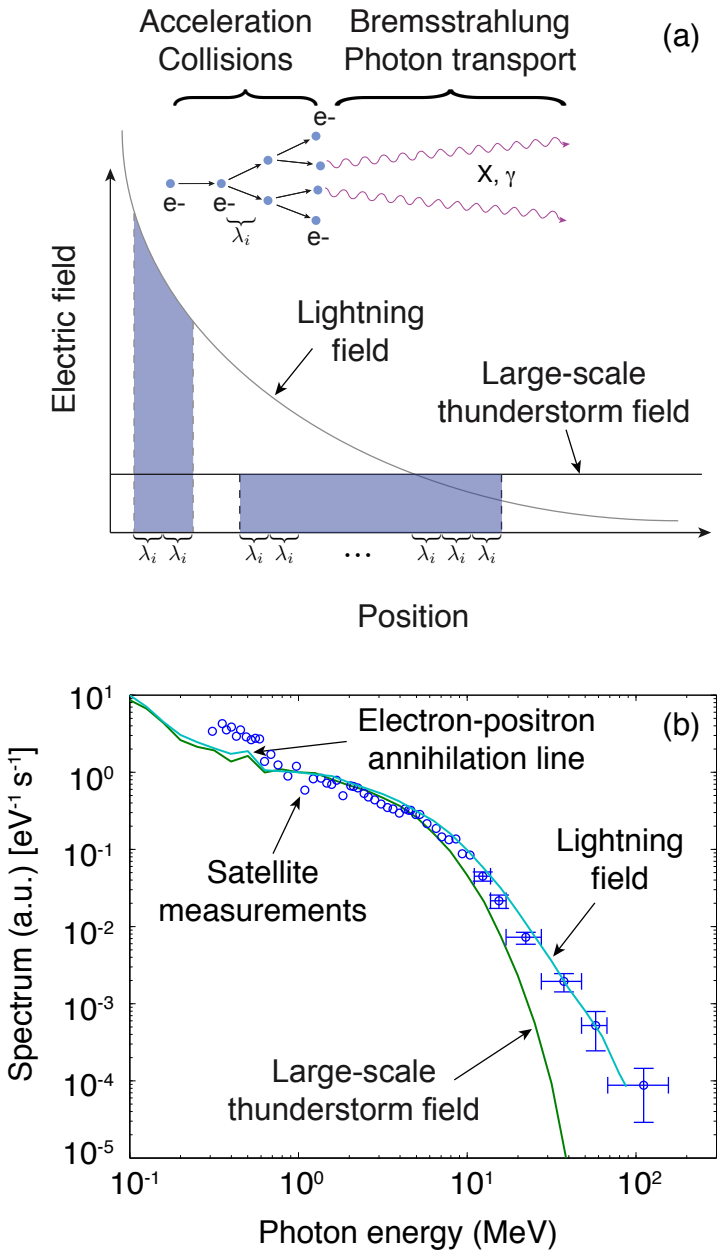

Figure 4.1. Acceleration and multiplication of electrons, and photon spectra. (a) Illustration of acceleration and multiplication of electrons in homogeneous and inhomogeneous fields. The electron-impact ionization process is represented with a characteristic length λ_i. For a given electron maximum energy (shaded areas), different energy distributions are expected between the homogeneous and inhomogeneous field cases because the corresponding distances of acceleration are associated with a different number of ionization lengths. (b) Spectrum at satellite altitude. Circles with error bars are reproduced from [*Tavani et al.*, 2011]. The spectrum produced in association with classical RREA process in weak thunderstorm fields (Figure 4.2(a)) is represented in green, and the spectrum obtained from the mechanism of direct acceleration in inhomogeneous lightning field (Figure 4.2(b)) is shown in blue.

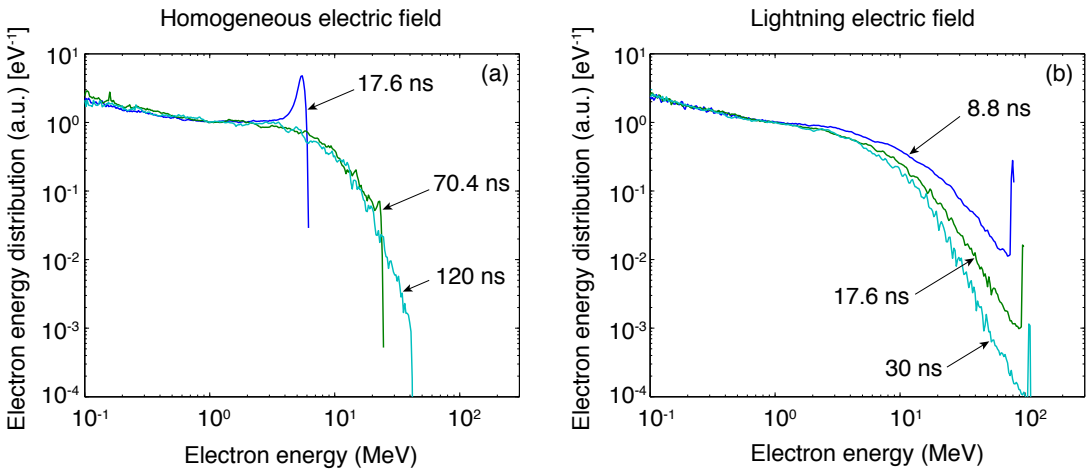

Figure 4.2. Electron energy distributions. (a) Electron energy distributions obtained for 3 different times in the case of propagation in a homogeneous electric field of magnitude 12.5 kV/cm. The progression of the energy distribution dynamically follows the classical steady-state RREA distribution, which weakly depends on the magnitude of the external electric field. (b) Electron energy distributions obtained for 3 different times in the case of propagation in a strongly inhomogeneous electric field produced in the vicinity of a lightning leader tip. The lightning length is taken as l=3.5 km and the ambient large-scale thunderstorm electric field has a magnitude E_0=2 × 10^5 V/m. The results have been obtained in air at ground level atmospheric density. The dynamics of the electron energy distribution in this case is very transient. Differences between electron distributions shown in panels (a) and (b) are associated with the deviation from the RREA spectrum observed by the AGILE satellite [*Tavani et al.*, 2011] and represented in Figure 4.1(b).

distribution, we normalized the distributions shown in Figure 4.2 so that they all yield unity at 1 MeV. Therefore, electron energy distributions are represented in arbitrary units (a.u.). Thermal runaway electrons are initiated with 65 keV in the anti-parallel direction of the electric field [*Celestin and Pasko*, 2011]. Figure 4.2(a) shows the typical dynamics of electrons in the case of RREAs, and we see that the energy distribution dynamically follows the shape of the steady state distribution over time. Figure 4.2(a) also clearly shows the formation of the classical RREA energy cutoff for energies higher than ∼10 MeV. It is important to note that even for transient states of the dynamics of the electrons, in the case of reasonable magnitude of large-scale electric fields in thunderstorms ($\ll E_k \simeq$30 kV/cm, where E_k is the conventional breakdown threshold field defined by the equality of the ionization and dissociative attachment coefficients in air), the electron energy dis-

tribution cannot significantly depart from the RREA equilibrium. The magnitude of 12.5 kV/cm is typical of the electric field in the streamer zone of negative leaders [*Bazelyan and Raizer*, 2000, Chapter 4.6]. Calculations with lower electric fields do not sensibly change this characteristic energy cutoff [*Babich et al.*, 2004; *Dwyer and Smith*, 2005; *Dwyer and Babich*, 2011]. We note that RREA electron energy distribution presented in Figure 4.2(a) is very similar to those already published in the literature [*Babich et al.*, 2004; *Celestin and Pasko*, 2010b; *Dwyer and Babich*, 2011].

Negative part of a bidirectional lightning leader representative of a +IC lightning discharge is believed to propagate by steps [*Rakov and Uman*, 2003, Chapter 9]. The electric field in the vicinity of the lightning leader tip is high only during negative corona flash processes [*Bazelyan and Raizer*, 2000, Chapter 4.6], that are associated with the development of a step of the negative leader. In order to calculate the electric field in the vicinity of a +IC lightning negative leader tip at the very beginning of the corona flash, we used the method of moments [*Balanis*, 1989, p. 670] (see Section 2.1). Physical parameters have been chosen to correspond to a realistic lightning discharge [*Rakov and Uman*, 2003] that would produce enough potential difference in its tip for electrons to reach ∼100 MeV. Electrons are launched 15 cm from the leader tip with 65 keV energy [*Celestin and Pasko*, 2011] in a region where the electric field is extremely inhomogeneous and high ($E \simeq A/r$). Given the extreme acceleration undergone by the electrons within a few nanoseconds, the exact energy of initial electrons does not influence any results of the present work. Figure 4.2(b) shows the normalized electron energy distribution of the ensemble of energetic electrons strongly accelerated in the highly inhomogeneous electric field of the lightning leader tip. The lower energy part (<10 MeV) of the electron distribution is very similar to that in the RREA case (Figure 4.2(a)). The higher energy part, however, differs significantly. Instead of a strong energy cutoff as obtained in the case of RREA (Figure 4.2(a)), the distribution takes a profile closer to a power law that gets steeper with time progression. As further discussed in Section 4.3, the related photons at satellite altitudes possess spectral features in good agreement with those reported by *Tavani et al.* [2011]. Eventually, for long times, the distribution will dynamically converge to a classical RREA distribution type (i.e., dominated by secondary-electrons and

exhibiting high energy cutoff).

4.3 Discussion

We have simulated the transport of bremsstrahlung photons produced by energetic electrons from a given source altitude (chosen to be 15 km) up to a satellite altitude of 500 km. Figure 4.1(b) shows the TGF spectrum for photons collected over a disk of 500 km radius and associated with the electron distributions obtained in the homogeneous field RREA case (Figure 4.2a) and in the case of acceleration of electrons in an inhomogeneous lightning leader field over a time of 30 ns (Figure 4.2(b)). As in Figure 4.2, the calculated spectra shown in Figure 4.1(b) are normalized so that they all yield unity at 1 MeV. We note that for times much greater than $t\sim$10 ns, the electron distribution gets closer to the RREA regime while the electron density produced becomes unphysically large. The very good agreement between measurements and model shown in Figure 4.1(b) provides evidence that the TGF spectrum may be produced during a very transient stage before the lightning field gets screened out by the tremendous production of low-energy electrons. The best agreement is found at t=30 ns, however a satisfying agreement is found for t=30$\pm$10 ns. We note that reported satellite measurements (Figure 4.1(b)) represent averaged data, and therefore a more accurate comparison could be achieved by averaging over different realistic lightning parameters. However, we found that these dependencies are weak compared to the deviation from the RREA regime, as long as a physical electron density produced is considered. Additionally, for a wide range of lightning parameters the electric field in the vicinity of the leader tip is found to keep the same $1/r$ shape, where r is the distance from the leader tip. Note that the collapse time t=30 $\pm$ 10 ns should be considered as an order of magnitude estimate. Indeed, as a reference, the simulation results on the electron distributions have been obtained at ground pressure. The air density drops by a factor of $\sim$7 at 15 km, and the collapse time of the electric field should increase accordingly. Nevertheless, this timescale corresponds well to measurements of current pulses associated with leader steps at low altitude [e.g., *Howard et al.*, 2011, and references therein].

The dispersion timescale of the photons due to Compton scattering in our re-

sults is on the order of ∼50 μs [*Celestin and Pasko*, 2012]. The timescale of TGFs as measured by AGILE [*Marisaldi et al.*, 2010; *Tavani et al.*, 2011] is on the order of 1–5 ms, although RHESSI and Fermi measurements [*Grefenstette et al.*, 2008; *Briggs et al.*, 2010] are closer to ∼0.2 ms or shorter [*Fishman et al.*, 2011]. These discrepancies in measurements prevent us from carrying out detailed comparison on TGF timescales. A possible cause of the differences in timescales is associated with the differences in instrumental dead time [*Grefenstette et al.*, 2008] and in methods different authors use in order to describe the TGF durations. Moreover, although it does not explain the differences between AGILE's and RHESSI/Fermi's results, based on the mechanism we suggest, a process related to lightning propagation might explain long TGF duration. *Lu et al.* [2011] have shown that TGFs were triggered in correlation with fast-rising high current pulses "superposed on a slow pulse that reflects a process raising considerable negative charge within 2–6 ms" and *Shao et al.* [2010] suggested that some of these fast-rising pulses are probably related to the stepping of the negative lightning leader. It is therefore conceivable that one TGF event be produced by several lightning leader steps (corresponding negative corona flashes) over a total period of 2–6 ms. If so, the time resolution of satellite detectors would not always enable possible discrimination between different gamma-ray bursts. This mechanism is also supported by the fact that multiple-pulse TGFs have been recorded since their discovery [*Briggs et al.*, 2010], and *Fishman et al.* [2011] have suggested that many of TGF pulses lasting between ∼200 μs and ∼1 ms might be attributed to overlapping of shorter pulses. A detailed analysis of the Compton dispersion times in the case of TGFs can be found in [*Celestin and Pasko*, 2012].

4.4 Conclusions

In the present study, we demonstrate that the recently observed significant deviation of the TGF spectrum from spectra associated with relativistic runaway electron avalanches in large-scale weak electric fields present in thunderstorms, for energies greater than ∼30 MeV, is the first direct evidence that TGFs are actually produced by runaway electrons strongly accelerated in highly inhomogeneous fields of +IC lightning leader. The presented mechanism is strongly supported by

the observations reported in [*Lu et al.*, 2010, 2011], which have conclusively shown that TGFs can be produced during the upward leader stage of a +IC lightning flash.

The described electron acceleration is a robust process that follows from strong electric field inhomogeneity around lightning leader tips. In order to obtain a good agreement with *Tavani et al.* [2011], we assumed that extremely high electric fields could be produced during the leader stepping process. Whether such a field magnitude is required in the mechanism we suggest, or can exist in the lightning leader tip region is an open question that requires further investigation. Additionally, a significant deviation from the typical RREA spectrum can be obtained only if a high electric potential is present in the lightning leader (we used 350 MV in the present study). More typical lightning producing TGF should involve electric potentials $U_{\rm l}{\sim}100$ MV (see Chapter 3). Using the formula for the non-equilibrium high-energy cutoff $q_{\rm e}A{\simeq}0.08 \times U_{\rm l}$ leads to $q_{\rm e}A{\simeq}8$ MeV that is close to the RREA high-energy cutoff $\mathcal{E}_{\rm c}{\simeq}7$ MeV. This would prevent a strong deviation from the classical RREA spectrum from happening over very short timescales in the case of typical TGFs.

The mechanism presented in this study introduces new outstanding implications on lightning propagation, production of energetic radiation from lightning [*Moore et al.*, 2001; *Dwyer et al.*, 2003, 2005] (see Chapters 7 and 8), production of intense beams of electrons and positrons observed from space [*Dwyer et al.*, 2008a; *Briggs et al.*, 2011], neutron production from lightning [*Shah et al.*, 1985; *Tavani et al.*, 2011] (see Chapter 6), and on radiation dose received by passengers and crew members on airplanes [*Dwyer et al.*, 2010], as the source of energetic electrons, X-rays, and gamma-rays, is much more compact and energetic than that deduced from previous studies.

Chapter 5

Optical Emissions Associated with Terrestrial Gamma-ray Flashes

In this chapter, in order to quantify theoretically optical emissions associated with TGFs, we first apply the Monte Carlo model of electrons introduced in Section 2.2 to simulate the dynamics of the large population of low- and high-energy electrons invoked in TGF events, and then use the optical emission model introduced in Section 2.4 to estimate the associated optical output. The purpose is to quantify optical emissions associated with TGFs. The results presented in this chapter have been published in [*Xu et al.*, 2015b].

5.1 Modeling Results

5.1.1 Distribution of Electrons in the Full Energy Range

We first discuss the distributions of electrons corresponding to relativistic runaway electron avalanches. Figure 5.1(a) shows the distribution of electrons in the full energy range produced by the acceleration and multiplication of relativistic runaway electrons in a large-scale homogeneous reduced electric field of E/N=70 Td ($\sim$18.8 kV/cm at $N\simeq 2.688\times10^{25}$ m^{-3}, 1Td=10^{-21} Vm^2) in air at ground pressure. This energy distribution is obtained after approximately 11 avalanche lengths, or e-folding times, and has already reached steady state. In spite of the noise in the energy distribution, it is clearly shown that over six orders of magnitude, the low-

energy part (<100 eV) is consistent with recently published full energy range model results [*Colman et al.*, 2010] and BOLSIG+ calculations [*Hagelaar and Pitchford*, 2005]. We also note that the ratio between the number of electrons with energy below and above 1 MeV is found to be 9365, in agreement with the analytical calculation documented in [*Dwyer and Babich*, 2011].

Steady-state electron energy distributions in RREAs, for the cases of homogeneous electric fields with magnitudes of 4.3 kV/cm and 12.5 kV/cm, are presented in Figures 5.2(a) and 5.2(b), respectively. The energy distributions are normalized so that the integration from 1 MeV to 100 MeV yields unity, following the assumption that the same number of electrons with energies above 1 MeV is produced in the source of TGFs. Also presented in Figure 5.2(a) are Monte Carlo simulation results (diamonds) documented in [*Dwyer and Babich*, 2011, Figure 3], representing the energy distribution of electrons in RREAs driven by an electric field of 4.3 kV/cm. For this electric field, the high-energy part of electron energy distribution shows excellent agreement between present study and results reported in [*Dwyer and Babich*, 2011, Figure 3]. We therefore estimate that our results on the electron energy distribution are validated over more than 15 orders of magnitude of the dynamic range. To our knowledge, such a result, over the full energy range of interest here, had never been obtained.

Unlike the case in Figure 2.10 (see discussion in Section 2.4.3), energy distribution for electrons propagating under external electric fields, like in RREAs, is governed by the competition between the energy gained from electric fields and energy losses in collisions with neutral gas molecules, which is characterized by the dynamic friction force. The avalanche of relativistic runaway electrons is characterized by abundant generation of large quantities of secondary electrons over the full energy range. Secondary electrons with energies below ∼2 eV are mostly trapped in this region because the applied electric field is insufficient to energize them to penetrate the vibrational barrier at ∼2 eV (see Figure 2.10). Secondary electrons in the "valley" of the dynamic friction force (between ∼3 eV and ∼15 eV) may be subject to an electric force that is capable of overcoming their energy losses in collisions with air molecules and therefore be accelerated to higher energies, depending on the strength of the driving electric field and the momentum of electrons. A driving electric field with a higher strength can surmount a larger

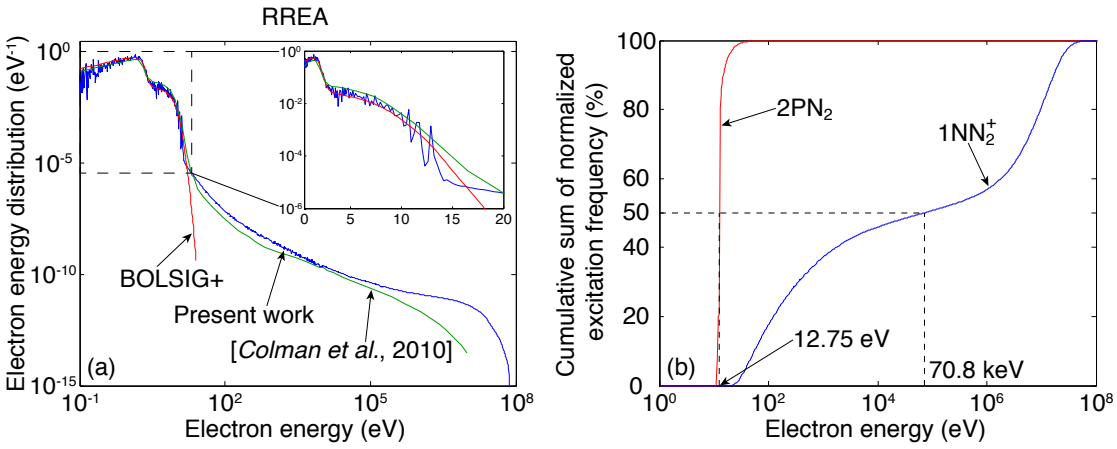

Figure 5.1. (a) Distribution of electrons in the full energy range produced by the acceleration and multiplication of relativistic runaway electrons in a large-scale homogeneous electric field of 70 Td (18.8 kV/cm). The results are obtained in air at ground level ($N_0 \simeq 2.688 \times 10^{25}$ m^{-3}). Also presented are BOLSIG+ calculations [*Hagelaar and Pitchford*, 2005] and modeling results presented in [*Colman et al.*, 2010, Figure 11]. The inset is a zoom-in view of the energy range below 20 eV. (b) Cumulative sum of the frequency for generating $N_2(C^3\Pi_u)$ and $N_2^+(B^2\Sigma_u^+)$ excited by the large population of electrons in RREA processes. 50% of the upper excited states of N_2 responsible for $2PN_2$ emissions is produced by electrons with energy below 12.75 eV while 50% of the upper excited states of N_2^+ responsible for $1NN_2^+$ emissions is produced by electrons with energy below 70.8 keV.

dynamic friction force, corresponding to higher electron energies in the sharp increase between ∼4 eV and ∼123 eV (see Figure 2.10). Therefore, one sees from Figure 5.2(a) that, under a stronger electric field, more electrons are sustained in the energy range between ∼2 eV and ∼15 eV. Also, in the high-energy part, the energy cutoff corresponding to the distribution obtained in 18.8 kV/cm is higher than that in 4.3 kV/cm.

For the purpose of understanding the energy dependence of the capability of electrons in generating the upper excited states responsible for optical emissions from $2PN_2$ and $1NN_2^+$, the excitation frequencies are calculated using equation (2.14) and the energy distribution obtained in the present study and shown in Figure 5.1(a). Figure 5.1(b) presents the cumulative sum of the frequency for generating $N_2(C^3\Pi_u)$ and $N_2^+(B^2\Sigma_u^+)$ in RREAs driven by an electric field of 70 Td. It is found that 50% of the upper excited states of N_2 responsible for $2PN_2$ emissions is produced by electrons with energy below 12.75 eV while 50% of the

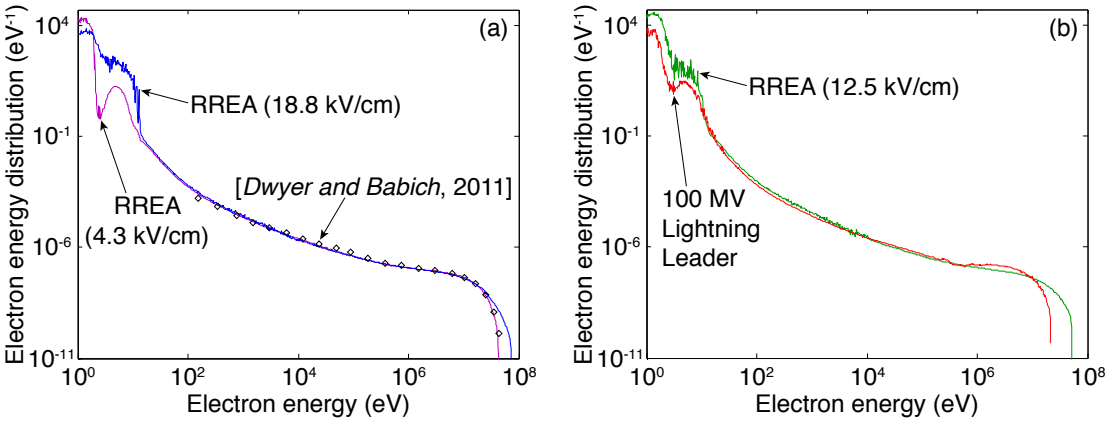

Figure 5.2. Distribution of electrons in the full energy range produced by the acceleration and multiplication of (a) relativistic runaway electrons in homogeneous electric fields: 4.3 kV/cm and 18.8 kV/cm and (b) relativistic runaway electrons in a homogeneous electric field of 12.5 kV/cm and thermal runaway electrons in the highly inhomogeneous electric field near the tip region of a 100 MV lightning leader during the negative corona flash. The energy distributions are normalized so that the integration over electron energy from 1 MeV to 100 MeV yields unity. The results are obtained in air at ground level atmospheric density. The diamonds in Figure 5.2(a) show Monte Carlo simulation results obtained from [*Dwyer and Babich*, 2011, Figure 3], representing the energy distribution of electrons in RREAs driven by the electric field of 4.3 kV/cm.

upper excited states of N_2^+ responsible for $1NN_2^+$ emissions is produced by electrons with energy below 70.8 keV. This feature can be attributed to the fundamental difference in cross sections. The excitation cross section for generating $N_2^+(B^2\Sigma_u^+)$ rises rapidly above the threshold, attains a maximum value around 114 eV, drops off slowly to around 1 MeV, and then reaches a relativistic plateau above 1 MeV. Consequently, even though they are in minority, relativistic electrons efficiently ionize nitrogen molecules, resulting in comparable amount of $N_2^+(B^2\Sigma_u^+)$ when compared to that produced by low-energy electrons. In contrast, the cross section of $2PN_2$ falls off much faster after the peak and becomes negligible in the region above 1 keV, where ionization and elastic collisions dominate. As a result, optical emissions from $2PN_2$ are mainly produced by low-energy electrons. We note that since the electron impact excitation cross section for generating $N_2(B^3\Pi_g)$ has a similar behavior as that of $2PN_2$, optical emissions from $1PN_2$ should also mainly come from low-energy electrons.

We now discuss the acceleration of thermal runaway electrons in lightning leader

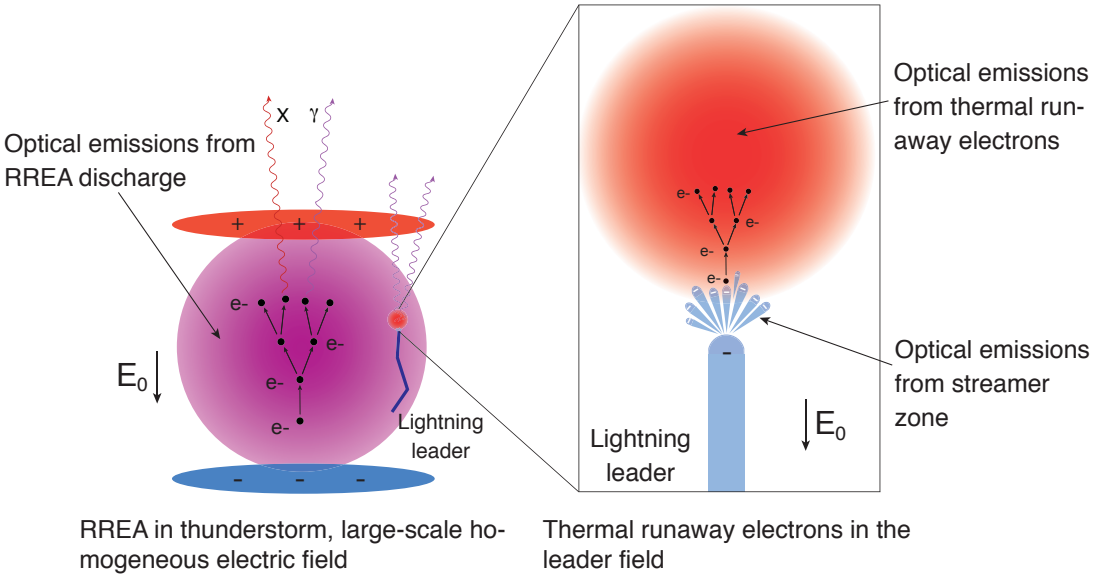

Figure 5.3. Illustration of optical emissions produced by two TGF production mechanisms: RREAs in the large-scale homogeneous electric field within thunderstorms and the acceleration of thermal runaway electrons under the highly inhomogeneous electric field produced around the tip region of +IC lightning leaders during the negative corona flash stage.

field. Figure 5.2(b) shows the full energy distribution for the ensemble of electrons accelerated in the highly inhomogeneous electric field around the tip region of a 100 MV lightning leader during the negative corona flash stage. The electron energy distributions are normalized so that the integration from 1 MeV to 100 MeV yields unity. Due to the differences in geometry and magnitude between the homogeneous and inhomogeneous electric fields, a deviation from the RREA energy distribution is observed: a much smaller amount of electrons is generated and accumulated in the low energy range and the high energy tail does not follow the typical energy cutoff observed in RREAs.

5.1.2 Characteristic Radial Dimensions

The characteristics of optical emissions that can be possibly generated during TGF production, including intensities and intensity ratios between optical band systems, mostly depend on the energy distribution and number density of electrons. In this study, we assume a uniform spherically symmetric spatial distribution for source

electrons involved in TGFs. The number density of electrons is then obtained using the full energy distribution and an assumption that 10^{17} electrons with energy above 1 MeV are generated during TGFs, that is believed to be required for explaining TGF fluences and spectra [*Dwyer and Smith*, 2005]. Although this assumption is not accurate enough for capturing the morphological characteristics of TGF-associated optical emissions, it does not have a significant impact on quantifying the total intensities of optical emissions. The characteristic radial dimensions representing processes of RREAs, acceleration and multiplication of thermal runaway electrons in the electric field produced by the 100 MV lightning leader, and the streamer zone of the 100 MV lightning leader are discussed below, schematically depicted in Figure 5.3, and summarized in Table 5.1.

In RREAs, propagation over multiple runaway avalanche lengths is required for relativistic electrons to gain energies on the order of tens of MeV and to explain TGF spectra, corresponding to an acceleration distance of approximately 1 km for typical electric fields in thunderclouds [e.g., *Dwyer et al.*, 2012]. In contrast, the electric field produced by a 100 MV lightning leader is highly inhomogeneous, intense close to the leader tip, and weak far from it [e.g., *Celestin et al.*, 2012]. This electric field is able to accelerate thermal runaway electrons over relatively short distances and a representative acceleration distance is found to be approximately 50 m for a 100 MV lightning leader.

Additionally, in this study, the intensity of optical emissions radiated from the streamer zone of the 100 MV lightning leader is also estimated using the streamer modeling results presented in [*Liu et al.*, 2008] and the knowledge about the number of streamers and the size of the streamer zone. Specifically, the number of streamers present in a streamer zone is N_s=Q_s/q_s [e.g., *Bazelyan and Raizer*, 2000; *Celestin and Pasko*, 2011], where Q_s is the total electric charge contained in the streamer zone, and q_s is the average charge carried by a streamer, typically on the order of 1 nC [*Bazelyan and Raizer*, 2000, p. 69-71]. As pointed out by *Celestin and Pasko* [2011], Q_s is quadratically dependent on the electric potential difference formed by lightning leaders with respect to ambient potential (U_l) and can be calculated as follows:

$$Q_s = \frac{\pi \varepsilon_0 {U_l}^2}{2E_s^-} \tag{5.1}$$

Table 5.1. Intensity of optical emissions from $2PN_2$ (column 3) and $1NN_2^+$ (column 4) in Rayleighs and intensity ratio between $2PN_2$ and $1NN_2^+$ (column 5) in the visible range with wavelengths between 390 nm and 700 nm for different acceleration processes (column 1) with different characteristic sizes (column 2) calculated at ground level.

	Radius, m	$2PN_2$, R	$1NN_2^+$, R	$\frac{2PN_2}{1NN_2^+}$
RREA (4.3 kV/cm)	1000	8.99×10^8	1.22×10^9	0.74
RREA (12.5 kV/cm)	1000	1.70×10^9	1.55×10^9	1.10
RREA (18.8 kV/cm)	1000	6.63×10^9	1.31×10^9	5.06
Thermal runaway electrons	50	8.28×10^{11}	5.23×10^{11}	1.58
Streamer zone	40	6.83×10^{10}	6.75×10^8	101.19

where E_s^- is the electric field in the steamer zone of negative leaders and its value is taken as 12.5 kV/cm [*Babaeva and Naidis*, 1997, Figure 7]. From these relations, we estimate that, for long unbranched lightning leaders with electric potential of 100 MV, the total number of streamers constituting the streamer zone should be on the order of 10^8. As for the characteristic size of the streamer zone, it is proportional to U_l: $R_{sz}^-=U_l/2E_s^-$. Therefore, the radius of the streamer zone in front of a 100 MV lightning leader is approximately 40 m at ground level.

5.1.3 Optical Emissions

Table 5.1 shows modeling results of optical emissions in the visible range with wavelengths between 390 nm and 700 nm from the following processes at ground level: RREAs, acceleration of thermal runaway electrons in the electric field produced by the 100 MV lightning leader, and the streamer zone of the 100 MV lightning leader. The columns 3 and 4 represent the maximum intensities of optical emissions from $2PN_2$ and $1NN_2^+$, respectively, during the listed processes. Column 5 shows the intensity ratio of $2PN_2$ to $1NN_2^+$ associated with the listed processes. Interestingly, both TGF production mechanisms produce considerable amount of optical emissions, with magnitudes comparable to those from the streamer zone associated with the 100 MV lightning leader. The intensities of optical emissions from the streamer zone of the 100 MV lightning leader are, respectively, 6.83×10^{10} R and 6.75×10^8 R for $2PN_2$ and $1NN_2^+$. The intensity ratio of $2PN_2$ to $1NN_2^+$ is $\sim$101.19 as indicated in [*Liu et al.*, 2008].

The optical emissions from $2PN_2$ (Column 3 of Table 5.1) associated with RREAs intensify with increases in driving electric fields. For instance, the intensities of optical emissions from $2PN_2$ corresponding to RREAs under electric fields of 4.3 kV/cm, 12.5 kV/cm, and 18.8 kV/cm are, respectively, 8.99×10^8 R, 1.70×10^9 R, and 6.63×10^9 R. However, optical emissions from $1NN_2^+$ associated with RREAs do not indicate the same tendency. The intensities for 4.3 kV/cm, 12.5 kV/cm, and 18.8 kV/cm are, respectively, 1.22×10^9 R, 1.55×10^9 R, and 1.31×10^9 R. To understand how these characteristic tendencies in optical output stem from the differences in electron energy distributions, it is helpful to look at the comparison presented in Figure 5.2(a). As the homogeneous electric field increases from 4.3 kV/cm to 18.8 kV/cm, a larger population of electrons is generated and sustained in the energy region from ∼11 eV (the threshold energy of producing $N_2(C^3\Pi_u)$) to ∼15 eV, which plays a critical role in generating optical emissions from $2PN_2$ (see Figure 5.1(b) and discussion in Section 5.1.1). Therefore, a stronger driving electric field should demonstrate more intense optical emissions from $2PN_2$ during RREAs as evident in Table 5.1. Contrary to $2PN_2$, optical emissions from $1NN_2^+$ can be as efficiently produced by high-energy electrons as by low-energy ones (see Figure 5.1(b) and discussion in Section 5.1.1). Considering that the energy distribution of electrons accelerating in 18.8 kV/cm only slightly differs from that of 4.3 kV/cm in the energy region above ∼20 eV (the threshold energy of producing $N_2^+(B^2\Sigma_u^+)$), optical emissions from $1NN_2^+$ should not change considerably, as also evident in Table 5.1.

The two TGF production mechanisms exhibit distinct optical features. First, when compared to RREA processes, optical emissions associated with the acceleration of thermal runaway electrons in lightning leader fields are more intense, with 8.28×10^{11} R from $2PN_2$, and 5.23×10^{11} R from $1NN_2^+$. This is primarily because the production of TGFs by thermal runaway electrons is naturally confined to a relatively compact region in the vicinity of negative leader tip [e.g., *Xu et al.*, 2012; *Celestin et al.*, 2012]. Second, the intensity ratios of $2PN_2$ to $1NN_2^+$ are different between the two production mechanisms. For example, the intensity ratios between $2PN_2$ and $1NN_2^+$ are, respectively, 0.74, 1.10, and 5.06 for RREAs driven by homogeneous electric fields of 4.3 kV/cm, 12.5 kV/cm, and 18.8 kV/cm while this ratio is 1.58 for thermal runaway electrons produced and accelerated in the

vicinity of the tip of a 100 MV lightning leader. This feature is also a reflection of the inherent differences in electron energy distributions. In the case of RREAs, as discussed in the previous paragraph, a stronger driving electric field enables more intense optical emissions from $2PN_2$ while producing similar levels of emissions from $1NN_2^+$. For this reason, one observes in Table 5.1 that the intensity ratio between $2PN_2$ and $1NN_2^+$ in RREAs becomes larger with the increase in driving electric fields. Nevertheless, as illustrated in Figure 5.2(b), the electron energy distributions are similar in the energy regions critical for optical emissions from $2PN_2$ and $1NN_2^+$ between the processes of thermal runaway electrons accelerating in a 100 MV lightning leader field and RREAs under an electric field of 12.5 kV/cm. Thus, the intensity ratios of $2PN_2$ to $1NN_2^+$ are close between these two processes as presented in Table 5.1.

5.2 Discussion

5.2.1 Fluorescence Efficiency

For the sake of comparison with previously published results [*Dwyer et al.*, 2013], we have calculated the fluorescence efficiency without collisional quenching corresponding to the steady state of RREAs for $2PN_2$ and $1NN_2^+$ using:

$$\epsilon_k = \frac{\nu_k \sum_\lambda E_\lambda I_k(\lambda)}{\int F_d(\varepsilon) v(\varepsilon) f(\varepsilon) d\varepsilon} \tag{5.2}$$

where E_λ is the energy of fluorescence photons with a wavelength of λ defined as $E_\lambda = hc/\lambda$, where h is the Planck's constant and c is the speed of light in free space, v is the speed of electron, $f(\varepsilon)$ is the electron energy distribution and is normalized so that $\int_0^{+\infty} f(\varepsilon) d\varepsilon = 1$, $I_k(\lambda)$ is the intensity distribution of optical emissions from either $2PN_2$ or $1NN_2^+$ obtained from [*Vallance Jones*, 1974] and is normalized so that $\sum_\lambda I_k(\lambda) = 1$, and $F_d(\varepsilon)$ is the dynamic friction force due to collisions with air molecules that electrons with an energy of ε experience at ground level (see Figure 2.10). The summation in the numerator is taken over wavelengths of major emission bands from $2PN_2$ or $1NN_2^+$.

It is found that the fluorescence efficiencies of $2PN_2$ for RREAs driven by elec-

tric fields of 4.3 kV/cm, 12.5 kV/cm, and 18.8 kV/cm, in the visible range of 390–700 nm without collisional quenching, are 0.0146%, 0.00287%, and 0.0208%, respectively. Fluorescence efficiencies of $1NN_2^+$, in the same wavelength range without collisional quenching, are 0.213%, 0.0283%, and 0.0443% for RREAs driven by electric fields of 4.3 kV/cm, 12.5 kV/cm, and 18.8 kV/cm, respectively. Based on these calculations, fluorescence efficiency in RREAs can vary notably for driving electric fields with different strengths.

More intense driving electric fields do not necessarily facilitate the capability of generating fluorescence photons, even if they result into more energy deposition. In RREAs, energy deposition and fluorescence emissions are caused by electrons in different energy regions. To illustrate this discussion, we have verified that, for the three electric fields studied in the present study, most of the energy deposited via collisions with air molecules is from the large ensemble of electrons in the energy region below ∼4 eV while optical emissions from $2PN_2$ are mostly produced by electrons with energies between ∼11 eV and 100 eV. As the driving electric field changes from 4.3 kV/cm to 12.5 kV/cm, although secondary electrons in the energy region between ∼11 eV and 100 eV are able to excite more $N_2(C^3\Pi_u)$ and release more energy to generate fluorescence photons from $2PN_2$, secondary electrons in the energy region below ∼4 eV much more efficiently deposit their kinetic energy into air molecules through vibrational and rotational excitations. As a consequence, the fluorescence efficiency for $2PN_2$ decreases. However, when the driving electric field further increases to 18.8 kV/cm, a much larger population of secondary electrons can be accelerated to and maintained in the energy region that is critical for $2PN_2$ and therefore, an enhancement in fluorescence efficiency is observed. Similarly, fluorescence efficiency for $1NN_2^+$ in RREAs is also highly dependent on the strength of the driving electric field.

Moreover, extensive air showers have also been studied through modeling of the deceleration of 1 MeV electrons [e.g., *Lafebre et al.*, 2009] in the atmosphere. Using the same model as described in Section 2.4, we have estimated the fluorescence efficiencies of 1 MeV electrons decelerating through air. It is interesting to note that the fluorescence efficiencies for $2PN_2$ and $1NN_2^+$, between 390 and 700 nm and without collisional quenching, are found to be 0.0405% and 0.554%, respectively, that are very close to the values reported and used in [*Dwyer et al.*, 2013]. We note

that the calculations of *Dwyer et al.* [2013] were based on experimentally measured fluorescence efficiency for 0–0 bands of $2PN_2$ and $1NN_2^+$ [*Keilhauer et al.*, 2006].

Additionally, we have estimated the amount of fluorescence photons that can be generated per runaway electron per meter in the wavelength range between 300 and 430 nm corresponding to the steady state of RREAs. This wavelength range is chosen in order to cover major emission bands of $2PN_2$ and $1NN_2^+$. The values are approximately 6.2, 10.7, and 38.3 m^{-1} for RREAs driven by electric fields of 4.3 kV/cm, 12.5 kV/cm, and 18.8 kV/cm at ground pressure, respectively. This shows that the stronger the sustaining electric fields, the more intense the associated fluorescence emissions. Note that this is consistent with the transition in electron energy distributions (i.e., enhancement in low energy part with increasing applied electric field) presented in Figure 5.2(a).

Fluorescence efficiency, in the study of air showers, describes a process in which primary electrons eventually deposit all their initial kinetic energy into air molecules of the Earth's atmosphere, which release it through emissions of fluorescence and collisional de-excitation. However, in TGFs, the acceleration of electrons and the subsequent deposition of energy into air molecules are supported by external electric fields. The amount of energy that is converted into fluorescence emissions mainly comes from that acquired from the driving external electric field. Therefore, we suggest that fluorescence efficiency in extensive air showers is not conceptually identical to that in TGFs. Also, the electron energy distribution obtained from the deceleration process in extensive air showers can be substantially different from that in TGFs. Due to the fact that the capability of electrons in generating fluorescence photons is strongly energy-dependent (see Figure 5.1(b) and discussion in Section 5.1.1), fluorescence efficiency measured in extensive air showers is different from that in TGFs.

5.2.2 Scaling of Optical Emissions

Modeling results of optical emissions from the acceleration of thermal runaway electrons in lightning leader field and the associated streamer zone (Table 5.1) are obtained at ground level air density and can be applied to the source altitudes of TGFs using similarity scaling. As indicated in equation (2.12), the intensity

of optical emissions is closely related to the number density of excited species n_k. For the two optical band systems considered in the present study, i.e., $2PN_2$ and $1NN_2^+$, n_k is populated by electron impact excitations (the term $\nu_k n_e$ in equation (2.13)) and depopulated by spontaneous emissions (the term A_k in τ_k) and collisional quenching processes with air molecules (the terms $\alpha_1 N_{N_2}$ and $\alpha_2 N_{O_2}$ in τ_k). At or below the source altitudes of TGFs [e.g., *Dwyer and Smith*, 2005; *Xu et al.*, 2012], collisional quenching processes dominate over spontaneous emissions and, in the steady state, n_k is obtained by the balance between direct excitations and collisional quenching. Since both collisional quenching ($\alpha_1 N_{N_2}+\alpha_2 N_{O_2}$) and electron impact excitation frequency (ν_k) scale as the number density of air molecules ($\sim N$), n_k scales as the number density of electrons n_e.

The total number of energetic electrons emitted from a lightning leader is invariant of air pressure [*Moss et al.*, 2006]. The characteristic radial dimension of the acceleration of thermal runaway electrons in the lightning leader field is defined as the region in which thermal runaway electrons accelerate and efficiently gain energy. This condition is fulfilled if the electric force exerted on thermal runaway electrons $q_e E$ exceeds the average friction force due to collisions with air molecules $q_e E_t$, where q_e is the electron elementary charge, E is the inhomogeneous electric field produced around lightning leader tips, and E_t is the relativistic runaway threshold field and its value at ground level atmospheric density is $\sim$2.8 kV/cm [e.g., *Dwyer et al.*, 2012]. As suggested by *Celestin et al.* [2012], the highly inhomogeneous electric field produced by high-potential lightning leaders can be approximated by a A/r function, where A=0.04$E_0 l$ is a fitting parameter and r is the distance from the leader tip. Since r varies inversely as E and E_t scales proportionally with the density of the neutral atmosphere, i.e., $E_t \sim N$, the characteristic radial dimension representing the acceleration of thermal runaway electrons in lightning leader field scales as $\sim 1/N$. Therefore, the number density of energetic electrons responsible for producing $N_2(C^3\Pi_u)$ or $N_2^+(B^2\Sigma_u^+)$ scales as $\sim N^3$. Furthermore, considering that the intensity of optical emissions in Rayleighs is obtained by integrating over the line of sight, which scales as $\sim 1/N$, and remembering that n_k scales in the same way as n_e, one sees that the cross-sectional intensity of optical emissions (number of photons per unit area) produced during the acceleration of thermal runaway electrons in lightning leader fields scales as $\sim N^2$. For

example, the characteristic radius for the acceleration process of thermal runaway electrons in the 100 MV lightning leader case is 50 m at ground level. However, this quantity at an altitude of 12 km, corresponding to the source altitude of TGFs [e.g., *Xu et al.*, 2012], would be ∼211 m. The intensity of $2PN_2$ is 8.28×10^{11} R at ground level and would be reduced to $\sim4.63\times10^{10}$ R at 12 km altitude.

As for the streamer zone associated with the 100 MV lightning leader, its radius R_{sz}^{-} is inversely proportional to E_s^{-}, which scales as $\sim N$. Thus, R_{sz^-} scales as $\sim 1/N$. For the same reason, the total amount of electric charge contained in the streamer zone Q_s also scales as $\sim 1/N$ (see equation (5.1)). Because the streamer electron density scales as $\sim N^2$ and the length scale of streamers scales as $\sim 1/N$ [e.g., *Liu and Pasko*, 2004], the average electric charge carried by a streamer q_s scales as $\sim 1/N$. Since both Q_s and q_s scale as $\sim 1/N$, the number of streamers constituting a streamer zone, which follows the relation $N_s{=}Q_s/q_s$, remains the same with respect to air pressure/altitude. In a single streamer, n_k scales as the streamer electron density, $\sim N^2$. As a result, by integrating over the line of sight, the total cross-sectional intensity of optical emissions from the streamer zone scales as $\sim N$.

5.2.3 Optical Emissions Associated with TGFs

Different driving electric fields can lead to different acceleration and multiplication processes undergone by runaway electrons, different energy distributions obtained, and thus different capabilities in generating optical emissions. In RREA mechanism, relativistic runaway electrons harvest energy from the homogeneous electric field and gradually build up a ∼7 MeV high-energy cutoff while producing a large ensemble of low energy electrons. Reciprocally, as demonstrated in [*Celestin et al.*, 2012], the intense inhomogeneous electric field naturally present in compact regions around negative lightning leader tips during stepping processes can accelerate thermal runaway electrons to high energy over much shorter distances, corresponding to much less low energy electrons generated and a significantly different high-energy cutoff. A direct consequence of these differences is that the optical output associated with these two mechanisms is different. For example, typical magnitude of the large spatial scale electric field measured within thunderstorms is lower than

4.3 kV/cm [e.g., *Marshall et al.*, 2001], the optical output from RREA processes driven by such field would be much weaker than that resulting from the acceleration process of thermal runaway electrons in lightning leader fields (see Table 5.1). In addition, the intensity ratio between $2PN_2$ and $1NN_2^+$ would be one order of magnitude smaller than that of the acceleration and multiplication processes of thermal runaway electrons in lightning leader fields.

TGF events are most likely accompanied with detectable levels of optical emissions for both existing models as suggested by modeling results (see Table 5.1). These optical emissions would be dominantly blue, or purplish blue (as previously noted by *Dwyer et al.* [2013]). Moreover, using modeling results of optical emissions produced during the acceleration of thermal runaway electrons in lightning leader fields (see Table 5.1) and considering that this process typically lasts from a few tens of microseconds to a fraction of millisecond, we estimate that the total amount of optical energy radiated in the visible wavelength range in this scenario would be on the order of a few tens of joules. It is comparable to that theoretically produced during relativistic feedback discharges [*Dwyer et al.*, 2013], but much smaller than that of normal lightning [*Uman*, 2001]. Further comparisons between modeling results and ground-based or space-based observations would require the transport of optical photons through the atmosphere to be taken into account. This is especially important in view of the future space missions ASIM (ESA) and TARANIS (CNES) that will detect TGFs from the nadir direction in association with photometric measurements. Furthermore, the temperature and humidity dependence of air fluorescence emissions have been recently observed [e.g., *Ave et al.*, 2008] and this effect should be accounted for to ensure accurate interpretation of optical observations.

The theory of TGFs produced by thermal runaway electrons initiated during negative corona flashes may introduce a chronological order between the optical emissions from molecules excited by thermal runaway electrons and their by-products, and those from the streamer zone. Indeed, one can expect that most thermal runaway electrons and their corresponding optical emissions would be produced during the early establishment of the streamer zone, the transient stage of negative corona flashes. According to discussion in [*Moss et al.*, 2006], the duration of the negative corona flash can be estimated from the speed of streamers and the

size of the eventual streamer zone. Considering that streamers, on average, have speeds on the order of $\sim 5 \times 10^6$ m/s [e.g., *Stenbaek-Nielsen et al.*, 2013] and the typical size of the streamer zone associated with high-potential lightning leaders in +ICs is in the range 50–600 m [e.g., *Winn et al.*, 2011], the corresponding negative corona flashes should have a duration in the range 10–200 μs. However, streamer zones could last for much longer time, typically in the range from 100 μs to 1 ms in +ICs [e.g., *Marshall et al.*, 2013; *Lu et al.*, 2011]. It is interesting to note that -CG lightning leaders emit intense bursts of X-rays during the stepping process [e.g., *Dwyer et al.*, 2005] and it is possible that fluorescence light is also observable around the tip of these leaders (see Chapter 7).

Even if difficult to observe, optical emissions from molecules excited by thermal runaway electrons accelerating in lightning leader fields are temporally and spatially separated from those normally produced by lightning leaders. Optical emissions associated with the acceleration process of thermal runaway electrons in lightning leader fields would be produced right after the establishment of a new leader step, while the lightning leader would emit light during the connection of the new step. Additionally, since thermal runaway electrons gain energy from the electric field produced in the vicinity of lightning leader tips, fluorescence photons caused by thermal runaway electrons would be emitted from a compact region in front of lightning leaders. Moreover, considering that electrons involved in lightning discharges and those produced by thermal runaway electrons can follow significantly different energy distributions, it is conceivable that the intensity ratio of optical emissions from $2PN_2$ and $1NN_2^+$ can be considerably different between the two processes. Finally, the production of optical emissions by RREAs sustained by large-scale thunderstorm electric fields is not necessarily associated with stepping lightning leaders and could therefore be observed in the absence of light from lightning discharges.

5.3 Conclusions

Principal contributions of this chapter can be summarized as follows:

1. We have studied the characteristics of electron energy distributions over the full energy range for two TGF production mechanisms: RREAs and production

of thermal runaway electrons during the negative corona flash stages of lightning leaders. The simulated energy distribution of RREAs is in good agreement with results presented in [e.g., *Dwyer and Babich*, 2011; *Dwyer et al.*, 2012] in the high energy region, and BOLSIG+ [*Hagelaar and Pitchford*, 2005] and [*Colman et al.*, 2010] simulation results in the lower energy part. The importance of the dynamic friction force in characterizing electron energy distribution has been discussed. Furthermore, we have emphasized that the capability of RREA electrons in generating the upper excited states responsible for optical emissions from $2PN_2$ and $1NN_2^+$ is strongly energy-dependent. It has also been illustrated how RREAs and the acceleration of thermal runaway electrons in the electric field produced by high-potential lightning leader lead to inherently different electron energy distributions.

2. From the knowledge of the electron energy distribution, we have quantified the optical emissions that are possibly generated during the production of TGFs. Modeling results indicate that TGFs are most likely accompanied with detectable levels of optical emissions. However, optical emissions produced during the acceleration of thermal runaway electrons in lightning leader fields are more intense than those associated with RREAs. The intensity ratio between $2PN_2$ and $1NN_2^+$ is intrinsically different between these two TGF production mechanisms.

3. Calculations of fluorescence efficiencies corresponding to air showers and the steady state of RREAs have shown that, although broadly used in the study of air showers to identify the energy of primary particles, fluorescence efficiencies are not appropriate for the study of optical emissions from TGF sources.

4. Optical emissions, appearing as the low-energy signature of the underlying electron acceleration mechanism, are a good probe to point to key ingredients in TGF production. The distinct optical features described in the present study can be used to constrain and validate existing TGF models. Moreover, since TGFs are closely associated with the initial development of +IC flashes [e.g., *Marshall et al.*, 2013], measurements of optical emissions also have important implications for studies of the initial breakdown stage of lightning flashes.

Chapter 6

Neutron Production by High-Potential Lightning Leaders

In this chapter, we apply the Monte Carlo models for electrons and photons introduced in Chapter 2 to study neutron generation via photonuclear interaction experienced by energetic bremsstrahlung photons produced during the acceleration of thermal runaway electrons in the electric field of high-potential lightning leaders. Modeling results in the case of RREAs are also compared to previously published calculations [*Carlson et al.*, 2010; *Drozdov et al.*, 2013] in order to validate the simulation. The work documented in this chapter has been presented in a conference talk [*Xu et al.*, 2013].

6.1 Modeling Results and Discussion

In order to validate current simulation of photonuclear interactions, we have calculated neutron production by photonuclear interactions experienced by energetic bremsstrahlung photons resulting from RREA processes. Specifically, it is assumed that the ensemble of electrons in RREAs follow the typical 7 MeV energy cutoff ($\sim\exp(-\varepsilon/\varepsilon_c)$, where ε is the electron energy and ε_c is the energy cutoff). With the knowledge of the differential cross section of bremsstrahlung radiation, we calculate the energy and angular distributions of bremsstrahlung photons. Using the Monte Carlo model described in Section 2.3, we further estimate corresponding neutron generation through photonuclear interaction between energetic photons and air

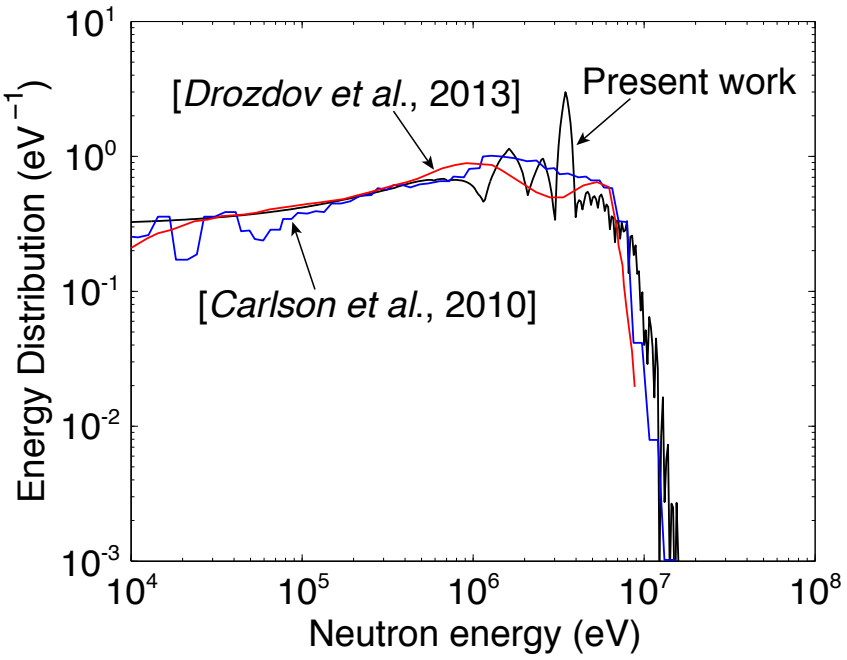

Figure 6.1. Comparison of distribution of neutrons in the energy range between 10 keV and 100 MeV produced by photonuclear interaction of energetic bremsstrahlung photons in RREAs between present calculation and modeling results presented in [*Carlson et al.*, 2010] and [*Drozdov et al.*, 2013].

molecules. Modeling results and the comparison with calculations documented in [*Drozdov et al.*, 2013; *Carlson et al.*, 2010] are presented in Figure 6.1. As shown in this figure, present modeling results are in good agreements with those reported by *Drozdov et al.* [2013] and *Carlson et al.* [2010] in terms of general shape. We note that the difference around 4 MeV is likely due to the use of different sets of differential cross sections for modeling photonuclear interactions.

After validation, the set of Monte Carlo models is employed to investigate neutron production by high-potential lightning leaders. Figure 6.2(a) shows the energy distribution of electrons caused by the acceleration in the electric field produced during the stepping of lightning leaders for two electrical potentials: 100 MV and 200 MV. As expected, one sees in this figure that an electrical potential of 200 MV accelerates thermal runaway electrons to higher energies. For both electrical potentials, electrons with energies above the threshold of 10.5 MeV are generated (see Figure 2.7). Figure 6.2(b) shows the energy spectra of photons

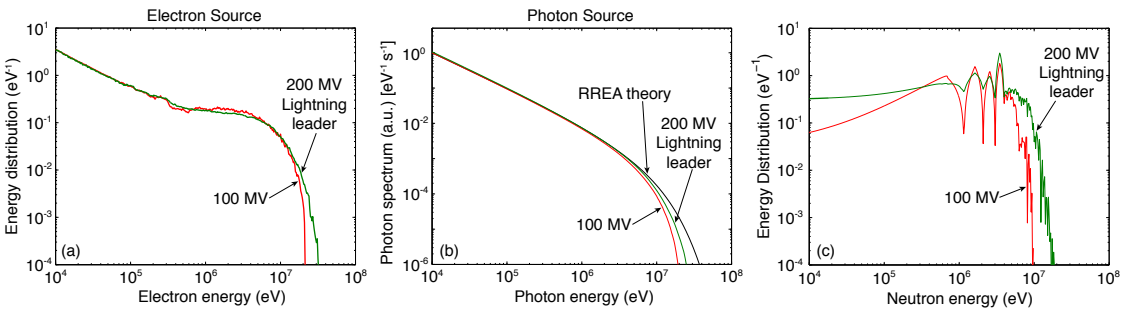

Figure 6.2. (a) Energy distributions of electrons accelerated in the electric field produced near the tip region by stepping lightning leaders with two electrical potentials: 100 MV and 200 MV. (b) Energy spectrum of corresponding bremsstrahlung photons. (c) Corresponding energy distributions of neutrons produced by photonuclear interaction.

resulting from bremsstrahlung radiation produced by those electrons presented in Figure 6.2(a) for two lightning leader electrical potential drops: 100 MV and 200 MV, and in RREAs. As clearly shown in this figure, a small portion of photons is produced with energies exceeding the threshold of photonuclear interactions (∼10.5 MeV). Figure 6.2(c) shows the corresponding normalized energy distributions of neutrons produced by photonuclear interactions in the energy range between 10 keV and 100 MeV. Both the 100 MV and 200 MV electrical potentials lead to production of neutrons, mostly with energies of a few MeVs.

Xu et al. [2012] have shown that the TGF energy spectrum measured by RHESSI can be explained by the acceleration of thermal runaway electrons in the electric field of a 100 MV lightning leader at an altitude of 12.4 km. From the theory described in [*Celestin and Pasko*, 2011], we estimate that a 100 MV lightning leader discharge produces a maximum of $\sim 10^{19}$ photons with energy above 10 keV. Approximately 0.3% of these photons have energies greater than the threshold of photonuclear interaction. Moreover, using the photon energy spectrum shown in Figure 6.2(b) and the cross section of photonuclear interaction, we estimate that approximately 6×10^{-3} neutrons can be generated per photon with energy >10.5 MeV produced by a 100 MV lightning leader. This indicates that $\sim 10^{14}$ neutrons could be produced during this process. We note that this value is an order of magnitude estimate and it is consistent with that calculated by *Babich and Roussel-Dupré* [2007].

Figure 6.3(a) is a sketch of the simulation domain used for modeling neutron

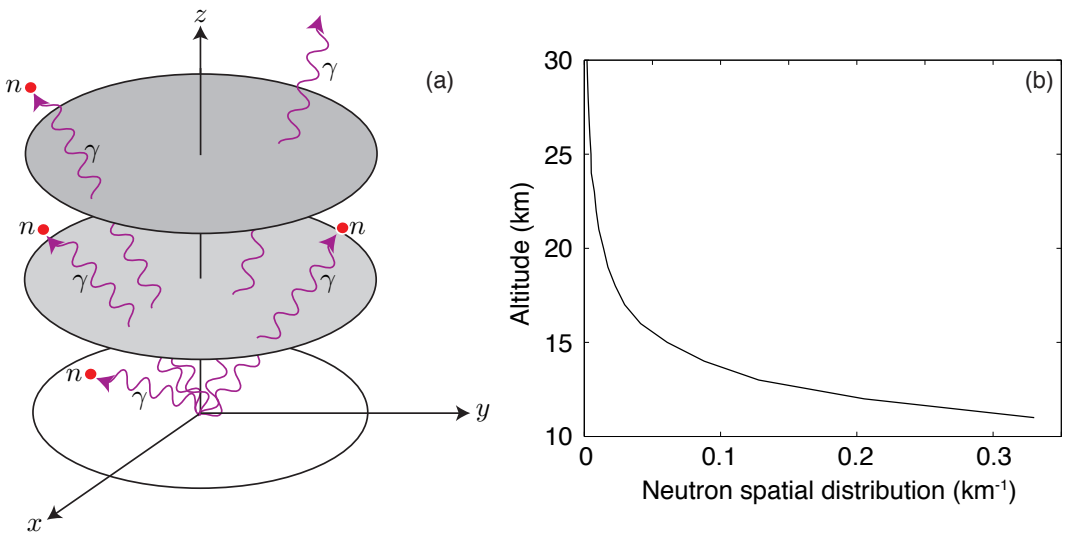

Figure 6.3. (a) Illustration of the simulation geometry used for modeling neutron production. (b) Spatial distribution of neutrons versus production altitude. The integration of the curve over altitude yields unity.

production in the Monte Carlo model. In particular, in order to investigate the spatial distribution of source neutrons, once high-energy photons experience photonuclear interactions, the altitudes at which the collisions take place are recorded. Figure 6.3(b) shows the spatial distribution of neutrons produced by a 100 MV lightning leader at an altitude of 12.4 km. The curve is normalized so that the integration over altitude yields unity. We note that the amount of neutrons produced with respect to the altitude follows an exponential distribution with an e-folding length of approximately 2.5 km. Interestingly, we observe that neutrons can be produced even a few tens of kilometers away from the photon source. This feature can be interpreted by the mean free path of those photons that give rise to neutron production. For instance, the mean free path of a 11 MeV photon in ambient air density corresponding to the altitude of 12.4 km is approximately 1.7 km. This means that such a photon can travel a few kilometers before experiencing a photonuclear interaction with an air molecule nucleus and generate a neutron.

6.2 Conclusions

The main contributions of this chapter can be summarized as follows:

1. We have simulated the neutron production through photonuclear interactions by the large ensemble of energetic photons generated during RREA processes. Present modeling results of neutron energy distribution are consistent with those reported in [*Carlson et al.*, 2010; *Drozdov et al.*, 2013].

2. Using Monte Carlo models simulating the dynamics of energetic electrons accelerated by the electric field produced during the stepping process of high-potential lightning leaders and the corresponding bremsstrahlung photons, we have quantified theoretically the energy distribution of neutrons that are produced through photonuclear interactions. We estimate that significant amount of neutrons, up to approximately $\sim 10^{14}$, can be generated by a TGF-producing lightning leader that creates an electrical potential drop of 100 MV in the tip region.

Chapter 7

X-ray Emissions Produced by Stepping Lightning Leaders

Both natural cloud-to-ground and rocket-triggered lightning flashes have been found to be associated with intense and brief bursts of X-ray emissions. In this chapter, using the Monte Carlo models for electrons and photons described in Chapter 2, we calculate theoretically the energy spectra of X-rays that would be measured on the ground and that would be produced by the bremsstrahlung emission of thermal runaway electrons accelerated in the inhomogeneous electric field produced around negative CG lightning leader tips forming potential drops of 5 and 10 MV. Moreover, we quantify the radial dependence of photon energy spectra and the effective radiation exposure that would be received by human beings in the nearby region. The results presented in this chapter have been published in [*Xu et al.*, 2014].

7.1 Modeling of Stepping Lightning Leaders

Similarly to the method employed in previous chapters, X-ray emissions produced by stepping -CG lightning leaders are modeled following a three step procedure. First, we use the method of moments described in Section 2.1 in order to calculate the electric field produced in the vicinity of lightning leader tips. Next, a Monte Carlo model of electrons, as described in Section 2.2, is employed to simulate the dynamics of thermal runaway electrons in the lightning leader field obtained

during the first step. Finally, from the differential cross section of bremsstrahlung radiation presented in Section 2.3.1, we model the production of X-rays and their subsequent propagation in the Earth's atmosphere using the Monte Carlo model for photons described in Section 2.3. The production mechanism and the relevant collisional processes are schematically depicted in Figure 7.1(a).

The method of moments (see Section 2.1) is used to solve the integral equations of the electric potential and compute the electric field produced in the vicinity of a -CG lightning leader tip. For the sake of simplicity, we consider one lightning leader as a straight perfectly conducting wire. Stepping lightning leaders typically produce an electrical potential drop on the order of 10 MV in the tip region [*Bazelyan and Raizer*, 2000, p. 106; *Rakov and Uman*, 2003, Section 4.2, p. 111]. In the present study, we mainly study lightning leaders with two representative electrical potentials: 5 MV and 10 MV. The electric potential of the lightning leader tip with respect to the ambient potential is approximately $V_0{=}E_0 l/2$ [*Bazelyan and Raizer*, 2000, p. 54], where E_0 is the ambient large-scale thunderstorm electric field, and l is the length of the unbranched leader channel. In particular, the quantity l is chosen as 1 km and the radius of the leader channel is 1 cm [*Rakov and Uman*, 2003, Section 4.4.6, p. 134]. For a 5 MV leader, we assume an ambient large-scale thunderstorm electric field of 0.1 kV/cm [e.g., *Marshall et al.*, 2001]. In the case of the 10 MV leader, the ambient electric field is considered to be 0.2 kV/cm [e.g., *Marshall et al.*, 2001].

The Monte Carlo model for electrons described in Section 2.2 to simulate the propagation and collisions of electrons in air under a given applied electric field. The initial location of the electrons is set at 15 cm from the leader tip in order to avoid the acceleration of electrons in unphysically high electric fields (see [*Celestin and Pasko*, 2011]). The initial energy of the thermal runaway electrons is defined as 65 keV because those are believed to be produced by streamer discharges developing in the vicinity of the leader tip during the negative corona flash process [*Celestin and Pasko*, 2011]. One notes that because of the strong acceleration of the thermal runaway electrons in the leader field, the exact value of the initial energy chosen is not important as long as the initial electrons are runaway in the leader field ($>$ a few keVs). Bremsstrahlung radiation resulting from the deflection of energetic electrons by air molecules is then calculated using the time evolution

of electron energy distributions. We note that X-rays produced by the initial production of thermal runaway electrons by streamers would have typical energies from a few tens of eVs to a few tens of keV and would therefore be absorbed over distances shorter than ∼40 m.

Besides the stepping of lightning leaders, the attachment process is known to be associated with X-ray emissions [e.g., *Howard et al.*, 2010]. Nevertheless, our present knowledge about the attachment processes is limited and, in this study, we mainly focus on X-rays generated by the stepping process of lightning leaders. Specifically, it is considered that lightning leaders descend vertically in a series of discrete steps [*Bazelyan and Raizer*, 2000, p. 84] towards the ground and X-rays are not emitted between steps [*Dwyer et al.*, 2005]. The step length of lightning leaders in -CGs typically measured in experiments is on the order of tens of meters [e.g., *Rakov and Uman*, 2003, Table 1.1]. It is chosen as 10 meters in the present study. Note that this particular length has little effect on calculations of X-ray energy spectra received on the ground for radial distances greater than ∼30 m. We also assume that X-rays produced after the formation of each new step follow the characteristic energy spectra of bremsstrahlung photons produced by the 5 MV or 10 MV lightning leaders as shown in Figure 7.1(c). As for the beaming of X-rays, it is assumed to be isotropic and within a downward solid angle of 2π, following the suggestion in [*Saleh et al.*, 2009].

Schaal et al. [2012] observed the X-ray radiation of a -CG lightning event (MSE 10-01) using TERA. The stepped leader in this event had an average speed of 4.22×10^5 m/s [e.g., *Howard et al.*, 2010; *Schaal et al.*, 2012]. They reported X-ray radiation in this event 891 μs prior to the time of return stroke, corresponding to the stepping of lightning leaders from approximately 376 m altitude to the ground [*Schaal et al.*, 2012]. This range of altitudes (i.e., from 376 m to the ground) is adopted as a reference for the source locations of X-rays in our modeling. However, we note that the choice of source altitudes is not critical in this study, especially for analyzing the energy spectra of X-rays received at ground level for radial distances greater than ∼30 m, because X-rays emitted from leader steps at relatively high altitudes are severely absorbed by the Earth's atmosphere due to the high air density that they experience during their propagation towards the ground. It is also supported by the fact that X-ray radiation significantly intensifies as lightning

leader approaches the ground [e.g., *Schaal et al.*, 2014].

When evaluating the significance of radiation to living organisms, the effective radiation dose with units of Sievert (Sv), which is expressed in terms of joules of absorbed energy per kilogram, is usually used [*Knoll*, 2010; *Dwyer et al.*, 2010]. In the present study, after obtaining the spatial energy distributions of X-rays at ground level, the effective radiation dose that can be received by the human body is calculated following the approach described in [*Dwyer et al.*, 2010, equation (2)]. In particular, the energy-dependent factors used for converting X-ray fluxes into radiation doses are obtained from [*Pelliccioni*, 2000].

7.2 Modeling Results

Figure 7.1(b) shows modeling results for the inhomogeneous electric field produced in the vicinity of a 5 MV lightning leader tip calculated using the method of moments and the positions of accelerating thermal runaway electrons at t=7.4 ns, which is chosen here for the sake of illustration, derived from the Monte Carlo simulation. As illustrated in this figure, the electric field produced by the 5 MV lightning leader is highly inhomogeneous [e.g., *Celestin et al.*, 2012] and the accelerating thermal runaway electrons are guided by the electric field lines. Figure 7.1(c) presents the time-averaged energy distribution functions obtained from the acceleration of thermal runaway electrons in the electric fields produced by stepping lightning leaders with two electric potentials: 5 MV and 10 MV. The curves are normalized by first integrating over the time evolution of energy distribution and then dividing by the duration of the acceleration process and the number of electrons initially injected in the simulation. The integration of the distributions presented in Figure 7.1(c) over the electron energy yields the amount of electrons generated per thermal runaway electron injected in the simulation. We note that the maximum kinetic energy that thermal runaway electrons get in the electric field of the 10 MV lightning leader is close to 2 MeV, resembling the most energetic X-rays typically observed by TERA [*Saleh et al.*, 2009; *Schaal et al.*, 2012]. In Figure 7.1(c), the small peaks around ∼70 keV are a consequence of the discreteness of the numerical integration of the spectra over time. Specifically, the peaks are produced by the initial group of thermal runaway electrons swiftly propagating in

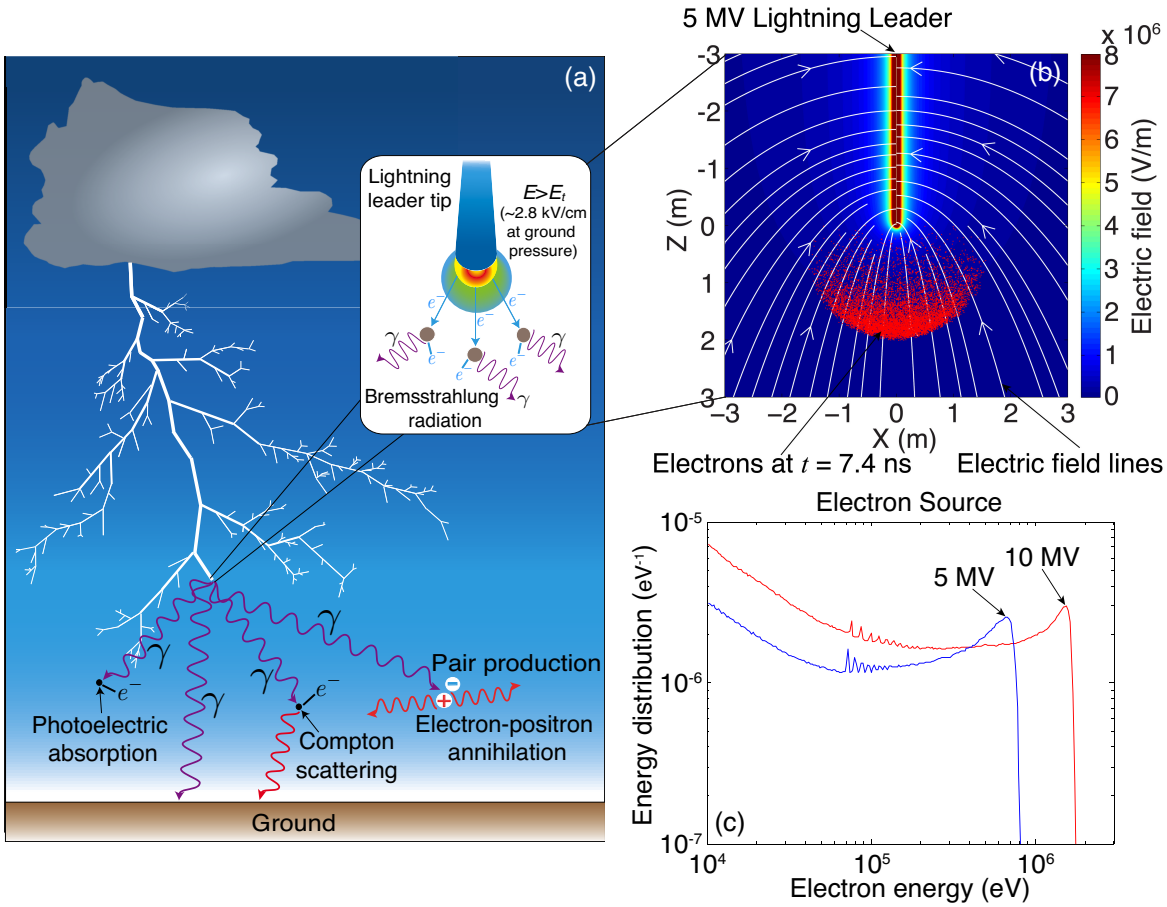

Figure 7.1. (a) Schematic of X-ray emissions produced during the stepping process of lightning leaders in a -CG flash, including acceleration of thermal runaway electrons in the electric field produced around lightning leader tip region, production of X-rays in bremsstrahlung emissions, and their further propagation in the Earth's atmosphere. (b) Simulation results for the inhomogeneous electric field produced by a 5 MV lightning leader calculated using the method of moments and the positions of accelerating thermal runaway electrons at t=7.4 ns derived from Monte Carlo simulation results. (c) Time-averaged electron energy distributions representing the acceleration of one thermal runaway electron in the electric fields produced around the tip region of stepping lightning leaders for two characteristic electrical potentials: 5 MV and 10 MV. The curves are normalized so that the integration over the electron energy yields the amount of electrons generated per thermal runaway electron injected in the simulation. The results are obtained in air at ground level atmospheric density.

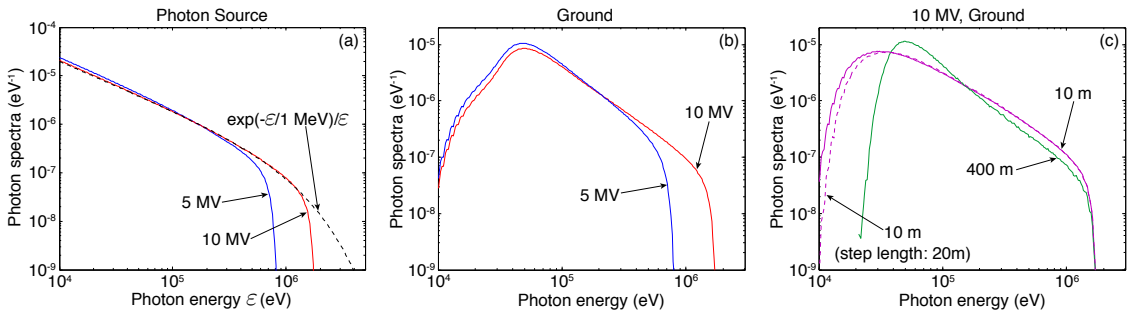

Figure 7.2. Photon energy spectra: (a) produced at the source by the bremsstrahlung radiation of energetic electrons accelerated in lightning leader fields for two electrical potentials: 5 MV and 10 MV (see Figure 1c); (b) collected at ground level within a radial distance of 500 m from the ground impact location of the lightning channel during the process of leader propagation from 376 m altitude to the ground. (c) Spectrum of photons collected at radial distances of 10 m and 400 m from the location of the 10 MV lightning channel ground impact. The dashed line represents the energy spectra of photons collected at a radial distance of 10 m but for step lengths of the lightning leader assumed to be 20 m long.

the energy space.

The energy spectra of bremsstrahlung photons at the source are further calculated and shown in Figure 7.2(a) using the time-averaged electron energy distributions presented in Figure 7.1(c). One sees that the predicted energy spectra of X-rays corresponding to both 5 MV and 10 MV lightning leaders do not follow the theoretical energy distribution of bremsstrahlung photons produced by electrons with an exponential energy distribution and an e-folding kinetic energy. As expected, we also observe that the energy spectra of bremsstrahlung photons produced by a 10 MV lightning leader are harder than those produced by a 5 MV lightning leader.

Figure 7.2(b) shows modeling results of the cumulative energy spectra of X-rays measured by ideal detectors (i.e., without detector response) during leader propagation from 376 m altitude to the ground in the case when detectors are uniformly placed per unit area within a radial distance of 500 m from the ground impact location of the lightning channel. A distance of 500 m is chosen to represent the detecting range covered by TERA [e.g., *Saleh et al.*, 2009]. Figure 7.2(c) presents the X-ray energy spectra in the 10 MV lightning leader case as measured by an ideal detector at radial distances of 10 m and 400 m from the lightning

channel ground impact. To obtain these results, we have accumulated the partial spectra of those shown in Figure 7.2(b) over concentric rings defined on the ground at radial distances between 8 and 12 m, and 398 and 402 m for the distances of 10 m and 400 m, respectively. We have verified that these rings are sufficiently thin so that the spectra vary only negligibly from the inner radius to the outer radius of each ring.

The dashed line in Figure 7.2(c) also shows the energy spectra of photons collected at the radial distance of 10 m but for the case that the step length of the lightning leader is 20 m, representing the uncertainties caused by the choice of leader step length (specifically, quantitative deviations observed in the energy range below 50 keV). For detectors located very close to the ground impact location of the lightning channel (i.e., within a few tens of meters), the recorded X-rays mostly originate from the last leader step, before the attachment process. This is mainly due to the fact that the decreased distance between the X-ray source and detectors enables a large population of photons with energies of a few tens of keV to reach detectors without being absorbed in the air. For instance, the mean free path for X-rays with an energy of 20 keV is ∼13 m at ground level air density and therefore, one should observe an enhancement in the spectrum in this range of energy for detectors located at a radial distance of 10 m if the step length decreases from 20 m (corresponding to the last step terminating at 16 m altitude) to 10 m (with the last step at 6 m altitude) as evident in Figure 7.2(c). However, we note that the choice of step length hardly changes the energy spectra of X-rays collected, for example, at a radial distance of 400 m. More interestingly, the X-ray energy spectra show a significant radial dependence: as the distances from lightning channel decrease, the energy spectrum extends to the range below ∼50 keV and becomes softer in the range above ∼50 keV.

7.3 Discussion

In this study, we have simulated the dynamics of thermal runaway electrons accelerating in the inhomogeneous electric field produced around lightning leader tips [*Celestin and Pasko*, 2011] to study the production of bremsstrahlung X-ray bursts from -CG lightning discharges. Using Monte Carlo simulations of the transport

of these X-rays down to the ground, we predict the characteristic time-averaged electron energy distributions produced by 5 MV and 10 MV lightning leaders and the corresponding raw energy spectra of X-rays that can be observed at ground level caused by the stepping process of lightning leaders in -CGs.

The energy distributions of electrons and resultant X-rays obtained from the acceleration of thermal runaway electrons in lightning leader fields are directly related to physical properties of the stepping lightning leaders [e.g., *Xu et al.*, 2012; *Celestin et al.*, 2012]. This point is illustrated by the comparison of source X-ray energy spectra presented in Figure 7.2(a). We observe that the maximum energy that electrons acquire from the 10 MV lightning leader potential, which is defined by the electric potential drop in the leader tip region, is approximately twice that of the 5 MV lightning leader. Moreover, the hardness of X-ray energy spectra is also different between the two leader potentials. In contrast to lightning leaders in -CGs, long unbranched lightning leaders in ICs generally possess larger electric potentials and have been demonstrated to be able to generate gamma-rays with energies up to a few tens of MeV and explain satellite measurements of TGF spectra [*Xu et al.*, 2012; *Celestin et al.*, 2012]. Due to the correlation between the photons and the causative potential drop produced by lightning leaders, measurements of X-ray radiation can bring a significant insight on the electrical properties of lightning leaders.

Modeling results of X-ray energy spectra received at ground by ideal detectors during the stepping process of lightning leaders are comparable to TERA measurements in terms of general shape and spectral hardness [*Schaal et al.*, 2012]. However, detailed comparison with TERA measurements should take detector response into consideration. As the radial distance from the lightning channel increases, photons received at ground, on average, experience more scatterings by air molecule. Given that photons are completely absorbed in photoelectric absorption and lose considerable energy in Compton scattering, energy spectra of photons received at ground should demonstrate significant radial dependence as evident in Figure 7.2(c). The sharp decrease in the energy region below $\sim$50 keV stems from the fact that the increased amount of photoelectric absorption hampers low-energy photons from reaching further radial distances. As for the change in the energy spectrum above $\sim$50 keV, it primarily reflects the fact that photons undergo more

Compton scatterings to reach further radial distances, which significantly softens the energy spectra. Furthermore, we note that the 511 keV positron annihilation line is not characteristics for the phenomenon discussed in this study. This is mainly because of the fact that photons typically produced by stepping lightning leaders in -CGs have energies up to a few MeV and electron-positron pair production is not the dominant collisional process in this energy range.

Based on XCAM measurements, *Schaal et al.* [2014] have suggested that the compact source region in front of lightning leaders, responsible for production of X-rays, has a radius between 2 and 3 m and the electric charge contained in this region is on the order of 10^{-4} C. From the quadratic relation between the electric charge contained in the streamer zone and the electrical potential of the associated lightning leader [*Celestin and Pasko*, 2011], we estimate from first principles that the charge in the streamer zone of a 5 MV lightning leader is approximately 2.8×10^{-4} C. As for the 10 MV lightning leader, the electric charge in the streamer zone is 1.1×10^{-3} C. We also estimate the radius of the streamer zone in front of the 5 MV and 10 MV lightning leaders and the values are, respectively, 2 m and 4 m. Both the electric charge contained in the X-ray source region and the size of the streamer zone are consistent with results in [*Schaal et al.*, 2014].

Using the singly differential cross section of bremsstrahlung emissions and estimating the number of initial thermal runaway electrons injected by steamers in the leader field from the method derived in [*Celestin and Pasko*, 2011], we find that a 10 MV lightning leader should produce less than 10^{15} photons (upper limit) with energies above 10 keV. From this number, we evaluate that the maximum radiation exposure at ground level, corresponding to the ground impact location of the lightning channel, as produced during the stepping process of a 10 MV lightning leader during its propagation down to the ground is $\sim$0.4 mSv. Given that $\sim$12 mSv is the effective radiation dose for a full-body CT scan [e.g., *Brenner and Elliston*, 2004], this amount of radiation is considered to be insignificant to humans.

7.4 Conclusions

The main contributions of this chapter can be summarized as follows:

1. We have predicted theoretically the energy spectra of X-rays produced by the bremsstrahlung radiation of thermal runaway electrons accelerated in the inhomogeneous electric field of lightning leader producing an electrical potential drop of 5 and 10 MV in the tip region.

2. The radial dependence of X-ray energy spectra received at ground level during the leader stepping process is quantified and discussed.

3. We have shown that the X-ray energy spectrum depends on the physical properties of the associated lightning leaders. Consequently, comparisons between modeling results and X-ray measurements could be used for diagnostics of electrical properties of lightning stepped-leaders that are still not well understood at present.

4. Modeling results suggest that the ground radiation generated during the stepping phase of CG lightning leaders is harmless to humans, with typical magnitudes on the order of a fraction of mSv.

Chapter 8

Optical Emissions Associated with Energetic Electrons Produced by Cloud-to-Ground Stepping Lightning Leaders

In Chapter 7, we have studied the X-ray emissions resulting from the acceleration process undergone by thermal runaway electrons under the electric field produced by stepping lightning leaders during CG flashes. In this chapter, we focus on the fluorescence optical emissions induced by this acceleration process. We apply the Monte Carlo model of electrons described in Section 2.2 to simulate consistently the dynamics of thermal runaway electrons and their lower-energy secondary electrons in the energy region above 10 eV. The optical emission model described in Section 2.4 is combined with these Monte Carlo simulations in order to explore the emission of fluorescence photons resulting from the excitation of air molecules produced by the large fluxes of low- and high-energy electrons involved. We also investigate the morphological and spectroscopic features of these optical emissions.

8.1 Modeling of Optical Emissions

In this chapter, we investigate optical emissions originating from the radiative relaxation of excited species of neutral and ionized nitrogen molecules, primarily the second positive band system of N_2 ($C^3\Pi_u \rightarrow B^3\Pi_g$) ($2PN_2$) and the first negative band system of N_2^+ ($B^2\Sigma_u^+ \rightarrow X^2\Sigma_g^+$) ($1NN_2^+$). Emissions from these two optical band systems are quantified, from first principles, in the framework of Monte Carlo simulations by modeling dynamically the evolution of excited species. Specifically, similarly to the modeling studies in Chapter 7, we first use the method of moments [*Balanis*, 1989, p. 670] described in Section 2.1 in order to calculate the electric field produced near the tip region of lightning leaders during the stepping process in CGs and the Monte Carlo model for electrons (see Section 2.2) is then employed to simulate thermal runaway electrons accelerating in this leader field. Finally, we use the optical emission model described in Section 2.4 to account for collisional quenching processes and evaluate the resultant optical emissions.

Based on detailed simulations of X-ray emissions during the stepping process of lightning leaders, *Xu et al.* [2014] have shown that typical ground-based measurements of X-rays are similar to those produced by lightning leaders with an electrical potential of 5 MV. This electrical potential drop is therefore chosen in the present study. For such a potential drop, we assume an ambient large-scale thunderstorm electric field of 0.1 kV/cm [e.g., *Marshall et al.*, 2001]. The length of the lightning leader is considered as 1 km and the radius of the leader channel is 1 cm [*Rakov and Uman*, 2003, Section 4.4.6, p. 134].

As for the simulation of electrons accelerating in the lightning leader electric field, the initial energy of thermal runaway electrons is taken as 65 keV [*Celestin and Pasko*, 2011] and the initial location of these electrons is set to be 15 cm from the leader tip, where the electric field is equal to 27.4 kV/cm. In addition to the high-energy electron simulations, we consistently simulate the production, dynamics, and collisions of low-energy secondary electrons with kinetic energy above 10 eV. We note that the lower limit in energy in our simulations (10 eV) is below the threshold energy for producing $N_2(C^3\Pi_u)$ or $N_2^+(B^2\Sigma_u^+)$, and thus, the generation of upper excited states responsible for optical emissions from $2PN_2$ and $1NN_2^+$ is fully modeled.

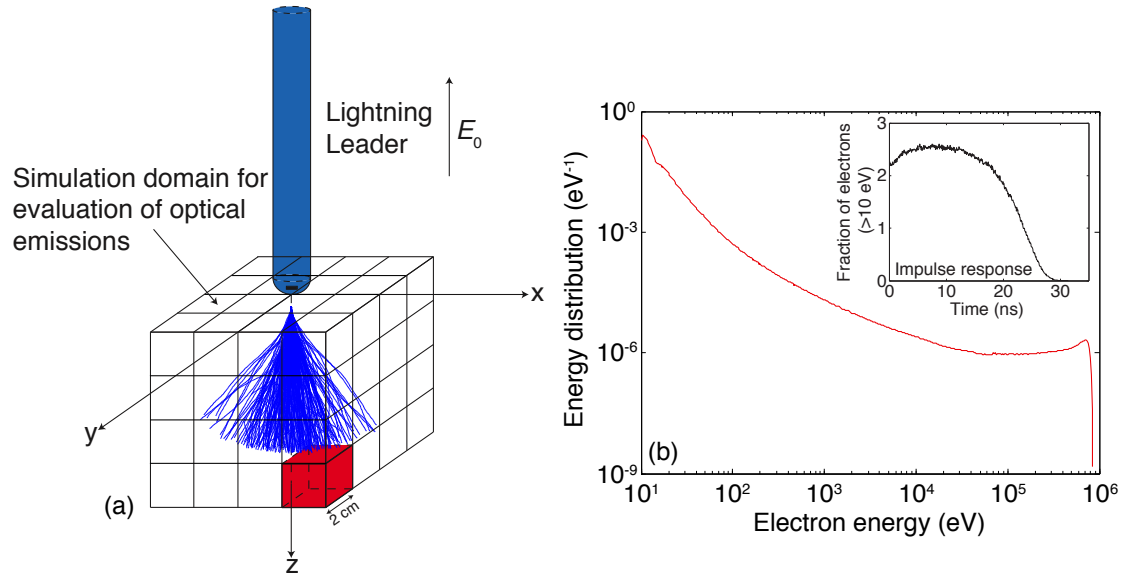

Figure 8.1. (a) Sketch of the simulation domain used for evaluation of optical emissions. The cube marked in red represents the numerical cell defined for keeping track of the evolution of excited species ($N_2(C^3\Pi_u)$ or $N_2^+(B^2\Sigma_u^+)$); (b) Time-averaged electron energy distribution representing the acceleration process of thermal runaway electrons in the 5 MV lightning leader field. The inset shows the fraction of electrons with energy above 10 eV per electron injected versus time. The results are obtained at ground-level atmospheric density.

For the two types of optical emissions considered in the present study, at typical altitudes of CG lightning leaders, the corresponding upper excited states are mostly populated via direct impact excitations by electrons and depopulated via collisional quenching with air molecules. In order to explore the morphological characteristics of the associated fluorescence emissions, the region in the vicinity of the lightning leader tip is discretized using a numerical cartesian grid of 401×401×401 grid points, corresponding to a total spatial extent of 8 m in x, y, and z directions. Figure 8.1(a) shows an illustration of the simulation domain used for the evaluation of optical emissions. The cube marked in red represents a numerical cell defined in Monte Carlo code for keeping track of the production of excited species ($N_2(C^3\Pi_u)$ or $N_2^+(B^2\Sigma_u^+)$). One of the advantages that is afforded by the Monte Carlo model is that it is capable of accurately describing and recording the spatial and temporal information on all collisions taking place in the system. Owing to this advantage, the excitation of $N_2(C^3\Pi_u)$ and $N_2^+(B^2\Sigma_u^+)$, especially their production location

and time, can be directly derived from Monte Carlo simulations. Concerning the deactivation of excited species, for each numerical cell defined in the simulation domain (see Figure 8.1(a)), we use an optical emission model, similar to that described by *Liu and Pasko* [2004], to account for the quenching processes with air molecules and generation of fluorescence photons via spontaneous emissions. Note that our present calculations do not take into account the effects of radiative transfer between the source of emission and the observer.

8.2 Modeling Results

Figure 8.1(b) shows the time-averaged electron energy distribution caused by thermal runaway electrons accelerating in the electric field produced in the tip region of a 5 MV lightning leader. The results are obtained in air at ground-level atmospheric density. The curve is normalized so that the integration over electron energy yields unity, representing the characteristic energy-gaining process undergone by every thermal runaway electron when propagating in the high field region. As shown in this figure, while primary electrons swiftly gain energy in the leader field, secondary electrons with energy between 10 eV and 10 keV are also generated along the trajectories, giving rise to emissions of fluorescence photons. The inset shows the fraction of electrons with energies above 10 eV per thermal runaway electron injected versus time, namely the impulse response function of the system. We note that the acceleration process of one thermal runaway electron can lead to a multiplication factor of approximately 2.5 for electrons with energy above 10 eV.

Figure 8.2(a) and 8.2(b) show optical emissions from $2PN_2$ and $1NN_2^+$, respectively, arising from the continuous production of thermal runaway electrons into the electric field of the 5 MV lightning leader by streamer discharges during the negative corona flash stage. Following *Schaal et al.* [2012], we assume that the continuous emissions rate of thermal runaway electrons is 10^{17} s^{-1}. Note that the "irregular" appearance of the emitted light is a numerical feature caused by the limitation in the number of particles modeled in the Monte Carlo simulation and can be improved by increasing the number of particles used. We expect that given the true fluxes of thermal runaway electrons, the shape of the fluorescence emissions

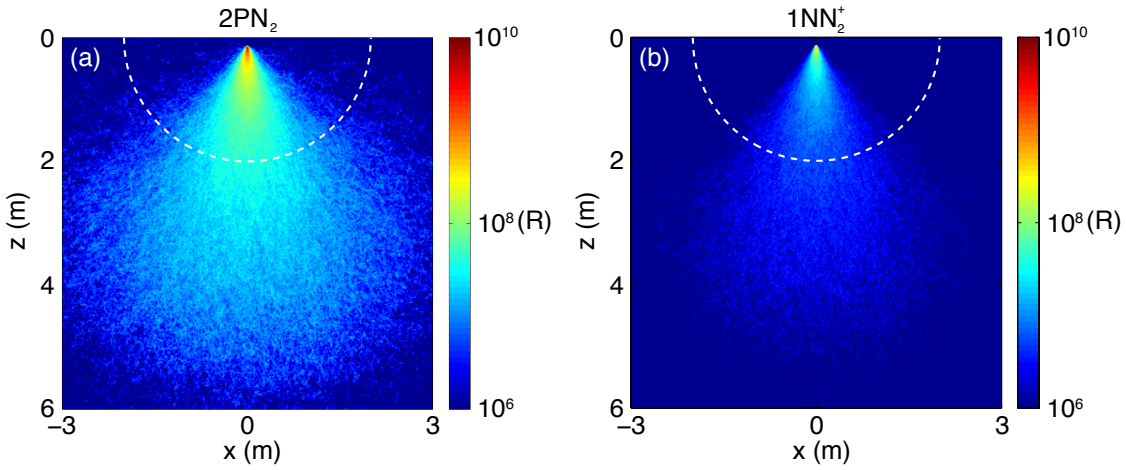

Figure 8.2. Optical emissions of (a) $2PN_2$ and (b) $1NN_2^+$ resulting from continuous emission of thermal runaway electrons into the 5 MV lightning leader field during the negative corona flash stage considering an emission rate of 10^{17} s^{-1} [*Schaal et al.*, 2012]. The results are calculated using a convolution technique at ground-level atmospheric density. The dashed lines represent the theoretical size of the associated streamer zone.

would be smooth. Moreover, from the linear relationship between the electrical potential possessed by lightning leaders and the radius of associated streamer zone [e.g., *Celestin and Pasko*, 2011], we estimate that the fan-shaped streamer zone present in front of the 5 MV lightning leader would have a radius of 2 m, as marked by dashed lines in Figure 8.2.

One can estimate that, for a continuous injection of thermal runaway electrons, steady state energy distribution is reached in a few tens of nanoseconds (see inset of Figure 8.1(b)). Directly modeling optical emissions associated with the continuous injection of thermal runaway electrons under lightning leader field is computationally expensive because of the large number of numerical cells defined in Monte Carlo simulation and the abundant generation of secondary electrons. One way to overcome this difficulty is to perform a time convolution of the impulse response of the system. Instead of directly modeling the continuous injection, we calculate the temporal evolution of the fluorescence beam corresponding to an ensemble of accelerating thermal runaway electrons in the case that they are all injected simultaneously at t=0 into the lightning leader field (impulse response). Considering that the number of injected thermal runaway electrons is steady and knowing the continuous emission rate, we derive the steady state optical emissions by convolv-

ing the impulse response over time. We have verified that directly modeling a continuous thermal runaway electron injection leads to results that are consistent with those derived from the convolution technique, but the computation time is significantly prolonged.

The fluorescence beams of $2PN_2$ and $1NN_2^+$, as shown in Figure 8.2, exhibit a conical shape, similar to laboratory observations of fluorescence from nitrogen at high pressures excited by energetic electrons [*Davidson and O'Neil*, 1964, Figure 1]. The diameter of this beam is approximately 6 m, as defined by the maximum distance that thermal runaway electrons travel in the leader tip region. Optical emissions of $2PN_2$ are more intense because the frequency of collisional quenching between $N_2^+(B^2\Sigma_u^+)$ and air molecules is much higher at this pressure, even if the excitation rates of $N_2(C^3\Pi_u)$ and $N_2^+(B^2\Sigma_u^+)$ are close.

8.3 Discussion

In this study, using Monte Carlo simulations for electrons with energy above 10 eV, we have studied the dynamics of thermal runaway electrons and their low-energy secondary electrons when propagating in the inhomogeneous electric field produced around the tip region of a 5 MV lightning leader. Using an optical emission model combined with these Monte Carlo simulations, we have predicted theoretically the optical emissions from $2PN_2$ and $1NN_2^+$ associated with the acceleration of runaway electrons.

We have also evaluated the optical emissions associated with thermal runaway electrons by assuming that the large population of high- and low-energy electrons involved in the acceleration process follows a uniform spherical spatial distribution, with a radius of 6 m. The total number of electrons with energy above 10 eV, in the steady state caused by continuous emission of thermal runaway electrons with a rate of 10^{17} s^{-1}, is obtained by convolving the impulse response shown in the inset of Figure 8.1(b). The value is approximately 5.5×10^9. With the knowledge about the energy and number density distributions of electrons, we find that the corresponding intensities of optical emissions from $2PN_2$ and $1NN_2^+$ are, respectively, 5.6×10^{10} R and 7.5×10^9 R, in good agreements with results derived using the set of numerical models described in the present study (see Figure 8.2).

This agreement validates the technique used by *Xu et al.* [2015] for quantifying optical emissions associated with TGFs.

The electrical potential possessed by lightning leaders can locally enhance the electric filed in the tip vicinity to an intensity exceeding the electrical breakdown field (∼30 kV/cm at ground level), leading to onset of streamer discharges [*Bazelyan and Raizer*, 2000]. A single streamer first rapidly ionizes air molecules and becomes a space charge wave. In principle, the streamer can then propagate long distances in the form of a narrow filament. When the radius of the streamer head reaches the magnitude of effective photoionization range, the streamer head may start branching and the streamer no longer propagates steadily [*Liu and Pasko*, 2004]. In view of the whole streamer zone, streamers are continuously initiated in the tip region and the number of streamers constituting the streamer zone at every moment of time can be estimated from the total charge contained in the streamer zone and the characteristic charge carried by a streamer [*Bazelyan and Raizer*, 2000, p. 70]. For the streamer zone associated with a 5 MV lightning leader, it is found that the total number of streamers it contains is approximately 2.8×10^5.

In order to estimate optical emissions radiated from the streamer zone of a 5 MV lightning leader, modeling results of a well-developed negative streamer propagating in a homogeneous field of 20 kV/cm at 14.5 ns after its ignition are used. These results had been obtained in a parametric study performed in preparation of the work reported by *Celestin and Pasko* [2010a] (see this work for further information on the model and the exact simulation configuration and parameters). Note that, at this stage, the radius of the streamer head is approximately 0.13 cm and excited species are mainly concentrated in the head. Knowing the number of streamers, we first calculate the total amount of excited species in the streamer zone. The number density of excited species in the streamer zone is further derived by assuming that they follow a uniform spherically symmetric spatial distribution with a characteristic radius representing the dimension of the streamer zone. Finally, we calculate corresponding optical emissions and found that the intensities are 4.8×10^{11} R and 7.6×10^8 R for $2PN_2$ and $1NN_2^+$, respectively. The results show that optical emissions from the streamer zone are one order of magnitude more intense than those associated with accelerating thermal runaway electrons.

There are notable differences between optical emissions associated with thermal runaway electrons and those from the streamer zone. First, fluorescence light initiated by thermal runaway electrons accelerating in lightning leader field is estimated to take place over a larger region than that occupied by the streamer zone (see Figure 8.2), although the luminosity is weaker. Furthermore, considering that the capability of electrons in generating fluorescence photons from $2PN_2$ and $1NN_2^+$ is highly energy-dependent [*Xu et al.,* 2015], the intensity ratio between $2PN_2$ and $1NN_2^+$ is related to the energetics of electrons involved. Since electrons in streamer discharges and those produced by thermal runaway electrons follow different energy distributions, this ratio would be inherently different between these two processes. For instance, using the time-averaged energy distribution shown in Figure 8.1(b) and formula described by *Celestin and Pasko* [2010a, equation (1)], we find that the ratio of the intensity of $2PN_2$ to that of $1NN_2^+$ corresponding to the acceleration process of thermal runaway electrons is approximately 7.5, whereas this quantity is approximately 900 for the associated streamer zone. Measurements of the above-mentioned differences, including size and intensity ratio, can provide insightful information to understand the stepping mechanism and associated X-ray production.

Besides the above-mentioned differences, optical emissions resulting from accelerating thermal runaway electrons are temporally separated from those of the streamer zone. Thermal runaway electrons are mostly produced during the early establishment of the streamer zone, namely the negative corona flashes. At this stage, thermal runaway electrons promptly propagate in the high field region and the associated fluorescence beam rapidly expands on a timescale of ∼30 ns. However, the streamer zone is not formed until the space charge field produced by propagating streamers lowers the electric field in the streamer zone down to the stability field of negative streamer propagation (∼12.5 kV/cm) [e.g., *Bazelyan and Raizer*, 2000; *Celestin and Pasko*, 2011]. The typical duration of this process is on the order of ∼1 μs [e.g., *Moss et al.*, 2006]. It is thus conceivable that, ideally, measurements would capture an expanding fluorescence beam from accelerating thermal runaway electrons followed by more intense optical emissions from propagating streamers.

8.4 Conclusions

The main contributions of this chapter can be summarized as follows:

1. We have quantified theoretically the optical emissions induced by the strong acceleration of thermal runaway electrons in the highly inhomogeneous electric field produced by stepping lightning leaders.

2. The results demonstrate that this acceleration process is accompanied with not only high-energy X-ray emissions but also detectable levels of low-energy optical emissions. The fluorescence beam exhibits a conical shape and appears to be larger than the streamer zone associated with the lightning leader.

3. We emphasize that the size of the fluorescence beam largely depends on the electrical properties of the associated lightning leader and the intensity ratio between $2PN_2$ and $1NN_2^+$ reflects the intrinsic difference in the energetics of electrons involved. Therefore, corresponding measurements would carry valuable information to understand the stepping mechanism and associated X-ray production processes.

Chapter 9

Suggestions for Future Research

In this chapter, we suggest possible research topics related to TGFs for the continuation to the work presented in this dissertation.

9.1 Radiation Exposure in TGFs

As presented in Chapter 3, a source altitude of 12.4 km is suggested for TGFs based on the production mechanism of thermal runaway electrons accelerated by the electric field of high-potential lightning leaders. This altitude happens to be closer to the altitudes of commercial flights than originally thought (e.g., 15 km as suggested by *Dwyer and Smith* [2005]). At the source altitude, it is widely believed that 10^{17} energetic electrons (MeV) are produced in order to explain satellite measurements of TGF fluence [*Dwyer and Smith*, 2005]. These fluxes of energetic electrons, along with the large population of resultant secondary electrons and bremsstrahlung photons, can create significant amount of radiation doses to aircraft passengers if the flight is close to the TGF source. As a result, the probability for an aircraft to be exposed to radiation from TGF source and the exposure of aircraft passengers should be estimated.

9.2 Comparison with Ground-based Measurements of X-ray Bursts

As we point out in Chapter 7, the energy spectrum of X-rays produced during the stepping process of CG lightning leaders that can be measured at ground highly depends on the electrical properties of associated lightning leaders. Comparison between modeling results and ground-based measurements represents an important task in future research as it may provide useful information to diagnose the electrical properties of stepping lightning leaders, constrain the underlying X-ray production mechanism, as well as improve our understanding of the stepping process. However, the energy spectra of X-rays produced during the stepping process of 5 MV and 10 MV lightning leaders presented in Chapter 7 (Figure 7.2(b)) are those that would be observed using ideal detectors. The energy spectra derived from ground-based measurements [e.g., *Saleh et al.*, 2009; *Schaal et al.*, 2012] are expected to be different from these results given that X-rays only partially deposit their energies in the detectors, for example, by Compton scattering. Therefore, to perform comparisons with TERA or XCAM measurements, the detector response needs to be taken into account and represents important future research topic.

9.3 Future Satellite Missions

Modeling results of optical emissions associated with TGFs presented in Chapter 5 are of great importance in view of the future space missions, Atmosphere-Space Interaction Monitor (ASIM) and Tool for the Analysis of RAdiation from lightNIng and Sprites (TARANIS), that will detect TGFs from the nadir direction with photometric measurements. ASIM is an observatory designed by the European Space Agency (ESA) that will be placed on the exterior of the International Space Station (ISS) in order to measure high-altitude lightning activities. TARANIS is designed by the Centre National d'Etudes Spatiales (CNES) for the studies of magnetosphere-ionosphere-atmosphere coupling via observations of transient luminous events. Measurements of optical emissions from the source region of TGFs can be compared with modeling results presented in the present dissertation in order to infer and constrain the production mechanism of TGFs. Recent satellite

measurements of TGF fluence usually suffer from dead time loses. With faster detectors and reduced dead time effects, future space missions would reveal the true fluence in TGFs, which is also critical in understanding the internal physics of this energetic phenomenon.

Moreover, as we conclude in Section 1.3 and Section 1.4.3, the energy spectrum of photons is weakly dependent on the magnitude of the homogeneous electric field driving RREAs [i.e., *Babich et al.*, 2004; *Dwyer and Smith*, 2005], whereas the energy spectrum of gamma-rays emitted during the acceleration of thermal runaway electrons in the highly inhomogeneous electric fields around lightning leader tips is defined by the physical properties and spatial electric field configurations associated with lightning stepped leaders [*Xu et al.*, 2012; *Celestin et al.*, 2012]. As a consequence, measuring the variation of TGF energy spectra can carry significant information to unveil the origin of TGFs.

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